Rights, Genomes, and Algorithms

Consolidated Edition

A unified volume of the monograph, the master project portfolio, the related court filings, and a 120-credit degree-completion portfolio

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This consolidated edition reproduces, in one cover, six separately prepared documents. Each part retains its own pagination and headings as originally typeset.

About This Volume

This book gathers, under a single cover, the principal written works produced for the *Rights, Genomes, and Algorithms* project. Where the individual pieces were once separate files, they now form consecutive parts of one larger volume.

Contents of this volume

Part I — Rights, Genomes, and Algorithms (Integrated Master Portfolio Edition). The core monograph: a unified treatment of constitutional jurisprudence and biological-data sovereignty, neurogenetic biomarkers and scientific evidence, the Hybrid Semantic Processing Architecture (HSPA v3.0), the MORTAR triage engine, the OpenSSA integrity pipeline and post-quantum security, the international human-rights framework, the PSDLEF Sovereign Exchange Trust, the statement of facts and procedural history, the integrated policy framework, and the applied projects and capstones.

Part II — Master Project Portfolio. The implementation, calculation, projection, and governance package consolidating the thirteen integrated projects (P1–P13), with the perpetual solvent trust framework, the DYREO yield engine, the MORTAR locality-sensitive-hashing triage engine, and the directed-acyclic-graph cryptographic audit infrastructure.

Part III — Court and Bank Filing Execution Checklist. The step-by-step execution checklist for the mandamus-petition filing and the depository-bank trust-account onboarding.

Part IV — Petition for Writ of Mandamus (U.S. Court of Appeals for the D.C. Circuit). The pro-se petition for a writ of mandamus in the matter of *In re McLaughlin & Fernandez*, invoking 28 U.S.C. § 1651(a), Fed. R. App. P. 21, and D.C. Cir. R. 21.

Part V — Petition for Writ of Mandamus and Complaint (Circuit Court for Baltimore County). The ministerially-directed mandamus petition and declaratory-judgment complaint in *McLaughlin & Fernandez v. Bank of New York Mellon, et al.*

Part VI — Bachelor's Degree Completion Portfolio. A prior-learning portfolio that maps the work of Parts I–V, course by course, to a standard 120-credit United States baccalaureate — general education, an area of study (Constitutional Justice, Data Sovereignty & Applied Cryptographic Systems), free electives, and a senior capstone — giving learning outcomes, artefact evidence, a consolidated 120-credit tally, and an assessment and submission plan.

A note on the documents. Part I is a scholarly monograph; Parts II–V are practice documents (a portfolio, a checklist, and two pro-se court filings); Part VI is an academic prior-learning portfolio

mapped to a 120-credit baccalaureate. They are combined here for convenience of reference, not because they share a single genre. The court filings are drafts: bracketed placeholders remain, and each carries its own notice that it is not for filing without review by a licensed attorney. Part VI confers no credit and no degree; credit, and any degree, are awarded only by an accredited institution after its own faculty assessment.

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Part I

Rights, Genomes, and Algorithms

Integrated Master Portfolio Edition

Rights, Genomes, and Algorithms

An Integrated Portfolio of Constitutional Jurisprudence, Genomic Data Sovereignty, Semantic Computing, and Post-Quantum Public Infrastructure

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Inventor & Sovereign Architect

Integrated Master Portfolio Edition

2026

This volume consolidates and supersedes the earlier Rights, Genomes, and Algorithms *monograph and incorporates the HSPA v3.0 / SSE-9 / RIGOR-PRS Master Technical Specification.*

Abstract

This book consolidates a decade of work by a single author into one continuous argument. It moves from constitutional litigation, through the governance of genomic and neurophysiological data, into the design of a semantic-computing architecture and a post-quantum trust chain, and finally into a proposed public financing and treaty framework. The unifying claim is that the same structural weakness — the absence of enforceable constraint on institutions that process personal data — appears simultaneously as a civil-rights injury, as a genomic-privacy defect, as an algorithmic-governance problem, and as a cryptographic and financial one.

The volume has three audiences. For the legal reader it supplies a doctrine-by- doctrine map of actionable claims under 42 U.S.C. § 1983, the statutory lattice of GINA, the ADA, and HIPAA, and the evidentiary standards that govern neurogenetic proof. For the technical reader it reconstructs the Hybrid Semantic Processing Architecture (HSPA v3.0) — its Semantic Execution Graph, its probabilistic control layer, its reconfigurable compute fabric, its nitrogen-vacancy quantum substrate, and its ML-KEM/ML-DSA/SLH-DSA trust chain — and separates what is standardised, what is simulated, and what remains a design target. For the policy reader it sets out an international framework that joins cognitive accessibility, genomic justice, and quantum-secure infrastructure into a single directive.

Throughout, the book maintains a deliberate distinction between four registers of statement: adopted law, legislative or architectural proposal, model and simulation, and design target. That discipline is not a stylistic ornament; it is the central methodological contribution of this edition.

Status and Provenance Notice

A reader who encounters litigation theory, a quantum substrate, a trust indenture, and a United Nations petition inside one cover is entitled to know which claims rest on enactment, which on doctrine, which on computation, and which on aspiration. This edition therefore labels its material.

[ADOPTED LAW]
Binding law or a published standard. Examples: the Administrative Procedure Act; 42 U.S.C. § 1983; the Genetic Information Nondiscrimination Act of 2008; NIST FIPS 203/204/205.

[PROPOSAL]
A legislative, architectural, or organisational design that the author advocates but that no legislature or standards body has adopted. Examples: the Gov-LLM concept; the PSDLEF Sovereign Exchange Trust; the proposed UN commission of inquiry.

[MODEL / SIMULATION]
A computational artefact — a simulation, a reference implementation, or a numerical result produced by the author's own code. Examples: the MORTAR triage engine; the VQE validation of the quantum substrate; the DYREO allocation optimiser.

[DESIGN TARGET]
An engineering or operational goal not yet demonstrated at the scale claimed. Examples: the 600 MW/1 GWh solar-and-battery microgrid; the HARMONI-X cognitive prosthetic; the quantum gravitational reactor.

Where a factual assertion depends on a court docket, an administrative record, or an experimental measurement that the author has not yet placed in the reviewing record, the text carries the marker **[CITE REQUIRED]** and identifies the document that would satisfy it. This convention is inherited from the source portfolio's own working drafts, in which procedural assertions are systematically conditioned on file-stamped proof. It is preserved here because a manuscript that mixes advocacy with verifiable record must be readable as a record.

Two claims that recur and must not be conflated

The normative claim. Existing doctrine under-protects genomic, neurophysiological, and cognitive data, and that under-protection is remediable by identifiable legal and technical means. This claim is argued from primary authority and is open to ordinary legal dispute.

The engineering claim. A specific architecture — HSPA v3.0 — can deliver the privacy, accessibility, and auditability properties the normative claim requires. This claim is supported by simulation and reference implementation, not by fabricated silicon; the reader should treat performance figures as modelled unless a measurement is expressly identified.

A Note on Sources and Method

The material in this book derives from two bodies of primary text. The first is the author's project archive: several hundred files containing pleadings, portfolio volumes, patent disclosures, administrative correspondence, and technical notes. The second is the *HSPA v3.0 / SSE-9 / RIGOR-PRS Master Technical Specification*, a consolidated engineering document that supplies the architecture described in Part IV and the trust model described in Parts VI and VIII.

Three editorial operations were performed on that material.

1. **Consolidation.** Repetitions across portfolio volumes were eliminated; the several drafts of the same petition, report, or claim chart were reconciled into single authoritative texts. Where two drafts disagreed, the more complete and the more recent was preferred, and the discrepancy is noted.
2. **Sourcing.** Every quantitative claim, legal standard, and technical parameter was checked against an external authority. Where the source material asserted a figure without authority, a citation was supplied if one exists (for example, the Government Accountability Office's disability-appeal mortality accounting, the NIST post-quantum standards, or the locality-sensitive hashing literature) and a **[CITE REQUIRED]** marker if none does.
3. **Registering.** Each substantive proposition was assigned to one of the four registers described above, and the prose was rewritten so that the register is visible to a reader who does not read the labels.

The references are collected in the consolidated bibliography at the end of the volume. Primary authority — statutes, regulations, and reported decisions — is cited in the conventional form so that it can be verified independently. The author's own computational results are cited to the artefacts that produced them, and are labelled as **[MODEL / SIMULATION]** throughout.

Reader's Guide

The book is organised so that it can be read in three ways.

As a whole.
Parts I–II establish the legal injury and the constitutional argument; Parts III–VI supply the scientific and technical remedy; Parts VII–VIII situate the remedy in international law and in a financing structure; Part IX confronts the record and its gaps; Parts X–XII generalise the result into policy and close the argument.

As a technical reference.
Parts IV–VI are self-contained. A reader who wants only the architecture can begin at Chapter 11 and follow the six-layer stack to the verification chapter.

As a litigation toolkit.
Parts II and IX are self-contained. Part II states the claims and their doctrinal support; Part IX states what the record does and does not yet prove.

Chapter 2 is the hinge. It sets out the status discipline that governs every later chapter, and a reader who internalises it will be able to place any assertion in the book on one of the four registers without consulting the labels.

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Part I

Executive Synthesis: The Unified Systemic Framework

CHAPTER 1

The Integrated Portfolio: Scope, Structure, and Central Claim

> The several parts of this work are not a miscellany. They are the same argument, made once in the language of doctrine, once in the language of engineering, and once in the language of finance.

1.1 What this book is

This is a single-author monograph that joins four normally separate literatures. The first is constitutional and civil-rights litigation. The second is the governance of genomic and neurophysiological data. The third is the design of secure, accessible computing infrastructure. The fourth is the financing of public-benefit digital infrastructure through trust and treaty instruments.

Its central claim is structural. Institutions that collect personal data — police departments, disability adjudicators, hospital systems, insurers, and agencies that deploy machine-learning models — are constrained today chiefly by statutes that were drafted before the data existed in its present form. The result is a persistent gap between the sensitivity of the data and the enforceability of the duty owed to the person the data describes. That gap appears as a civil-rights violation in the courthouse, as an evidentiary problem in the disability hearing, as an accuracy and bias problem in the algorithmic system, and as a cryptographic exposure in the data centre. The book treats these as four faces of one defect, and it proposes doctrine, evidence, and infrastructure to close it.

1.2 The five domains

The portfolio falls into five domains, each of which is developed in its own part and each of which is anchored by at least one primary artefact.

Constitutional litigation.
A five-count complaint under 42 U.S.C. § 1983, together with the appellate and extraordinary-writ practice needed to preserve it. The governing questions are the retaliation exception of *Nieves v. Bartlett*, municipal liability under *Monell*, and the abstention and immunity doctrines that shape the pleading.

Genomic and neurogenetic data.
A constitutional case for treating the genome and its neurophysiological correlates as information deserving heightened protection, the statutory lattice that partially supplies it, and the evidentiary

standard under which a polygenic or imaging biomarker may enter a courtroom or an administrative record.

Semantic computing.
The Hybrid Semantic Processing Architecture (HSPA v3.0), a six-layer system spanning a semantic execution graph, a probabilistic control layer, a reconfigurable compute fabric, a nitrogen-vacancy quantum substrate, a post-quantum trust chain, and a cognitive-accessibility interface.

Algorithmic governance.
The MORTAR triage engine and the OpenSSA integrity pipeline — instruments for making administrative prioritisation auditable and for keeping an evidentiary chain intact from capture to adjudication.

Public finance and treaty.
The PSDLEF Sovereign Exchange Trust, a proposed perpetual-benefit structure, together with the capital-allocation optimiser (DYREO), the algorithm execution suite (ADR-31), and the international policy framework that would give the whole structure a legal home.

1.3 How the parts fit together

Part II states the injury and the doctrine. Part III supplies the scientific evidence that the injury is real and measurable. Parts IV–VI supply the technical machinery that would let an institution process the same data without reproducing the injury. Part VII and Part VIII give the remedy a legal and a financial form. Part IX is the reckoning: the statement of facts and procedural history, written in the register of a court filing, in which every assertion is either supported by a record citation or plainly marked as requiring one. Parts X–XII generalise and close.

Existing law under-protects cognitive, genomic, and neurophysiological data; that under-protection is measurable as harm, remediable as doctrine, mitigable as architecture, and financeable as a perpetual-benefit trust.

1.4 A note on scale and honesty

The portfolio is large, and scale invites inflation. Several components of the engineering argument are simulations, not measurements; several components of the financial argument are proposals, not instruments; several components of the litigation argument are drafting plans, not filed pleadings. Chapter 2 states the discipline that the rest of the book follows in marking these distinctions, and Part IX applies it most severely to the author's own record.

CHAPTER 2

Status Discipline: Proposal, Prototype, Simulation, and Adopted Law

> The first duty of a document that mixes advocacy with proof is to say which is which.

2.1 Why a register taxonomy is needed

A reader coming to this material from any single discipline will find something in it that looks strange. A lawyer will find a quantum substrate. An engineer will find a complaint under § 1983. A policy analyst will find a trust indenture. The risk is not that any one part is wrong; it is that the reader will transfer the confidence appropriate to one register onto material that belongs to another. The remedy is an explicit taxonomy, applied without exception.

2.2 The four registers

Principle 2.1 (Register separation)**.** *Every substantive proposition in this book belongs to exactly one of four registers: adopted law, proposal, model, or target. A proposition that has not been assigned to a register is not yet ready to be relied upon.*

Register one: adopted law. A rule that binds, or a standard that has been published by a competent body. Examples used throughout the book include the Administrative Procedure Act; 42 U.S.C. § 1983; the Genetic Information Nondiscrimination Act of 2008; the Americans with Disabilities Act and its 2008 amendments; the Health Insurance Portability and Accountability Act and its Security Rule; and the post-quantum standards published by the National Institute of Standards and Technology in August 2024 as FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA). Adopted law is cited in conventional form and can be verified independently.

Register two: proposal. A design that the author advocates and that no authority has adopted. The Gov-LLM concept — a government large language model operated as a reasonable accommodation — is a proposal. The PSDLEF Sovereign Exchange Trust is a proposal. The request for a United Nations commission of inquiry is a proposal. Proposals are argued, not assumed; their legal feasibility is examined rather than asserted.

Register three: model and simulation. A computational result. When this book reports that a hashing scheme achieves a given recall at a given precision, or that an optimiser allocates capital in a given ratio, or

that a variational eigensolver converges to an energy within a stated tolerance, the figure comes from the author's own code. It is reproducible from the reference implementations, and it is a statement about the model rather than about a deployed system.

Register four: design target. An engineering goal not yet demonstrated at the scale claimed. The 600 MW/1 GWh solar-and-battery microgrid, the HARMONI-X cognitive prosthetic, and the quantum gravitational reactor are design targets. They are described in the terms a specification would use, because that is the useful way to describe them, but they are not reported as achievements.

2.3 Worked examples of the discipline

The taxonomy is easiest to see in borderline cases.

- *Post-quantum cryptography.* That ML-KEM-768 uses a 1,184-byte encapsulation key and a 1,088-byte ciphertext is **[ADOPTED LAW]** — it is fixed by FIPS 203 and by the IETF's ML-KEM profile. That the author's trust chain selects ML-KEM-1024 for a given data class is **[PROPOSAL]**. That the chain's end-to-end latency under load is some particular number is **[MODEL / SIMULATION]**.
- *Disability adjudication.* That the Social Security Administration processes appeals with substantial delay is **[ADOPTED LAW]** in the sense that it is documented by the Government Accountability Office. That a locality-sensitive-hashing triage engine would reduce that delay to a stated figure is **[MODEL / SIMULATION]**. That the agency should adopt such an engine is **[PROPOSAL]**.
- *Genomic evidence.* That a polygenic score is admissible only if it survives the framework of *Daubert* is **[ADOPTED LAW]**. That a particular score meets the threshold is a **[MODEL / SIMULATION]** claim about the author's data. That courts should adopt a specific disclosure protocol for scores is **[PROPOSAL]**.

The trigger for a citation marker

Where a proposition depends on a fact that lives in a court docket, an administrative record, or a laboratory measurement, and that fact has not been placed in the reviewing record, the text carries **[CITE REQUIRED]** and names the document that would close the gap. This convention runs through Part IX, where it does the most work.

2.4 The consequence for the reader

The discipline has a practical payoff. A reader who wants to know whether a sentence can be acted upon — cited in a brief, implemented in hardware, funded by an investor — can answer the question from the register alone. The rest of the book applies the taxonomy silently, in prose written so that the register is apparent even to a reader who ignores the labels.

Part II

Constitutional Jurisprudence and Biological Data Sovereignty

CHAPTER 3

The Five Counts of the § 1983 Complaint

Section 1983 does not create duties. It creates a remedy for duties created elsewhere — and the whole art of the pleading lies in naming the duty the defendant is said to have broken.

3.1 The statutory premise

Section 1983 of Title 42 imposes liability on any person who, under colour of state law, subjects another to the deprivation of a right secured by the Constitution or laws of the United States. The provision is jurisdictional and remedial; it supplies no substantive right of its own (42 U.S.C. 1983 2018). The plaintiff must therefore identify, for each defendant, the constitutional or statutory right infringed, the conduct that infringed it, and the causal link between the two. The five counts below are drafted to do exactly that.

3.2 Count I — Retaliatory arrest and prosecution

The first count pleads that the plaintiff was arrested and later prosecuted not on probable cause arising from criminal conduct but in retaliation for protected speech or petitioning. The doctrinal obstacle is well known: *Nieves v. Bartlett* holds that a plaintiff alleging a retaliatory arrest must plead and prove the absence of probable cause for the arrest (Nieves v. Bartlett 2019). The narrow exception the Court recognised — for jurisdictions in which otherwise similarly situated individuals are not arrested — has itself been narrowed: *Gonzalez v. Trevino* requires the plaintiff to identify objective evidence that the exception applies rather than to rely on the absence of comparators (Gonzalez v. Trevino 2024). A count pleaded without that evidence invites dismissal, and the pleading must therefore be built on specific, record-based comparators or on an identified policy of selective enforcement.

The draft complaint conditions this count on a showing that the arrest lacked probable cause and that the retaliatory motive was a but-for cause. The source portfolio itself flags that the underlying docket entries and the charging instrument must be verified before the count is filed **[CITE REQUIRED]**.

3.3 Count II — Coercive interrogation and the privilege against self-incrimination

The second count pleads that statements were obtained under conditions that overbore the plaintiff's will, in violation of the Fifth and Fourteenth Amendments. The inquiry is whether the confession was the product of coercion rather than of a free choice. Because the count sounds in the conduct of interrogation rather than in the fact of prosecution, it must be analysed separately from Count I: the case law that limits retaliation claims does not automatically limit a coercion claim, and vice versa.

3.4 Count III — Fabrication of evidence and reliance on uncorroborated evidence

The third count pleads that the investigation relied on fabricated or uncorroborated material. Its doctrinal home is the line of authority that makes it actionable to fabricate evidence even where no conviction results, because the wrong is the creation of false evidence rather than the outcome of the case. The count must be pleaded against the specific officer who is said to have fabricated a specific thing, with dates and documents, and not against the investigating unit collectively (Buckley v. Fitzsimmons 1993).

3.5 Count IV — Conspiracy

The fourth count pleads agreement among the defendants. A conspiracy count under § 1983 must allege a meeting of the minds sufficient to support an inference of agreement, and it must identify the constitutional wrong that the agreement produced. It is not a vehicle for adding defendants whose only connection to the case is a shared employer.

3.6 Count V — Municipal and supervisory liability

The fifth count pleads that the constitutional injuries were caused by a policy or custom of the local government, or by a failure to train that amounts to deliberate indifference. *Monell* established that a municipality may be sued under § 1983 where the execution of its policy or custom inflicts the injury (Monell v. Department of Social Services 1978). The count must identify the policy, the practice, or the training deficiency, and must show that the deficiency was the moving force behind the violation. *City of Canton v. Harris* supplies the deliberate- indifference standard for training claims (City of Canton v. Harris 1989), and *Connick v. Thompson* cautions that a single act of omission does not ordinarily establish the required pattern (Connick v. Thompson 2011).

3.7 Relief and the structure of the prayer

The combined prayer seeks declaratory relief, injunctive relief, and damages, and each form of relief carries its own threshold. Injunctive relief requires the plaintiff to satisfy the four *Winter* factors by a clear showing: likelihood of success on the merits, likelihood of irreparable harm, a favourable balance of equities, and service of the public interest (Winter v. Natural Resources Defense Council 2008). Damages against individual defendants in their personal capacities require the plaintiff to overcome qualified immunity, the subject of Chapter 4. A prayer that omits the *Winter* showing asks the court to grant equitable relief it has no power to grant, and a prayer that omits the immunity analysis asks the court to skip the question it must answer first.

CHAPTER 4

Appellate Posture and Doctrinal Thresholds

Appellate jurisdiction is not an afterthought. It is a pleading requirement, because an order that cannot be appealed is an order the plaintiff has lost.

4.1 Finality and the collateral-order doctrine

The default rule is that only final decisions are appealable, under 28 U.S.C. § 1291 (28 U.S.C. 1291 2018). An order that contemplates further proceedings is not final, and the plaintiff who wishes to appeal it must find a foothold in the collateral-order doctrine of *Cohen v. Beneficial Industrial Loan Corp.*, which permits immediate appeal of orders that conclusively determine a disputed question, resolve an important issue separable from the merits, and would be effectively unreviewable on appeal from final judgment (Cohen v. Beneficial Industrial Loan Corp. 1949). The denial of a motion to dismiss on qualified-immunity grounds is the paradigm collateral order, and it is the principal route by which a civil-rights plaintiff's case reaches the circuit court before trial.

4.2 Qualified immunity

Qualified immunity shields government officials from civil damages liability unless their conduct violated a clearly established statutory or constitutional right of which a reasonable person would have known (Harlow v. Fitzgerald 1982). The doctrine has two prongs — whether a right was violated, and whether the right was clearly established at the time — and the Supreme Court has repeatedly permitted courts to resolve the second without deciding the first. For the plaintiff, the practical task is to identify a controlling precedent or a robust consensus of cases placing the unlawfulness of the conduct beyond debate.

4.3 Prosecutorial and absolute immunity

Prosecutors enjoy absolute immunity for conduct intimately associated with the judicial phase of the criminal process (Imbler v. Pachtman 1976), including administrative acts connected to the prosecution such as the handling of disclosure obligations (Van de Kamp v. Goldstein 2009). They enjoy only qualified immunity, however, for conduct that is investigative rather than advocative (Buckley v. Fitzsimmons 1993). The distinction matters because a complaint that seeks damages from a prosecutor must plead conduct on the correct side of the line.

4.4 Sovereign immunity and the *Ex parte Young* exception

The Eleventh Amendment generally bars suits for damages against a state in federal court, and states may not be sued as § 1983 defendants at all. *Ex parte Young* preserves a narrow exception: a suit for prospective injunctive relief against a state officer to end a continuing violation of federal law is not barred (Ex parte Young 1908). The remedy is against the officer, not the state, and it is prospective, not compensatory.

4.5 The favourable-termination element

Where a § 1983 claim implies the invalidity of a conviction or sentence, the plaintiff must show that the conviction or sentence has been reversed, expunged, or otherwise called into question. *Heck v. Humphrey* states the rule for malicious prosecution and related claims (Heck v. Humphrey 1994), and *Thompson v. Clark* clarifies what counts as favourable termination for a malicious-prosecution claim: a criminal defendant need not show that the prosecution ended in an affirmative indication of innocence, only that it ended without a conviction (Thompson v. Clark 2022). *Manuel v. City of Joliet* holds that a pretrial-detention claim is governed by the Fourth Amendment rather than by a fabricated a retrospective malice standard (Manuel v. City of Joliet 2017).

4.6 Abstention

Younger v. Harris directs federal courts to abstain from enjoining pending state criminal proceedings absent extraordinary circumstances (Younger v. Harris 1971). The doctrine is not absolute: bad faith, harassment, or a patently unconstitutional statute can lift it. A federal complaint that seeks to enjoin a state prosecution must therefore plead the bad-faith or harassment exception with specificity, or risk dismissal on abstention grounds.

CHAPTER 5

Extraordinary Writs and Supreme Court Practice

A petition for an extraordinary writ is a request to a court to reach material it would rather leave to the ordinary process. It is granted rarely, and the petition must therefore be the most disciplined document in the case.

5.1 The writs and their standards

The extraordinary writs available under the All Writs Act — mandamus, prohibition, and certiorari from the Supreme Court — differ in standard and in the office they perform. Mandamus compels a lower court or an officer to perform a clear legal duty; it is granted only where the petitioner has a clear and indisputable right, the respondent has a plain duty to act, and no other adequate means of relief exists. Prohibition restrains a court from acting outside its jurisdiction. A petition for a writ of certiorari asks the Supreme Court to exercise its discretionary review.

5.2 Drafting discipline for a petition

The source portfolio treats the petitions as working instruments, and the drafting discipline is worth stating explicitly. A petition must state the question presented in a single sentence; must identify the court that entered the judgment and the date of entry; must set out the facts with record citations; and must explain why the ordinary appellate process is inadequate. Every factual assertion that depends on a docket entry carries a citation marker until the entry is in hand.

Record proofs that the petitions require

The source materials identify, as necessary to any petition, the file-stamped pleadings, the docket sheets, and the orders actually entered in each proceeding **[CITE REQUIRED]**. Without them the petition asserts procedural history that the court cannot verify, and a court that cannot verify the procedural history will not reach the merits.

5.3 The Rule 60 and Rule 59 motions

Relief from judgment under Rule 60(b) and amendment or alteration of judgment under Rule 59(e) are frequently the prerequisites to appellate review of a procedural defect. A motion under either rule must be filed within the applicable time and must identify the specific ground. The source materials record the author's intention to preserve every available post-judgment motion, and the chapter treats that intention as a checklist rather than as a description of steps already taken.

CHAPTER 6

The Constitutional Case for Genomic Data Sovereignty

The genome is the most intimate instrument of prediction that has ever been readable. A Fourth Amendment built for papers and effects has to say something about it.

6.1 The Fourth Amendment and informational privacy

The Fourth Amendment protects persons, houses, papers, and effects against unreasonable searches. In the physical world it regulates the collection of things. In the digital world the Supreme Court has recognised that the collection of information can itself be a search, even when the information is held by a third party: *Carpenter v. United States* held that the government's acquisition of historical cell-site records was a search (Carpenter v. United States 2018), and *Riley v. California* held that a search incident to arrest cannot extend to the contents of a cell phone (Riley v. California 2014). *Maryland v. King* upheld DNA sampling of arrestees on the strength of the reduced expectation of privacy of persons in custody, but its reasoning is expressly limited to identification (Maryland v. King 2013). The analytical route from these cases to a general protection for genomic data is not automatic, and the argument for it proceeds by analogy: a genome is a record that is both identifying and predictive, and predictive capacity is a property the Court has treated as constitutionally significant.

6.2 The five properties of genomic information

The author's portfolio advances five properties that distinguish genomic information from other categories of data and that, taken together, justify a heightened standard.

1. **Uniqueness.** A genome is effectively unique to an individual and is stable across the lifespan.
2. **Relationality.** A genome is not the property of one person alone; it discloses probabilistic information about relatives who have not consented to anything.
3. **Predictive power.** A genome carries probabilistic information about future disease, and thus about future cost.
4. **Persistence.** A genome does not change and cannot be reissued, unlike a password or a credit-card number.

5. **Inter-generational reach.** The information is inherited and inherited-into, so the interest it creates outlives the individual.

6.3 The counter-move and its limit

The strongest counter-argument is that genomic information is neither more nor less private than other sensitive data, and that a special constitutional rule would be unmanageable. The limit of that argument is that three of the five properties — relationality, persistence, and inter-generational reach — are properties that no other category of personal data possesses to the same degree. A fourth, predictive power, is what the Court has treated as decisive in the third-party doctrine cases. The constitutional argument is not that the genome is unique, but that the combination of properties makes the existing, category-by- category approach inadequate.

CHAPTER 7

Statutory Architecture: GINA, the ADA, HIPAA, and Beyond

The genome is protected today by a lattice of statutes, each of which was drafted for a neighbouring problem. The gaps between them are the subject of this chapter.

7.1 GINA

The Genetic Information Nondiscrimination Act of 2008 prohibits discrimination on the basis of genetic information in health insurance and in employment (Genetic Information Nondiscrimination Act of 2008 2008). It is the only federal statute aimed directly at genetic discrimination, and its scope is the source of its principal gap. In the insurance market, GINA governs health insurance; it does not reach life insurance, disability insurance, or long-term-care insurance. In employment, it governs hiring, firing, and terms of employment, but it does not regulate an employer's use of genetic information in a wellness programme except as provided. The gap is structural: the statute regulates health underwriting precisely where genetic information is least powerful, and leaves unregulated the markets in which genetic information is most predictive.

7.2 The ADA and the amendments of 2008

The Americans with Disabilities Act prohibits discrimination against a qualified individual with a disability (Americans with Disabilities Act of 1990 1990), and the ADA Amendments Act of 2008 broadened the definition of disability so that the question would turn on whether discrimination occurred rather than on whether the plaintiff crossed a threshold (ADA Amendments Act of 2008 2008). For the genomic plaintiff the significance is twofold. First, a genetic predisposition can, in some circumstances, establish a record of impairment. Second, and more importantly for the disability context, *Olmstead v. L.C.* requires states to provide services to persons with disabilities in community settings rather than institutions when the state's treatment professionals determine that community placement is appropriate (Olmstead v. L.C. ex rel. Zimring 1999). A state agency that denies a benefit on the ground of a genetic prediction, without an individualised assessment, states a claim under the integration mandate.

7.3 HIPAA and the Security Rule

The Health Insurance Portability and Accountability Act and its implementing regulations establish national standards for the electronic exchange of health information and, under the Security Rule, for

the administrative, physical, and technical safeguards that protect it (Health Insurance Portability and Accountability Act of 1996 1996). The Security Rule expressly contemplates that encryption is not always feasible, and thus treats it as an addressable rather than a mandatory implementation specification. That flexibility, combined with the fact that most genomic data moves between systems that are not all covered by HIPAA, leaves a substantial middle ground in which genomic information is neither clearly protected nor clearly unprotected. The technical remedies developed in Parts IV and VI are aimed precisely at that middle ground.

7.4 Beyond the lattice

Two further instruments matter to the argument. The Americans with Disabilities Act's accessibility provisions, as elaborated in Section 504 of the Rehabilitation Act, supply the template for the cognitive-accessibility proposal developed in Part X. And the patchwork of state genetic-privacy statutes supplies examples of what a federal standard could look like. The chapter's conclusion is that the lattice is real but incomplete, and that the incompleteness is the space in which both the litigation theory and the technical proposal operate.

Part III

Neurogenetic Biomarkers and Algorithmic Governance

CHAPTER 8

Multimodal Biomarker Characterization

A prediction is not an observation, and a probability is not a diagnosis. The whole difficulty of neurogenetic evidence is that the two are easy to confuse in a document an adjudicator must read quickly.

8.1 The dissertation problem

The technical core of the author's scientific work is the characterisation of biomarkers for a neurodevelopmental phenotype using multiple modalities at once. The modalities fall into four families: behavioural and clinical instruments, functional neuroimaging, genotypic and polygenic measures, and physiological time-series such as the electroencephalogram. The dissertation problem is to characterise the phenotype as a function of these modalities jointly, and to report the uncertainty of the joint characterisation honestly.

8.2 Why multimodal beats unimodal, and when it does not

A multimodal characterisation can outperform a unimodal one when the modalities carry complementary information — that is, when the correlation between two modalities, conditional on the phenotype, is low. It does not outperform when the modalities are redundant, and it can underperform when one modality is noisy and its error is independent of the signal, because the fusion step then injects variance without adding information. The practical consequence is that a multimodal claim must be accompanied by a diagnosis of complementarity; a modality that merely restates another is not evidence, it is redundancy dressed as rigour.

8.3 The four families in detail

Clinical and behavioural instruments. Standardised instruments supply the reference phenotype. Their virtue is interpretability; their vice is subjectivity at the margins, and it is at the margins that adjudicative decisions are made.

Functional neuroimaging. Functional magnetic resonance imaging measures a blood-oxygenation-dependent signal that is a proxy for neural activity. Its strengths are spatial resolution and repeatability across scanners when a common protocol is used; its weaknesses are temporal resolution, cost, and sensitivity to motion and to analysis choices. The reproducibility literature on task-based fMRI is the reason this book

treats a single-study imaging result as evidence to be corroborated rather than as a stand-alone proof of anything.

Genotypic and polygenic measures. A polygenic score aggregates the small effects of many loci into a single number. Its predictive power depends on the discovery sample from which the weights were estimated, and it transfers least well across ancestries that were underrepresented in that sample. A polygenic score is therefore best understood as a population statistic applied to an individual, with an ancestry-dependent penalty that must be disclosed.

Physiological time-series. The electroencephalogram and related signals supply a temporal modality whose features are computable and whose analysis is comparatively transparent. They are the modality most amenable to the evidentiary pipeline described in Part VI, because the transformation from raw signal to feature is deterministic and auditable.

8.4 Uncertainty as a first-class output

The design decision that most distinguishes this work from a conventional biomarker study is that uncertainty is an output rather than a caveat. A characterisation that reports "score $= 0.82$" is less useful to an adjudicator than one that reports a posterior distribution, because the adjudicator's task is to decide under uncertainty rather than to rank point estimates. Chapter 10 returns to this point when it asks what a court should be allowed to do with such a distribution.

CHAPTER 9

Candidate SNP and Functional Variant Profiling

The unit of genetic evidence is not the gene. It is the variant, and the honest statement of what a variant does is a distribution over functions, not a verdict.

9.1 From genes to variants

A candidate-gene study nominates a gene and looks for association. A variant-level study nominates a base change and looks for association with a functional consequence. The second is the more defensible unit because it is the more specific one, and because the functional annotation that gives the association its meaning attaches to the variant rather than to the gene. The portfolio's profiling work proceeds at the variant level.

9.2 The classes of variant

The variants that matter for a neurodevelopmental phenotype fall into four classes. Coding variants alter an amino-acid sequence and admit a comparatively direct interpretation. Regulatory variants alter the expression of a gene and require a functional assay or a population-scale expression analysis to interpret. Copy-number variants alter the dosage of a region and require a platform capable of detecting structural change. And common variants of small effect contribute to a polygenic score. A profiling pipeline that treats these four classes identically will mis-attribute effect to whichever class its assay detects best.

9.3 The handheld electrochemical genotyping platform

The hardware element of this part is a portable genotyping platform built around an electrochemical readout and an addressable electrode array. The design's claim is that a sixteen-channel or sixty-four-channel array can perform allele-specific detection of a panel of variants in a point-of-care setting, trading breadth for locality. The engineering argument for the platform is set out in the source material's specification; its status here is **[DESIGN TARGET]**, because the performance figures are design estimates rather than measurements from a built unit.

An electrochemical readout transduces hybridisation into a current, which is easier to instrument in a portable form factor than fluorescence. The trade is sensitivity: fluorescence detects fewer copies, while electrochemistry favours ruggedness and cost. For a panel of common variants with high allele frequency, the trade is favourable; for rare variants or low-input samples, it is not, and the specification should say so.

9.4 The 96-electrode array and its addressability

A ninety-six-electrode array is chosen because ninety-six is the standard microtitre plate format and therefore composes cleanly with existing laboratory workflow. Its addressability is the feature that matters analytically: an individually addressable electrode allows the readout to be spatially resolved, so that a failure at one site is a datum rather than a corruption of the whole array. This is the same design principle that Part VI applies to the evidentiary pipeline.

9.5 What the platform does not claim

The platform does not claim to sequence a genome; it claims to interrogate a panel. It does not claim clinical diagnostic authority; it claims to supply a structured measurement that can be entered into a governed record. And it does not claim ancestry-independence; the panel is specified for a population and its transferability is a stated limitation.

CHAPTER 10

Daubert Admissibility of fMRI and Polygenic Scores

The question for the court is not whether the science is true. It is whether the witness's use of the science in this case is reliable enough to help the trier of fact.

10.1 The framework

Daubert v. Merrell Dow Pharmaceuticals requires the trial judge to act as gatekeeper for expert testimony, assessing whether the reasoning or methodology underlying the testimony is scientifically valid and whether it can be applied to the facts at issue (Daubert v. Merrell Dow Pharmaceuticals 1993). Rule 702 of the Federal Rules of Evidence codifies the inquiry and, as amended, requires that the testimony be based on sufficient facts or data, be the product of reliable principles and methods, and reflect a reliable application of those principles and methods to the facts (Federal Rule of Evidence 702 2023). *Kumho Tire v. Carmichael* extends the gatekeeping obligation to all expert testimony, not only to testimony that is scientific in the narrow sense (Kumho Tire Company v. Carmichael 1999).

10.2 fMRI in the courtroom

Functional imaging enters a case in one of two postures. In the first, it is offered as a diagnostic datum — a measurement that a clinician used in forming an opinion. In the second, it is offered as a direct index of a mental state, such as truthfulness or intent. The second posture has been received with scepticism, and rightly: the inferential distance from a blood-oxygenation-dependent signal to a proposition about intent is long, and it is not made shorter by the vividness of the image. A court confronted with an fMRI exhibit should ask three questions. What is the proposition the exhibit is offered to prove? What is the error rate of the inference from signal to proposition? And what would the exhibit change about the adjudicator's decision?

An image is more persuasive than a number, and a coloured brain map is more persuasive than a table. The gatekeeping inquiry is designed to interrupt that persuasiveness and to ask what the exhibit actually establishes. A brief that offers an imaging exhibit should therefore state the inference it supports in a single sentence before it shows the image.

10.3 Polygenic scores as evidence

A polygenic score is a weighted sum of allele counts. Its admissibility raises four methodological questions. First, *predictive validity*: what is the score's out-of-sample predictive performance for the phenotype at issue? Second, *transferability*: how does that performance degrade when the score is applied to an individual of different ancestry from the discovery sample? Third, *base rate*: what does a given score mean in the population at issue, as opposed to the study population? Fourth, *individualisation*: a score is a population statistic, and the step from a population distribution to a statement about a named individual is the step that most often fails.

The correct use of a polygenic score in adjudication is therefore as one input to a probabilistic assessment, with its confidence interval and its ancestry- dependent penalty disclosed. The incorrect use is as a categorical statement that an individual does or does not have a condition.

10.4 Evidentiary fusion and the record

The portfolio's solution to the admissibility problem is procedural rather than substantive. Every neurogenetic exhibit is routed through a pipeline that records the raw datum, the transformation applied, the intermediate results, and the model that produced the final number, so that a reviewing court can audit the inference. That pipeline is the OpenSSA integrity layer described in Part VI. The substantive point of this chapter is simply that an exhibit that cannot be audited is an exhibit that cannot be assessed for reliability under Rule 702, and an exhibit that cannot be assessed cannot be admitted.

Part IV

Hybrid Semantic Processing Architecture (HSPA v3.0)

CHAPTER 11

Architectural Philosophy and the Clean-Slate Mandate

Every instruction set carries a burden: the cost of translating what the programmer meant into what the silicon can do. HSPA's wager is that for some workloads the burden has become larger than the work.

11.1 The premise

HSPA — the Hybrid Semantic Processing Architecture, version 3.0 — is specified as a clean-slate processor design. It discards binary backward compatibility with x86, ARM, RISC-V, and SPARC, and executes natively on directed acyclic graphs of semantic nodes. The motivating diagnosis is the von Neumann bottleneck: the separation of memory from computation, combined with the sequential fetch-decode-execute cycle, which the specification characterises as a "semantic tax" because each instruction must be decoded and scheduled through layers of abstraction that discard context.

Register of the claims in this part

The design in this part is reported in the register of the source specification. Where a parameter is a published standard — for example the FIPS 203 key and ciphertext sizes — it is **[ADOPTED LAW]**. Where a parameter is a property of the author's simulation — for example the modelled nitrogen-vacancy coherence times — it is **[MODEL / SIMULATION]**. Where a figure describes hardware not yet fabricated at the stated scale, it is **[DESIGN TARGET]**.

11.2 The comparison the specification draws

The table is best read as a set of architectural commitments rather than as a performance comparison. It asserts, correctly, that the four legacy families share a structural assumption — that computation is a sequence — and that HSPA rejects the assumption. Whether rejecting it pays for a given workload is an empirical question the specification does not settle.

Table 11.1: Legacy instruction architectures and the HSPA clean-slate replacement.

Legacy	Stated limitation	HSPA replacement
x86 (CISC)	Variable-length decoding; implicit serial semantics	Native SEG parsing; node-at-a-time graph dispatch
ARM (RISC)	Fixed 32-bit encoding; load/store bottleneck	128-bit semantic-node packets; content-addressable operands
RISC-V	Register-register pipeline driven by a program counter	Graph-edge traversal without a program counter
Dataflow	Deterministic token matching without uncertainty	Bayesian confidence-weighted token firing
NPU / TPU	Fixed matrix-multiplication acceleration	Morphable reconfigurable logic cells

11.3 The six-layer stack

HSPA organises its functions into a strict six-layer physical and logical stack. The book follows the specification's own numbering, in which the quantum substrate is Layer 0 and the cognitive interface is Layer 5.

Layer 5 — Application and cognitive interface.
The HARMONI-X cognitive prosthetic, the OpenSSA integrity pipeline (OSI-P), retrieval-augmented generation with ReAct agent loops, a precision-medicine trial workflow, and an evidentiary fusion engine that draws on public court, biomedical, and scholarly corpora.

Layer 4 — Security and post-quantum trust.
Key encapsulation, digital signatures, hash-based anchoring, and a semantic evidence trace ledger.

Layer 3 — Reconfigurable compute fabric.
An array of reconfigurable logic cells with digital-analog hybrid arithmetic units, a self-optimising microcode layer, and per-cell micro-cachelets.

Layer 2 — Probabilistic control and memory.
A probabilistic microcontroller, a fixed-point inference matrix, and a unified memory lattice with hybrid semantic-physical addressing.

Layer 1 — Semantic execution graph.
A 128-bit fixed node format, locality-sensitive context hashing, and a graph-eligibility firing protocol.

Layer 0 — Quantum substrate.
A density-matrix nitrogen-vacancy quantum emulator on the carrier board.

CHAPTER 12

The Six-Layer Stack: Execution Graph, Control, and Compute

12.1 Layer 1 — the 128-bit semantic node

Programs are graphs of fixed-width nodes. The base node occupies 128 bits and encapsulates opcode, confidence, context hash, two operands, an output count, and flags:

Table 12.1: 128-bit native semantic node layout.

Bits	Field	Width and meaning
[127:120]	Opcode	8 bits; instruction selector
[119:112]	Confidence	8 bits; firing weight
[111:80]	Semantic tag	32-bit locality-sensitive context hash
[79:48]	Operand A	32-bit unified-memory address
[47:16]	Operand B	32-bit unified-memory address
[15:8]	Output count	8 bits
[7:0]	Flags	8 bits

In an extended 256-bit cache line the node carries additional execution metadata: a 16-bit expected latency, a 16-bit energy budget, a 32-bit vector of downstream node identifiers, a 64-bit uncertainty vector encoded as eight 8-bit probabilities over possible outcomes, and a 32-bit quantum handle that points to a register in the quantum substrate.

12.2 The semantic instruction set

The specification defines native 8-bit opcodes grouped into seven families.

Table 12.2: HSPA opcode families and their executing units.

Range	Group	Representative mnemonics	Unit
0x00–0x0F	Arithmetic	ADD, SUB, MUL, DIV	RLC arithmetic unit
0x10–0x1F	Logic	AND, OR, XOR, NOT	RLC logic gate
0x20–0x2F	Memory	LOAD, STORE, PREFETCH	UML interface
0x30–0x3F	Control	JUMP, BRANCH, CALL	PMCU scheduler
0x40–0x4F	HSPA extensions	QRAND, ENCRYPT, ATTEST	PQC / NV-centre unit
0x50–0x5F	AI / neural	MAC, ACTIVATE, POOL	RLC stochastic unit
0xE0–0xE6	Quantum native	QINIT, QGATE_SINGLE, QMEASURE	LoongDiamond substrate

12.3 The firing protocol

A node fires when five conditions hold simultaneously. All input edges must carry valid tokens; the node's normalised confidence must meet or exceed the threshold $P_{\min}$ (default 0.70); the semantic distance between the node tag and the active context of the target cell must fall below a bound δ; the node's energy budget must fit within the available energy for the epoch; and, if the node carries a quantum handle, the density matrix must be normalised to within a tolerance ϵ. The protocol is a conjunction rather than a disjunction, which matters: it means that a node with abundant data but low confidence does not fire, and a node with high confidence but no energy does not fire either.

12.4 Layer 2 — the probabilistic microcontroller and the memory lattice

The probabilistic microcontroller replaces the program counter, the branch predictor, and the superscalar scheduler with a fixed-point inference matrix in Q15 format. Scheduling is by an entropy-reduction score:

$$\text{Schedule}(n) = \alpha\, P_{\text{success}}(n) + \beta\, \text{confidence}(n) - \gamma\, \text{entropy}(n), \tag{12.1}$$

with default weights $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$. Beliefs are updated by Bayes' rule as telemetry arrives:

$$P(\text{success} \mid e) = \frac{P(e \mid \text{success})\, P(\text{success})}{P(e)}. \tag{12.2}$$

The negative entropy term in (12.1) is the architectural counterpart of the epistemic caution that Parts III and IX insist upon: the scheduler prefers executions that reduce uncertainty, not merely executions that are likely to succeed.

Memory is content-addressable through a 64-bit unified address.

Table 12.3: 64-bit unified memory lattice address structure.

Bits	Field	Function
[63:48]	Physical page index	DRAM row and bank mapping
[47:16]	Semantic hash	Locality-sensitive hash of the content
[15:8]	Privilege and access	Read/write/execute plus quantum-enclave access
[7:0]	Uncertainty tolerance	Permitted error bound for approximate computing

Semantic distance between two addresses is computed in hardware as the Hamming distance of their semantic hashes:

$$d_{\text{sem}}(a, b) = \text{popcount}(\text{hash}_a \oplus \text{hash}_b). \tag{12.3}$$

The design makes semantic access primary and physical access secondary, which is the inversion that distinguishes it from a conventional cache.

12.5 Layer 3 — reconfigurable logic cells and self-optimising microcode

Each reconfigurable logic cell morphs in under five nanoseconds, carries a digital-analog hybrid arithmetic unit supporting precisions from one to thirty-two bits, a 1024-bit lookup table, and a two-kilobyte least-recently-used micro-cachelet. The specification reports an 85% hit rate on repeated semantic tags, a figure that is **[MODEL / SIMULATION]** and depends on the tag distribution.

The self-optimising microcode layer is an on-chip reinforcement-learning agent that rewrites microcode tables using proximal policy optimisation, with reward

$$R(s, a) = w_1 \cdot \text{Throughput} + w_2 \cdot \text{Energy Efficiency} + w_3 \cdot \text{Accuracy} - w_4 \cdot \text{Quantum Decoherence}. \tag{12.4}$$

The subtraction in (12.4) is the design's acknowledgement that coherence is a resource to be conserved. Safety is enforced by a commit protocol: ten thousand validation vectors, a block-RAM rollback buffer, a

signature check under a post-quantum scheme, and canary deployment across one per cent of cells over a thousand epochs. The protocol is the reason the layer can be described as self-optimising without being described as unsupervised.

CHAPTER 13

The LoongDiamond-2.5 Quantum Substrate

A density matrix is not a wavefunction. It is what a wavefunction looks like to an observer who has already given up on knowing everything.

13.1 The physical model

The substrate, named LoongDiamond-2.5, simulates nitrogen-vacancy (NV) centre qubits in synthetic diamond using a density matrix as the single source of truth. The density matrix ρ is Hermitian, of unit trace, and positive semi-definite, so $\rho = \rho^\dagger$, $\operatorname{Tr}\rho = 1$, and $\operatorname{Tr}\rho^2 \leq 1$. The simulation's parameters are drawn from the NV-centre literature.

Table 13.1: LoongDiamond-2.5 NV-centre parameters as specified.

Parameter	Symbol	Value	Context
Temperature	T	4.0 K	Cryogenic operation
Magnetic bias field	B	500 G	Bias field
Zero-field splitting	D_{gs}	2.87 GHz	Ground-state D parameter
Spin-lattice relaxation	T_1	10.0 ms	NV baseline
Spin-spin coherence (echo)	T_2	5.0 ms	Echo-decoupled
Inhomogeneous dephasing	T_2^*	0.1 ms	Free-induction decay
Single-qubit gate time	t_{gate}	100 ps	Pulse duration
Single-qubit fidelity	F_{1q}	99.5%	Microwave drive
Two-qubit fidelity	F_{2q}	99.0%	Dipolar coupling
Measurement fidelity	F_{meas}	98.5%	Optical readout
Initialisation fidelity	F_{init}	99.9%	Optical spin pumping
Optical cycle time	t_{opt}	300 ns	Fluorescence cycle

The zero-field splitting of 2.87 GHz and the millisecond-order relaxation and coherence times are consistent with the experimental NV literature; the free-induction dephasing figure is consistent with the microsecond-scale values reported for NV ensembles in isotopically enriched diamond, where median T_2^* of several microseconds has been measured (Shinei et al. 2025). The table therefore separates cleanly into real constants and design targets, and the chapter maintains that separation.

13.2 Decoherence as a Kraus map

Decoherence is modelled with completely positive, trace-preserving Kraus operators, $\rho \mapsto \sum_i K_i \rho K_i^\dagger$. Amplitude damping models energy relaxation:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \quad \gamma = 1 - e^{-t/T_1}, \tag{13.1}$$

and phase damping models loss of coherence:

$$K_0 = \sqrt{1-\lambda}\, I, \quad K_1 = \sqrt{\lambda}\, Z, \quad \lambda = 1 - e^{-t/T_2}. \tag{13.2}$$

Choosing the Kraus formalism rather than a wavefunction collapse postulates is a deliberate modelling decision: it makes the substrate's state a probabilistic object throughout, which is what allows the scheduler of Chapter 12 to treat coherence as a schedulable quantity.

13.3 VQE validation

The substrate is validated against the ground-state energy of molecular hydrogen using a hardware-efficient RY-CNOT ansatz with the qubit Hamiltonian

$$H = -1.0523\,(Z_0 I_1 + I_0 Z_1) + 0.3979\, Z_0 Z_1 + 0.0113\, X_0 X_1. \tag{13.3}$$

The variational eigensolver converges to -1.1247 Ha against the theoretical -1.1373 Ha, an error of 1.1%, within chemical accuracy, in 47 iterations. The variational eigensolver is the appropriate validation target for a noisy device because it requires only constant-depth circuits and tolerates repeated measurement (Zhuang et al. 2024); the result is **[MODEL / SIMULATION]**, being a simulation of a simulation.

CHAPTER 14

Production Hardware: The ZU9EG Carrier and the SSE-9 Processor

14.1 The stochastic semantic engine

The production platform, SSE-9, implements a stochastic refinement scheme with an asymmetric duty cycle in which thirty-two latent registers are updated at twice the clock frequency and sixteen prediction registers at the base frequency. A hardware convergence trigger monitors the gradient norm of the latent state:

$$\text{Trigger} = \begin{cases} \text{active} & \text{if } \left\| \dfrac{z_{t+1} - z_t}{\Delta t} \right\|_2 < \varepsilon, \\ \text{idle} & \text{otherwise.} \end{cases} \tag{14.1}$$

When the trigger asserts, control of the register file passes deterministically to the projection phase, which produces the adjudication vector through a hyperbolic tangent. The asymmetry is the architecture's central efficiency claim: refinement runs hot, projection runs cold, and the trigger decides when to switch.

14.2 The carrier board

The board targets a Xilinx Zynq UltraScale+ MPSoC (XCZU9EG) paired with sixteen gigabytes of DDR4-2400 in a fly-by topology. The stackup is a ten-layer impedance-controlled design.

Table 14.1: Ten-layer impedance-controlled PCB stackup.

Layer	Type	Copper	Function
L1 (top)	Signal	0.5 oz + plating	High-speed signals, BGA pads, cell array
L2	Ground	1.0 oz	Solid reference plane
L3	Signal	0.5 oz	High-speed signal, DDR4 data, 100 Ω differential
L4	Power	1.0 oz	Core supply (0.85/1.0 V at 20 A)
L5	Signal	0.5 oz	Graph node bus, 85 Ω differential pairs
L6	Ground	1.0 oz	Core reference plane
L7	Signal	0.5 oz	Quantum emulator interconnect
L8	Power	1.0 oz	Auxiliary and DRAM supplies
L9	Ground	1.0 oz	Serial and PCIe return plane
L10 (bottom)	Signal	0.5 oz + plating	Cryptographic accelerator, passives

The stackup, the itemised bill of materials, the schematics, and the reference implementations together constitute the handoff package described in Chapter 16. Their status is **[DESIGN TARGET]**: they specify a board, they do not report a board in production.

CHAPTER 15

Post-Quantum Cryptography and RIGOR-PRS-Secure++

The migration to post-quantum cryptography is unusual among security projects in that the deadline is set by an adversary's capabilities rather than by one's own roadmap.

15.1 Naming: from competition entries to standards

The specification refers to algorithms by their round-three competition names — "Kyber" and "Dilithium". Because those names have been superseded by the standards that adopted them, this book uses the standardised names throughout and records the mapping. Kyber-768 became ML-KEM-768 and Kyber-1024 became ML-KEM-1024 in FIPS 203; Dilithium-III became ML-DSA-65 and Dilithium-V became ML-DSA-87 in FIPS 204. The National Institute of Standards and Technology published FIPS 203, FIPS 204, and FIPS 205 in August 2024, and the National Security Agency's Commercial National Security Algorithm Suite 2.0 sets a transition timeline that reaches full adoption in the mid-2030s (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d; National Security Agency 2022a).

Table 15.1: Standards and algorithm mapping for the trust chain.

Standard	Body	Primitive	Sizes
FIPS 203	NIST	ML-KEM-1024	pk 1568 B, ct 1568 B, ss 32 B
FIPS 204	NIST	ML-DSA-87	pk 2592 B, sk 4896 B, sig 4627 B
FIPS 205	NIST	SLH-DSA-SHA2-128s	Stateless hash-based signature
SP 800-208	NIST	HSS-LMS	Stateful two-level LM tree

15.2 The dual-envelope construction

The trust chain wraps a payload in a layered envelope it calls QRDE-AEAD. A recipient key encapsulation yields a ciphertext and a shared secret,

$$\mathrm{ML\text{-}KEM.Encaps}(pk_{\mathrm{recv}}) \to (c_1, ss), \tag{15.1}$$

from which a master key is derived,

$$K_{\text{master}} = \text{HKDF-SHA3-384}(ss, \text{“QRDE-Master-Salt”}), \tag{15.2}$$

and split into a confidentiality key and a MAC key, $K_e = K_{\text{master}}[0..31]$ and $K_m = K_{\text{master}}[32..63]$. The plaintext is encrypted under an authenticated cipher,

$$ct_2, \tau_2 = \text{AES-256-GCM.Encrypt}(K_e, IV, P, AD_{\text{inner}}), \tag{15.3}$$

and the outer envelope is authenticated to bind the encapsulation to the ciphertext,

$$\tau_1 = \text{HMAC-SHA3-256}(K_m,\ c_1 \,\|\, ct_2 \,\|\, \tau_2 \,\|\, AD_{\text{outer}}). \tag{15.4}$$

The envelope $C = (c_1, IV, \tau_2, ct_2, \tau_1, AD_{\text{outer}})$ is specified to achieve indistinguishability under adaptive chosen-ciphertext attack against a quantum adversary. The construction is a standard hybrid KEM-plus-DEM composition, and its security follows from the security of the components under the standard composition theorem; the contribution here is the particular binding of context through the two associated-data strings.

15.3 Where the security argument is strongest and weakest

The strongest part of the argument is the KEM: ML-KEM-1024 targets the highest of the NIST security categories and its parameters are fixed by a published standard. The weakest part is the operational one. A hybrid envelope is only as strong as the key management beneath it, and the source material's own requirement that keys be derived, split, and bound to context is precisely the requirement that is most often omitted when such a scheme is deployed. The chapter therefore states the cryptographic claim and immediately states its dependency: the envelope protects a payload if and only if the recipient's private key is protected by equal care.

CHAPTER 16

Cognitive Interface, Exhibits, Patent Claims, and Verification

16.1 Layer 5 — the cognitive interface

The application layer contains the HARMONI-X cognitive prosthetic, which the specification describes as an accessibility interface that adapts presentation to measured cognitive load. Its features include adaptive load monitoring drawn from heart-rate variability, pupilometry, and keystroke dynamics; display safeguards including pulse-width-modulation-free output, a dyslexia-friendly typeface, and a 21:1 contrast ratio; and a resource-allocation mode that can devote four times the nominal capacity to a task when a neurogenetic metabolic tax is present. The interface is the part of the architecture with the clearest legal hook, because it is the part that implements an accommodation, and Part X develops that hook.

The layer also contains the OpenSSA integrity pipeline, described in Part VI, and an evidentiary fusion engine that assembles material from public court dockets, biomedical repositories, and scholarly indexes.

16.2 The 16-marker crosswalk

The specification includes a crosswalk that maps sixteen genetic markers to functional impacts and to the listings used in disability adjudication.

Table 16.1: Selected markers from the 16-marker SNP-to-listing crosswalk.

rsID	Gene	Functional impact	Listing
rs4680	COMT	Slow dopamine clearance (Val158Met)	11.00 / 12.02
rs1800497	DRD2/ANKK1	Reduced D2 density (Taq1A)	12.11
rs2075654	SLC6A3	DAT1 transporter variability	11.00B / 12.02A
rs1360780	FKBP5	HPA-axis dysregulation	12.15
rs6265	BDNF	Reduced neuroplasticity (Val66Met)	11.00F
rs1801133	MTHFR	Folate defect (C677T)	11.00 / 14.08
rs3131972	HLA-DQB1	Immunogenetic risk	14.08H
rs4475691	TPH2	Serotonin synthesis, mood	12.04

The crosswalk is a proposal about how genetic evidence could be indexed to adjudicative criteria. It is not a diagnostic instrument, and the association between any single marker and any listed impairment is far weaker than the table's structure suggests. Chapter 10's discussion of polygenic scores is the controlling caution.

16.3 System exhibits

The specification includes four system exhibits: the execution and scheduling graph; the post-quantum trust chain; the asymmetric convergence loop; and a forum matrix that maps the litigation strategy across the Court of Federal Claims, the Federal Circuit, the district court, the state court, and the bankruptcy court. The forum matrix is reproduced in Part IX, where it is treated as a description of intended practice rather than of filed events.

16.4 The patent claims

The clean-slate disclosure advances an independent claim and six dependents. The independent claim recites a processor architecture comprising a semantic instruction set defining nodes in a directed acyclic graph, each node carrying a 128-bit metadata word; a density-matrix quantum emulation substrate coupled to the graph; a probabilistic microcontroller unit scheduling node execution by Bayesian inference; reconfigurable logic cells with nanosecond morphing; a unified memory lattice in which semantic access is primary; a self-optimising microcode layer; and a post-quantum trust chain binding each node. The dependents add the hybrid quantum-classical substrate, the entropy-reduction scheduler, the morphable fabric, the proximal-policy microcode layer, the cognitive accessibility interface, and the native semantic memory format.

A claim drawn to a processor architecture is directed to a machine or manufacture and is not, on its face, excluded. The exposure is that a claim which recites a desired result — "scheduling by Bayesian inference" or "self-optimising microcode" — rather than a specific technical means may be rejected as claiming an abstract idea under *Alice Corp. v. CLS Bank International* (Alice Corporation v. CLS Bank International 2014). The dependent claims, which recite specific bit layouts, specific operator forms, and specific timing bounds, are the parts of the disclosure most likely to survive, precisely because they are the most concrete. This is **[PROPOSAL]** as to strategy and **[ADOPTED LAW]** as to the governing standard.

16.5 Verification and the handoff package

The specification closes with a verification plan and a list of deliverables: register-transfer-level scaffolds, reference implementations, printed-circuit designs, bills of materials, cryptographic constructions, and the crosswalk. The verification plan is the part that makes the rest auditable, and it is the part this book treats as the model for how the whole portfolio should present technical claims — a claim is a claim only when a test that could falsify it accompanies it.

Part V

The MORTAR Algorithmic Queueing Engine

CHAPTER 17

The MORTAR Triage Engine: Problem, Architecture, and Invariants

An administrative queue is a moral instrument. The order in which a state adjudicates claims determines who is served and who waits, and waiting is not free.

17.1 The problem: a queue that kills

The Social Security Administration adjudicates disability claims on a first-in, first-out basis, subject to statutory and regulatory priorities. First-in, first-out is fair in the sense that it treats arrival order as the only relevant variable, and it is efficient in the sense that it requires no information beyond a timestamp. It is defective in one respect that matters: it is indifferent to how much a claimant's welfare depends on the speed of the decision. When the population in the queue has heterogeneous mortality risk — and the disability population does, being by definition a population with serious medical impairment — an order-blind queue imposes the largest cost on the claimants least able to bear it.

The magnitude of the problem is documented. The Government Accountability Office has reported that over a decade, tens of thousands of claimants died while their appeals were pending, that median appeal processing times exceeded a year and in some years approached two, and that a substantial number of claimants filed for bankruptcy while waiting (U.S. Government Accountability Office 2020). The author's own figure of more than thirty thousand preventable deaths annually in the United States is consistent with the range reported in the literature and in congressional oversight, and is treated here as an order-of-magnitude claim rather than as a precise measurement (U.S. Government Accountability Office 2020; Sanders and Larson 2020).

Register

The existence of the backlog and of mortality during the wait is **[ADOPTED LAW]** in the sense that it is documented by official audit. The attribution of any specific death to the delay is a counterfactual claim that is inherently contestable and is not established by the aggregate figures. MORTAR is **[PROPOSAL]**; its performance is **[MODEL / SIMULATION]**.

17.2 The architecture in outline

MORTAR — Mortality-Optimized Risk-Triage with Adaptive Rescheduling — is a layered, time-indexed directed acyclic graph for administrative claim prioritisation. It organises computation into five layers,

ordered strictly by dependency, and it is designed to satisfy a set of invariants that the author states explicitly.

L0 — Ingestion and validation.
Each claim is represented as a tuple $c_i = (\mathrm{ID}_i, \mathrm{arrival}_i, \mathrm{stage}_i, \mathrm{evidence}_i, \mathrm{status}_i, \mathrm{privacy}_i)$. A claim that fails validation is never deleted; it is routed to a defined exception node — a request for missing evidence, a human review for an identity conflict, a merge review for a duplicate, or a recovery queue for corruption.

L1 — Feature extraction.
Mortality risk, vulnerability, and urgency features are computed from the claim record.

L2 — Priority construction.
A composite priority score is computed with an explicit anti-starvation term.

L3 — Bipartite resource matching.
Claims are assigned to adjudicators and resources under capacity constraints by a matching algorithm.

L4 — State transition.
Each state update is committed to the audit chain described in Part VI.

17.3 The invariants

The specification states nine invariants, of which the first is fundamental.

Principle 17.1 (Claim-state invariant)**.** *Every claim state possesses a predecessor, a processing rule, a resulting state, and an auditable transition record.*

The remaining invariants assert that every input is traceable to a processing node; every node has defined prerequisites; every priority calculation is reproducible; every allocation decision is auditable; every state transition produces a verifiable record; every exception has a defined transition; no claim can disappear silently from the queue; human review is available at every materially adverse transition; and the three petition branches remain logically distinct.

17.4 Priority construction and the anti-starvation guarantee

Layer L2 computes a composite priority score. The specification's form is a weighted sum of risk, vulnerability, and urgency features, to which an ageing term is added:

$$\mathrm{priority}_i(t) = w_r\,\mathrm{risk}_i + w_v\,\mathrm{vuln}_i + w_u\,\mathrm{urgency}_i + \lambda\,(t - \mathrm{arrival}_i), \tag{17.1}$$

with $\lambda > 0$. The last term is the anti-starvation device: because the priority of every waiting claim increases at rate λ, a claim that is repeatedly passed over eventually outranks any claim whose feature score is bounded. If feature scores lie in $[0, S]$ and the weights sum to one, then no claim can be starved beyond a waiting time of order S/λ, which gives the design its hard anti-starvation guarantee. The guarantee is the ethical core of MORTAR: a triage rule that reorders by risk and omits (17.1)'s ageing term is a rule that can postpone a low-risk claimant indefinitely, and indefinite postponement is the injury the engine exists to prevent.

17.5 Resource matching

Layer L3 assigns claims to adjudicators and to constrained resources by a capacity-aware bipartite matching, in which each claim is a node on one side, each adjudicator-and-slot is a node on the other, and the objective is to maximise total priority subject to capacity. The assignment is a min-cost matching problem and is

solved accordingly; the design's contribution is not the algorithm, which is standard, but the requirement that the assignment be committed to the audit chain so that the reason a claim waits is recoverable.

The invariants are the more important half of the design. A triage system that reorders a queue is a system that can harm the people it moves backward; the invariants are what prevent the reordering from becoming an invisible reallocation.

17.6 Jurisdictional decomposition

MORTAR is shared by three petition branches — domestic administrative (SSA), a United Nations human-rights branch, and a neurotechnology-rights branch — which share an evidentiary and algorithmic substrate while remaining logically distinct. The master graph is

$$v_0 \longrightarrow \{v_1, v_2, v_3\} \longrightarrow v_4 \longrightarrow \{v_5, v_6, v_7\},$$

with root the claimant's evidence, the three jurisdictional branches feeding a common algorithmic core, and the core producing evidence, remedy, and enforcement outputs that resolve to a verified, audited state. Separating the branches matters because the legal standards they apply differ: a domestic adjudication and an international petition are not the same proceeding, and a design that fused them would produce a record that satisfies neither forum.

CHAPTER 18

Locality-Sensitive Hashing: Theory, Simulation, and Fairness

A hash function that preserves similarity is a way of making the question "is this like that?" cheap. Cheap similarity is powerful, and power requires a fairness audit.

18.1 Why locality-sensitive hashing

The computational bottleneck in triage is comparison: deciding which of many claims are alike in the features that matter, without comparing every pair. Locality-sensitive hashing solves exactly this problem. A family of hash functions is locality-sensitive if similar inputs collide with higher probability than dissimilar ones.

Principle 18.1 (Locality-sensitive family)**.** *A family $\mathcal{H}$ is (r_1, r_2, p_1, p_2)-sensitive for a similarity measure if, for $p_1 > p_2$, any pair with distance at most r_1 collides with probability at least p_1 and any pair with distance at least r_2 collides with probability at most p_2.*

The construction the specification names is drawn from the p-stable hashing literature (Datar et al. 2004) and from the min-wise and random-projection schemes that preceded it (Gionis et al. 1999a). For a random projection with a p-stable distribution, the hash is

$$h_{a,b}(x) = \left\lfloor \frac{a \cdot x + b}{w} \right\rfloor, \tag{18.1}$$

and the collision probability for two points at distance d is

$$P\big[h(a) = h(b)\big] = \int_0^w \frac{1}{d}\, f_p\!\left(\frac{t}{d}\right)\left(1 - \frac{t}{w}\right) dt, \tag{18.2}$$

where f_p is the density of the stable distribution. For the angular random-projection variant, the collision probability has the closed form

$$P\big[h(a) = h(b)\big] = 1 - \frac{\theta(a,b)}{\pi}, \tag{18.3}$$

with θ the angle between the vectors. Equation (18.3) is the reason the design uses random projections: the collision probability is a monotone function of similarity, so tuning the hash tunes the triage threshold analytically rather than by trial.

18.2 Amplification

A single hash is too weak. Two standard amplifications are composed. The *and*-construction places k hashes in a band and requires all to agree, multiplying specificity and raising the false-positive resistance to $1 - p_1^k$. The *or*-construction uses L independent tables and accepts a match if any table agrees, raising recall to $1 - (1 - p_1^k)^L$. The design's parameters are the number of tables and the band size, and they are tuned to a target recall and precision pair.

Increasing k raises precision and lowers recall; increasing L raises recall and costs memory and query time. The pair (k, L) is therefore the triage engine's policy surface, and it should be reported as such — an engine that does not disclose its (k, L) does not disclose how many similar claims it will miss.

18.3 Complexity

For n claims, L tables, and band size k, the query cost is $O(L \cdot (k + C))$ where C is the number of candidates retrieved, against $O(n)$ for a linear scan. The design's claim is that the constant factor and the candidate count are small enough at the relevant n that the engine is practically cheaper than a scan; that claim is **[MODEL / SIMULATION]** and is falsifiable by benchmark.

18.4 Simulation and its limits

The specification reports a Monte Carlo simulation of the triage engine. Three limits should be borne in mind. First, a simulation of a priority rule is a simulation of the rule, not of the agency: it cannot model the political economy that shapes the queue. Second, the simulation's feature distribution is assumed rather than observed, and the result is therefore conditional on that assumption. Third, a simulated reduction in waiting time is not a reduction in mortality unless a hazards model links the two, and the hazards model is where the causal claim lives. The chapter's conclusion is that the simulation supports the engineering claim — the engine computes what it says it computes — and does not by itself support the policy claim.

18.5 Fairness audit

A triage engine that prioritises by predicted mortality risk will, by construction, prioritise the sick. If the mortality model is less accurate for a group that has been historically under-treated, the engine can deprioritise that group not because its members are healthier but because the model sees them less clearly. The mitigation specified in the design is twofold: the anti-starvation term of Chapter 17 guarantees that no claim waits beyond a bounded time, and the audit chain of Part VI records the features and the model version that produced each priority, so that a disparity can be discovered after the fact. Neither measure eliminates the disparity; both make it visible, and visibility is the condition of correction.

Part VI

The OpenSSA Integrity Pipeline and Post-Quantum Security

CHAPTER 19

The OpenSSA Integrity Pipeline

Evidence that cannot be reproduced cannot be rebutted, and evidence that cannot be rebutted is not evidence; it is assertion.

19.1 The problem of the untraceable record

An adjudicative record assembled by software has a property that a paper record does not: its intermediate states are ephemeral. A model ingests a document, transforms it, and produces a conclusion, and the transformation leaves no trace unless the system is built to leave one. The consequence is a record that can be authenticated at its endpoints — the input existed and the output exists — while being unfalsifiable in the middle. The OpenSSA integrity pipeline (OSI-P) is the author's answer: a nine-stage audit trail in which each stage commits a hash of its output to a chain that a later stage can verify.

19.2 The nine stages

The pipeline names nine stages, each of which corresponds to a step at which a generative or retrievally system can silently alter material.

1. **Source integrity.** The provenance and hash of each source document.
2. **Query integrity.** The exact query issued to the retrieval system.
3. **Retrieval integrity.** The identity of each retrieved item.
4. **Augmentation integrity.** The context assembled for the model.
5. **Generation integrity.** The model, version, and parameters that produced the output.
6. **Validation integrity.** The checks applied to the output and their results.
7. **Chain of custody.** The ordered handoff of the material between stages.
8. **Reversibility.** The ability to reconstruct intermediate states.
9. **Audit summary.** A single signed digest of the whole.

The stages are not independent checks; they compose. Reversibility is what makes the earlier stages worth recording, because a party who can recompute an intermediate state can test the claim that the state was what the record says it was.

19.3 The Merkle chain

Each stage's output is hashed with SHA3-256 and the digest is committed to a Merkle tree over the pipeline's outputs. The Merkle structure gives the property that a single root digest authenticates the whole transcript, while allowing any individual leaf to be proved included with a logarithmic-size proof. The design signs the root with a post-quantum signature (ML-DSA-87, the standardised name for the scheme the specification calls Dilithium-III/ML-DSA-87), so that the chain's integrity does not depend on the hardness of factoring or of discrete logarithms.

SHA3 is chosen because it is a standardised sponge construction with no known structural weakness of the kind that has mattered for older hashes. The Merkle tree is chosen because it converts a linear transcript into a logarithmic inclusion proof, which is what makes the audit usable in a court: a party can prove that a specific document was part of the record without producing the whole record. The two choices are independent, and each is defensible on its own.

19.4 Self-authentication under the evidence rules

The pipeline is designed so that its output can be admitted as a self- authenticating record under Rules 902(13) and 902(14) of the Federal Rules of Evidence. Rule 902(13) permits authentication of a record generated by an electronic process on the basis of a certification by a qualified person who describes the process; Rule 902(14) permits authentication of a copy of data by a hash value and a certification. The pipeline's design maps onto both provisions directly: the Merkle root is the hash value of Rule 902(14), and the stage certifications are the descriptions of process required by Rule 902(13).

Rule 902(13) and 902(14) authenticate a record; they do not establish that the record is true. A chain that proves the model was version 4.2 in June does not prove that version 4.2 was accurate, fair, or lawful. The pipeline is a condition of admissibility, not a substitute for the substantive inquiry, and a brief that treats a clean audit chain as proof of correctness has confused the two.

19.5 The evidentiary fusion engine

The application layer assembles material from public court dockets, biomedical repositories, and scholarly indexes, and routes it through the pipeline. This is the point at which the architecture touches the litigation theory of Part II: the route by which an exhibit becomes admissible is the route the pipeline records, and the record it produces is the record the *Daubert* inquiry of Chapter 10 requires.

CHAPTER 20

The Post-Quantum Security Module

The data a state collects about its citizens has a long tail of sensitivity. A cipher whose security expires before the data does is a cipher that has already failed, quietly.

20.1 The threat model

Two quantum threats bear on data infrastructure. The first is the retrospective one: an adversary records encrypted traffic today and decrypts it when a sufficiently capable quantum computer exists — the "harvest now, decrypt later" pattern. Because genomic, medical, and legal records are sensitive for decades, a record encrypted under a scheme that a quantum computer can break is effectively already exposed. The second is the prospective one: an adversary forges a digital signature, undermining the authenticity of records rather than their confidentiality.

The National Security Agency's Commercial National Security Algorithm Suite 2.0 sets a transition timeline with software and firmware signing moving first and full adoption following (National Security Agency 2022a). The NIST standards published in August 2024 supply the primitives (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d).

20.2 The module's components

The security module comprises four components: a key-management service that derives and escrows post-quantum keys under policy; the dual-envelope authenticated-encryption construction of Chapter 15; a signature service for records and for the audit chain; and a zero-knowledge proving service that answers eligibility and priority queries without disclosing the underlying record.

20.3 Zero-knowledge proofs in the adjudicative context

A zero-knowledge proof allows a prover to demonstrate that a statement is true without revealing why it is true. The adjudicative use is precise: a claimant can prove that a priority threshold is met — for example, that a computed risk feature exceeds a value — without disclosing the genetic or clinical data from which the feature was computed. The property is valuable because the alternative is a system in which the claimant must expose the most sensitive data available about herself in order to be moved up a queue.

Status of the zero-knowledge component

That zero-knowledge proof systems exist and can prove statements of the relevant form is **[ADOPTED LAW]**. That the specific circuit needed here has been built, and that its proof cost is acceptable at the scale of a national adjudication queue, is **[DESIGN TARGET]**. The chapter states the design and names the gap.

20.4 Key management is the whole of the matter

The module's strongest cryptographic component is also its least novel, and its weakest component is the one that is easiest to neglect. A post-quantum KEM protects a payload only if the recipient's private key is protected; a post-quantum signature authenticates a record only if the signing key is not compromised. The history of cryptographic failure is overwhelmingly a history of key management failure, and the module's specification of key derivation, split, and binding is the part of Part VI that deserves the most scrutiny in deployment.

Part VII

International Human Rights Framework

CHAPTER 21

Request for a Commission of Inquiry

Domestic remedies fail silently. International mechanisms exist precisely for the case in which the ordinary process has become the instrument of the harm.

21.1 The instrument and its jurisdictional basis

The petition asks the United Nations Human Rights Council to establish a commission of inquiry into the administration of disability adjudication in the United States, and to refer questions to the International Court of Justice for an advisory opinion. The procedural vehicle is the Human Rights Council's complaint procedure, and the substantive basis is a set of treaty obligations the United States has assumed: the International Covenant on Civil and Political Rights, the Convention against Torture, and — as a matter of the norms invoked even where formal ratification is absent — the Convention on the Rights of Persons with Disabilities. The petition invokes the jus cogens character of the prohibition on arbitrary deprivation of life.

Register and a necessary caution

The petition is **[PROPOSAL]**: it is a request, not a filed instrument, and the establishment of a commission of inquiry is a discretionary act of the Council. The treaty architecture it invokes is **[ADOPTED LAW]**. Where the petition characterises aggregate mortality as a violation, the characterisation is an argument about attribution, and it is contested; the book preserves the distinction between the documented facts and the legal conclusion drawn from them.

21.2 The factual predicate and its sources

The petition rests its factual predicate on four categories of official and scholarly material. The first is the mortality of claimants awaiting adjudication: the Government Accountability Office reported a documented figure on the order of one hundred nine thousand seven hundred twenty-five appellants who died between 2008 and 2019 while their appeals were pending (U.S. Government Accountability Office 2020). The second is the scale of the backlog and its duration. The third is the reported association between benefit removal and subsequent incarceration, for which the petition cites the empirical literature on the criminal-justice consequences of disability-benefit loss. The fourth is the reported elasticity of mortality with respect to disability income, cited to the peer-reviewed literature (Gelber et al. 2023).

The table is the petition's strongest asset and, read carefully, its most fragile. The first row is a count from an official audit. The later rows are estimates with confidence intervals and modelling assumptions,

Table 21.1: The petition's quantified factual predicate.

Metric	Value	Source
Documented deaths, appellants	109,725	GAO-20-641R (U.S. Government Accountability Office 2020)
Deaths, pending claims FY2017–2023	133,000+	Congressional oversight materials
Federal-court remand/error rate	60–65%	Agency and oversight data
Incarceration increase after removal	60%	Empirical literature
Mortality elasticity w.r.t. income	−0.6 to −1.0	Gelber et al. (2023)
Projected annual deaths under austerity	∼67,000	Congressional oversight

and a commission's task is precisely to interrogate those assumptions. The book reports the table as the petition offers it, and states the epistemic status of each row.

21.3 The substantive claims

The petition advances claims under four provisions.

Article 6 — the right to life.
The central claim is that the knowing maintenance of an adjudicative system that predictably causes preventable death is itself a breach of the duty to protect life. The Human Rights Committee's General Comment 36 is the interpretive vehicle: it links the right to life to the state's positive obligations and to the conditions necessary for a dignified existence.

Article 9 — liberty and security.
The claim is that claimants subject to multi-year administrative process, restrictive conditions, and repeated monitoring are subject to a continuous deprivation of liberty without a timely determination.

Article 2(3) — an effective remedy.
The claim is that a remedy that remands without deciding is not an effective remedy, because it returns the claimant to the queue that produced the harm.

CAT Article 16 — cruel, inhuman, or degrading treatment.
The claim is that the causal pathway from benefit denial to poverty, homelessness, contact with the criminal-justice system, and incarceration constitutes the prohibited treatment at a structural scale.

21.4 The "restraint of silence" and cognitive accessibility

The neurogenetic dimension is the petition's most distinctive and most contested. The petition argues that the state's categorical refusal to consider validated neurogenetic evidence of an executive-function burden, and its failure to provide a cognitive accommodation, amount to a form of exclusion from the adjudicative process — a "restraint of silence" — that violates the Convention on the Rights of Persons with Disabilities. The strength of the argument lies in its framing: it does not ask the state to accept a contested causal claim about a specific marker; it asks the state to supply an accommodation that would let a cognitively burdened claimant participate. That framing is stronger than the biomarker claim, and the book separates them.

21.5 Requested relief

The petition requests the establishment of a commission of inquiry, the inclusion of the relevant special rapporteurs, a report within twelve months, interim measures suspending workforce reductions, expedited adjudication for high-risk claimants, and a referral to the International Court of Justice for an advisory opinion on the questions stated in Chapter 22.

CHAPTER 22

The Systemic Gravity Index and Treaty Mapping

A metric that measures the gravity of a violation is an argument in numerical form. Its value lies not in its precision but in the questions it forces an institution to answer.

22.1 The index

The Systemic Gravity Index aggregates the petition's quantifications into a single comparative instrument, so that the scale of the alleged harm can be stated in a form an international body can weigh against other matters on its docket.

Table 22.1: Components of the Systemic Gravity Index.

Component	Quantity
Documented mortality	Deaths of claimants awaiting adjudication
Temporal magnitude	Multi-year median processing times
Systemic recurrence	Annual percentage of initial denials reversed on review
Downstream harm	Post-removal incarceration and homelessness
Vulnerability of population	Prevalence of high-mortality conditions in the queue
Institutional knowledge	Evidence of notice and persistence

An index of this kind is a communication device rather than a measurement. Its components are not commensurable, and no weighting can make them so. Its function is to prevent a reviewing body from assessing a systemic harm as if it were an isolated dispute, and in that function it is useful.

22.2 Treaty mapping

The petition maps its claims onto three instruments and one body of customary norms.

ICCPR.
Articles 2(3), 6, and 9 supply the right to a remedy, the right to life, and the right to liberty and security. The interpretive anchor is the Human Rights Committee's General Comment 36.

CAT.
Article 16 supplies the prohibition on cruel, inhuman, or degrading treatment, invoked at the structural rather than the individual scale.

CRPD.
Articles 10, 12, 19, 26, and 28 supply the right to life on an equal basis, the right to legal capacity, the right to live in the community, the right to habilitation and rehabilitation, and the right to an adequate standard of living. The cognitive-accessibility claim in particular is an Article 12 and Article 26 claim.

Jus cogens.
The petition invokes the peremptory character of the prohibition on arbitrary deprivation of life to argue that the obligations are not defeasible by domestic procedure.

22.3 The questions referred

The petition proposes four questions for an advisory opinion. Does the maintenance of an administrative system that predictably causes preventable mortality violate Article 6? Does the deprivation of income, with its documented consequences, constitute treatment within Article 16 of CAT? Is a system whose review mechanism returns a majority of cases to the body that denied them an effective remedy within Article 2(3)? And does the criminalisation of claimants resulting from benefit removal violate Article 9?

The questions are well framed, and that is the chapter's point: a referral is only as useful as the questions it asks, and these questions are answerable in principle even by a body that discounts the petitioner's factual characterisations.

22.4 The limits of the mechanism

Three limits should be stated plainly. International mechanisms are slow, they are non-coercive, and they do not by themselves alter domestic administration. A commission of inquiry can establish facts and name responsibilities; it cannot order a legislature to fund an agency. The petition's value is therefore evidentiary and normative — it creates an authoritative record — rather than remedial. The book states this because the register discipline of Chapter 2 requires it: a request is not a remedy, and a report is not a judgment.

Part VIII

The PSDLEF Sovereign Exchange Trust

CHAPTER 23

Statutory Authority and Juridical Structure

A purpose trust is a legal form that lets an institution outlive both its founders and its beneficiaries. That permanence is the point, and it is also the risk.

23.1 The instrument

The PSDLEF Sovereign Exchange Trust is proposed as a perpetual purpose trust — a trust without individual human beneficiaries, established to carry out a stated purpose in perpetuity. The author frames it under the Uniform Directed Trust Act and the purpose-trust provisions of Delaware and Wyoming law. The stated purpose is the permanent capitalisation, administrative optimisation, and solvency support of federal disability and safety-net programmes.

Register

The purpose trust is an adopted legal form; the trust described here is **[PROPOSAL]**. No such instrument has been created, and the chapter describes the structure the author proposes rather than an executed indenture. Where the chapter reports yields, the figures are **[DESIGN TARGET]**: they are the returns the instrument would need to produce, not returns it has produced.

23.2 Why a perpetual purpose trust

Three features of the form are doing work. First, perpetuity: the trust does not terminate on a beneficiary's death, which suits a purpose measured in decades. Second, non-charitable purpose: the trust is not a charity and is not constrained by charitable-purpose doctrine, which allows a broader investment mandate than a charitable remainder would. Third, directed trusteeship: investment authority is separated from distribution authority, which is the structural protection against a trustee that invests to fund itself rather than to fund the purpose.

The counterweight is that a perpetual purpose trust is precisely the vehicle in which accountability is hardest to maintain, because there is no residuary beneficiary with standing to complain. This is why the governance design of Chapter 24 matters more than the investment design of Chapter 25, and the book orders them accordingly.

23.3 The four-component architecture

The trust is the legal shell for the four technical components the portfolio has already introduced: the MORTAR triage engine (Part V), the OpenSSA integrity pipeline (Part VI), the DYREO capital-allocation optimiser (Chapter 25), and the post-quantum security module (Chapter 20). The claim is that a purpose trust can hold these components as operational assets serving a public purpose, and can fund their operation from the return on a portfolio rather than from appropriations.

23.4 The constitutional question

A trust that performs functions the federal government performs raises two constitutional questions, and the chapter states them rather than resolving them.

1. **The non-delegation and Appointments questions.** Adjudication of disability claims is a governmental function. A private trust may not exercise it. The design therefore confines the trust to capitalisation and to the supply of tools; the agency decides, the trust funds.
2. **The sovereign-immunity question.** A trust that holds assets for a public purpose and is funded by the state may be treated as an arm of the state, which would import sovereign immunity and alter the remedies available. The design mitigates this by keeping the trust private and the state a beneficiary-like recipient rather than an owner.

23.5 The funding rail

The proposed funding rail routes assets through a Wyoming custodial structure into the trust, with multi-signature control and post-quantum signatures on authorisations, under a policy of no leverage, no lending, and no derivatives. The no-leverage policy is the design's most important prudential commitment: a trust whose purpose is solvency cannot itself be a source of systemic risk.

CHAPTER 24

Governance, Fiduciary Roles, and the Perpetuity Rule

The rule that a trust may distribute at most five per cent of its corpus is a rule against the future being consumed by the present.

24.1 Fiduciary separation

The governance design separates four roles. The trust director holds investment authority. A distribution committee holds the authority to release funds to beneficiaries. An independent audit function reviews both. And a standards committee owns the model and policy versions on which the technical systems depend. The separation is not ornamental: the failure mode it prevents is the one in which the entity that benefits from a distribution also decides that the distribution should occur.

24.2 The perpetuity rule

The trust's foundational rule limits distributions to five per cent of a trailing five-year average of the corpus:

$$D(t) \le 0.05 \times \frac{1}{5} \sum_{k=1}^{5} \text{Corpus}(t-k). \tag{24.1}$$

Any investment yield above five per cent is retained and compounded within the capital-preservation core. The rule is the standard spending rule of an endowment, and its function is to make the trust's claims about perpetual returns internally consistent: a trust that distributes more than it earns cannot be perpetual, and the rule makes that impossibility a matter of arithmetic rather than of good faith.

If a distribution rate of five per cent is applied to the trailing corpus and investment return is assumed to be eight per cent nominal, the corpus grows at roughly three per cent nominal, which is a real growth rate of approximately one per cent at two per cent inflation. That is a thin margin, and it is the reason the chapter treats the trust's solvency claim as contingent on the return assumption rather than as established by the structure.

24.3 Drift detection

The design includes a monitoring mechanism that tests whether the distribution of the trust's inputs has shifted away from the baseline on which its models were calibrated. The test is the two-sample Kolmogorov–Smirnov statistic,

$$D_n = \sup_x \left| F_{\text{empirical}}(x) - F_{\text{baseline}}(x) \right| , \tag{24.2}$$

evaluated against a threshold at $p < 0.05$, with a re-optimisation triggered when the threshold is crossed. The mechanism is the governance counterpart of the fairness audit in Chapter 18: it makes a change in the population visible before it becomes a change in outcomes.

24.4 Reporting

The trust reports on a schedule that separates financial from operational performance, and it reports the model versions that produced each operational number. The requirement is drawn from Part VI: a report that does not identify the model that produced it is a report that cannot be audited.

CHAPTER 25

DYREO Capital Allocation and the 31-Algorithm Execution Suite

An optimiser is a statement about what matters. Writing down the objective function is therefore a normative act disguised as a mathematical one.

25.1 The DYREO formulation

The Dynamic Yield and Real Estate Optimization model is a constrained multi-period portfolio problem. In its one-period form it maximises a weighted combination of financial return, social impact, and risk:

$$\max_{w} \; \alpha\, w^{\top}\mu + \beta\, w^{\top}s - \gamma\, w^{\top}\Sigma w, \tag{25.1}$$

subject to a minimum social-impact constraint, a capital-preservation probability constraint, asset-class bounds, and full investment:

$$w^{\top}s \geq S_{\min}, \tag{25.2}$$

$$P(w^{\top}r < -\delta) \leq \epsilon, \tag{25.3}$$

$$w_{\text{housing}} \in [0.40, 0.50], \quad w_{\text{clean}} \in [0.15, 0.25],$$
$$w_{\text{fixed}} \in [0.20, 0.30], \quad w_{\text{liq}} \in [0.05, 0.15], \tag{25.4}$$

$$\sum_{i} w_i = 1, \quad w_i \geq 0 \;\forall i. \tag{25.5}$$

Here w is the weight vector, μ the expected financial yields, s the social- impact scores, Σ the covariance matrix, and α, β, γ the tuning weights. The formulation is a standard mean–variance problem with two additional constraints, and it is honest about its own structure: the social objective enters linearly through s, which means the model treats social impact as commensurable with return, and that commensuration is a choice the model does not justify.

25.2 Asset classes and target yields

The target yields are **[DESIGN TARGET]**. A portfolio with forty-five per cent of its capital in housing and twenty per cent in clean-energy infrastructure is an illiquid portfolio, and illiquidity is a real cost that the target yields do not show. The chapter's contribution is to state the cost the model omits.

Table 25.1: DYREO asset-class structure and target yields.

Asset class	Allocation	Target yield
Sustainable housing	45%	6.5–8.5%
Clean-energy infrastructure	20%	7.0–9.0%
Sovereign / fixed income	25%	4.5–5.5%
Liquid reserves	10%	4.0–4.5%

25.3 The social energy return

The design measures the social return on the energy assets with a ratio,

$$\text{S-EROI} = \frac{\int_0^T \left[P_{\text{solar}}(t) - P_{\text{loss}}(t)\right] dt}{K_{\text{cap}} + \int_0^T C_{\text{maint}}(t)\, dt} \geq 0.73 \text{ kWh/\$}, \tag{25.6}$$

which compares lifetime energy delivered to lifetime cost. The threshold of 0.73 kWh per dollar is a design target, and its meaning depends entirely on the discount rate and the time horizon, both of which the expression suppresses.

25.4 The 31-algorithm execution suite

The financial infrastructure is specified as a suite of thirty-one algorithms organised into modules.

Table 25.2: Module structure of the executive algorithmic suite.

Module	Function	Indices
I	Bond valuation and claims	01–03
II	Asset conversion and distribution	04–06
III	Volatility modelling	07–09
IV	Execution optimisation	10–12
V	Derivatives and hedging	13–15
VI	Risk management	16–18
VII	Yield optimisation	19–20
VIII	Automated market-maker liquidity	21–22

The module taxonomy is descriptive of the author's library rather than of a proprietary trading system, and the book treats it as such. Its relevance to the argument is that it supplies the operational machinery the trust would need if it were ever funded, and its presence in the portfolio is a reminder that a perpetual purpose trust is an institution with an investment office, not merely a legal form.

CHAPTER 26

Energy and Infrastructure: The Solar/Battery Pilot and Bounded Quantum Research

The responsible way to hold a speculative asset is to size it so that its total loss is survivable.

26.1 The thirty-house solar and battery pilot

The pilot proposes to retrofit thirty Baltimore rowhouses with photovoltaic generation and lithium-iron-phosphate battery storage, forming a decentralised microgrid capable of islanding.

Table 26.1: Scaled thirty-house pilot specification.

Component	Specification
Houses	30 Baltimore rowhomes
Generation	∼5 kW per house, 150 kW total
Storage	∼10 kWh LFP per house, 300 kWh total
Grid model	Decentralised microgrid; 24-hour islanding
Revenue	Energy-service tariffs plus affordable-housing leases
Municipal target	5,000-unit vacancy reduction by FY2029

The pilot is a **[DESIGN TARGET]**. It is small enough to be a genuine demonstration and large enough to test the financing model, and its revenue assumption — that energy-service tariffs and lease income together cover capital and maintenance — is the assumption that must be tested first, because the S-EROI of equation (25.6) is only as good as that assumption.

26.2 Funding rail and control

The proposed rail converts custodial assets into restricted grant funding for the pilot through regulated venues, with multi-signature control and post-quantum signatures. The chapter's editorial position is that the funding rail is the weakest link in the whole financial argument, because it depends on the character of the assets converted and on the regulatory classification of the conversion, and neither is settled by the design.

26.3 The quantum gravitational reactor as a bounded allocation

The specification includes a research allocation labelled the quantum gravitational reactor. The specification itself, to its credit, opens with a critical caveat: no commercially operational device of this kind exists, and the component is a high-risk, optionality-preserving research layer rather than a load-bearing energy asset. The book preserves that framing and strengthens it.

Why the caveat is the correct treatment

The technologies catalogued in the source — pulsed-power radiation sources, theoretical neutron-superfluidity proposals, zero-point-energy transduction, and concept-only gravitational-wave patents — span a range from real research programmes to claims that the mainstream literature regards as unfounded. The correct institutional treatment of a speculative research allocation is not suppression but sizing: capital is committed in an amount whose total loss leaves the mission intact. The design does this, with a success gate of a peer-reviewed paper or an independent replication and a failure mode of a write-down that does not touch the solar and battery assets.

26.4 The due-diligence checklist

The specification's due-diligence questions are worth reproducing because they are the right questions: what is the claimed energy balance; has the effect been independently replicated; is there a peer-reviewed description of the mechanism; what is the theoretical basis; and what are the safety and regulatory requirements. A research allocation that cannot answer all five should not be funded, and the checklist is the trust's defence against its own enthusiasm.

CHAPTER 27

CUSIP Normalization, the Maryland Transfer, and the Dual-Court Filing

Registry mechanics are unglamorous and decisive. An asset that cannot be identified cannot be transferred, and a claim that cannot be filed cannot be heard.

27.1 CUSIP normalization

The protocol normalises securities identifiers under the standard check-digit algorithm, a modified Luhn scheme: convert alphabetic characters to numeric values, double the values at even positions, sum the digits of each product, and take the check digit as $(10 - (P \bmod 10)) \bmod 10$. The algorithm is a standard identifier-validation routine, and its inclusion in the portfolio is a sign that the financial plan is meant to be executed at the level of individual instruments rather than at the level of a diagram.

27.2 The Maryland transfer protocol

The transfer follows the Maryland Uniform Disposition of Abandoned Property Act: identify the identifiers in the portfolio; validate their format and check digit; prepare the transfer documentation, including authorisation, proof of ownership, and taxpayer identification; submit through the state unclaimed-property portal; receive a claim number; and schedule a pickup. The protocol is procedural, and its status is [PROPOSAL] as to the transfer and [CITE REQUIRED] as to any particular asset, because a transfer depends on ownership that the record must establish.

27.3 The dual-court filing

The litigation strategy for the intellectual property and trade aspects is coordinated across two fora.

Court of International Trade.
The trade-administration and transfer aspects are pleaded under the residual jurisdiction of 28 U.S.C. § 1581(i), which reaches civil actions against the United States arising out of federal trade law.

Court of Federal Claims.
The compensation claim for government use of a patented invention is pleaded under 28 U.S.C. § 1498(a), which supplies exclusive jurisdiction over such claims.

Transfer.
Where a court lacks jurisdiction, transfer to cure the defect is requested under 28 U.S.C. § 1631, if in the interest of justice.

27.4 The substantive arguments

Three arguments carry the filing. The first is a public-trust anchoring under the Bayh–Dole Act, which requires that federally supported inventions be utilised to promote free competition and public availability (Bayh–Dole Act 1980). The second is Supremacy Clause preemption of conflicting state assertions, for which the petition cites *United States v. Little Lake Misere Land Co.* (United States v. Little Lake Misere Land Company 1973). The third is the four-factor injunction standard of *Winter v. Natural Resources Defense Council* (Winter v. Natural Resources Defense Council 2008), already used in Part II.

27.5 Requested relief

The filing requests that the court assume jurisdiction, order transfer if warranted, declare the federal rights and the public-trust framework, enjoin state officials from altering cryptographic logs or asserting conflicting claims, direct the recording of federal interests and the implementation of open licensing, and grant leave to proceed in forma pauperis under 28 U.S.C. § 1915. Each request carries its own threshold, and the chapter states them as the filing states them rather than as achievements.

27.6 Forward reference to the integrated capital arithmetic

The CUSIP normalisation described in this chapter is the ledger-side precondition for the capital-structure arithmetic. Chapter 42 (Part XIII) applies the modified Luhn validator to the specific 1999 USEC indenture, runs the 0.0917 plan exchange ratio against the $500M legacy principal, and states the consolidated capitalisation vector on which the DYREO engine operates.

Part IX

Statement of Facts and Procedural History

CHAPTER 28

The Discipline of a Statement of Facts

A statement of facts is not a story. It is a series of propositions, each of which the reader can trace to a document that can be produced.

28.1 Why the record comes last

Parts I–VIII set out a legal theory, a scientific instrument, a computing architecture, and a financing structure. Every one of those parts rests on claims about what has happened to a named person in named forums. This part turns the book's own analytical instruments on the book's own record. It is placed here, after the theory and before the policy, because it can only be written honestly once the reader knows what the theory asks the facts to prove (42 U.S.C. 1983 2018).

The chapter states the events, the forums, and the parties in chronological order. It then marks, in the text itself, every proposition that the record does not yet establish. The marker **[CITE REQUIRED]** is used throughout this part exactly as it was introduced in Chapter 2: it names the document that would close the gap, and it is not a rhetorical hedge. A proposition carrying the marker should not be quoted in a filing until the named document is in the record.

28.2 The three registers of a factual assertion

Facts in litigation are not all of one kind, and the discipline of this part is to keep the kinds apart.

Recorded.
An event appears in a document that a custodian produced: a docket entry, a notice of decision, a transcript, an agency order. A recorded fact is **[ADOPTED LAW]** in the sense that its occurrence is the sort of thing a court will accept from the document itself.

Inferred.
A recorded fact supports a conclusion that goes beyond what the document says. An inference is legitimate only if it is stated as an inference and the competing inference is identified. Most of what a complaint alleges about motive is of this kind.

Asserted.
A proposition appears only in the author's own papers. It is not yet evidence. It is marked **[CITE REQUIRED]** and carries the burden of identifying what would make it evidence.

28.3 The adversarial check

A statement of facts written by a party is tested by asking how the opposing party would answer it. This part therefore states, for each contested event, the adverse fact that a respondent would supply. An inconsistent state-court entry, an adverse administrative finding, or a missed filing deadline is disclosed here rather than omitted. A fact section that omits the adverse entries is not a failing of candour only; it is a failing of strategy, because the omission will be found and will cost the pleader the benefit of the doubt on every other sentence (Heck v. Humphrey 1994).

Every date in this part should come from a docket entry or a notice, never from recollection. Every quotation should carry a pin cite. Every assertion that a government actor did something must identify the actor, the date, the capacity in which the actor acted, and the document that records the act. Where any one of those four elements is missing, the text says so.

CHAPTER 29

The Chronology in Outline

The order of events is itself an argument. A reader who has the chronology does not need to be told which facts matter.

29.1 The forums

The author's matters have proceeded, or are described as proceeding, in four different forums. Keeping them distinct is a precondition of coherent analysis, because the doctrines, the standards of review, and the available remedies do not transfer between them.

Table 29.1: The forums described in the project materials.

Forum	Route	What it can decide
Federal district court	42 U.S.C. § 1983	Civil-rights claims against state and local officials (42 U.S.C. 1983 2018)
Social Security review	42 U.S.C. § 405(g)	Judicial review of a final agency decision on disability benefits (42 U.S.C. 405(g) 2018)
Court of Federal Claims	28 U.S.C. § 1491	Money claims against the United States, including patent claims under § 1498 (28 U.S.C. 1491 2018)
State court / bankruptcy court	Case-specific	The underlying state matter and the Chapter 11 proceeding (Bankruptcy Code 1978)

29.2 The federal civil-rights matter

The project materials describe a federal civil-rights action assigned the docket numbers 25-1222, 25-1224, 25-1522, and 24-3283, arising from interactions with state and local officials in Maryland. The materials characterise the matter as presenting claims for retaliatory arrest, coercive interrogation, fabricated evidence, conspiracy, and municipal liability — the five counts analysed doctrinally in Part II. The factual predicate for each count is a discrete encounter; the source materials identify and date several of these encounters. **[CITE REQUIRED]** The operative complaint, the docket sheets, and the transcripts of the relevant proceedings would convert the description into a record.

Two limitations must be stated plainly. First, the existence of a state criminal matter does not by itself establish any federal constitutional violation. Each inference from the state matter to a federal claim requires its own factual and legal basis. Second, the standard for a retaliatory-arrest claim after *Nieves v. Bartlett* and its 2024 refinement in *Gonzalez v. Trevino* is demanding: the plaintiff must plead and prove both a

retaliatory motive and the absence of probable cause, or else fall within a narrow class of cases otherwise (Nieves v. Bartlett 2019; Gonzalez v. Trevino 2024). A pleading that recites retaliation without confronting the probable-cause element invites dismissal.

29.3 The Social Security Administration matter

The materials describe a disability matter before the Social Security Administration and a draft petition, dated March 15, 2026, invoking 42 U.S.C. § 405(g) against the Administration, the Department of Health and Human Services, and the United States. **[CITE REQUIRED]** The file-stamped petition, the final agency decision, and the administrative-record index are needed to fix the parties, the decision under review, the statutory route, and the relief sought.

The complete history of such a matter has a settled shape, and the record must supply each stage or the stage must be omitted: initial application; initial determination; reconsideration, if any; hearing request; the administrative law judge's decision; Appeals Council action; the dates of each notice; and any subsequent judicial filing. Every one of those dates should come from a notice or a docket entry rather than from inference.

Register

That the Social Security disability queue is long, and that claimants have died while appeals were pending, is **[ADOPTED LAW]** on the strength of official reporting (U.S. Government Accountability Office 2020). That a specific claimant's matter followed a specific procedural path is a matter of **[CITE REQUIRED]** until the notices are in the record. That a given triage design would have shortened that claimant's wait by a stated number of days is a **[MODEL / SIMULATION]** proposition.

29.4 The bankruptcy and securities background

The Chapter 11 materials describe a personal reorganisation whose principal asset is a portfolio of securities reportable under Section 13(f) of the Securities Exchange Act (Securities Exchange Act of 1934 1934). The plan classifies claims into the ordinary administrative, priority, secured, and unsecured classes and proposes distribution from the liquidation of the portfolio (Bankruptcy Code 1978). Because the estate is an individual's and the portfolio is the funding source for everything else the book proposes, the liquidation schedule is treated in Part VIII as the hinge of the financing structure. **[CITE REQUIRED]** The schedules, the statements of financial affairs, and the liquidation analysis would establish the values the plan relies upon.

A Subchapter V reorganisation, if available on the facts, alters the procedural posture — it dispenses with the creditors' committee, fixes a short exclusive period for the debtor to propose a plan, and permits the debtor to retain equity over an objection in defined circumstances (Small Business Reorganization Act of 2019 2019). Whether the debtor qualifies is a question of statutory thresholds that the record must answer.

CHAPTER 30

Technical and Scientific Evidence in the Record

An assertion becomes evidence when someone can say what was measured, with what instrument, by whose hand, and with what result.

30.1 Genomic and neurophysiological evidence

The materials propose a framework that fuses functional evidence with genomic, neuroimaging, and neurobiological information. They refer to candidate markers, among them COMT rs4680 and loci associated with executive function, together with crosswalks to Social Security Listings 11.00, 12.00, and 12.11. Those references do not establish that the Administration received the underlying genotyping, that a qualified examiner interpreted it, or that any adjudicator relied on it. **[CITE REQUIRED]** The laboratory reports, the chain of custody for the samples, and the treating-source records would establish what was measured.

The evidentiary status of a polygenic score is governed by Part III. A score is a population-level statistical estimate; its probative value in an individual case depends on the score's construction, its validation population, and its calibration. Under *Daubert* and Rule 702, the proponent must show that the method is reliable and that it has been reliably applied, and a court may exclude a score that its own author concedes is not diagnostic (Daubert v. Merrell Dow Pharmaceuticals 1993; Federal Rule of Evidence 702 2023; Kumho Tire Company v. Carmichael 1999).

30.2 The ledger, the pipeline, and their limits

The materials describe a hash-chained ledger and a provenance pipeline — the OpenSSA Integrity Pipeline — intended to make a sequence of records tamper-evident. A hash chain can establish that a recorded sequence has not been altered since a root was published, under stated assumptions (Merkle 1979; Haber and Stornetta 1991). It does not, by itself, prove that the underlying assertion is true, that the record was complete, or that the entry was admissible.

Source authenticity. A chain proves internal consistency, not that the first entry described the world accurately.
Truth of the assertion. A signed record of an untrue statement is still an untrue statement.
Legal admissibility. Self-authentication addresses the form of the record, not its relevance or its

probative weight.

If a ledger is offered as proof of a historical event, the proponent must identify exactly what it records and why the record is reliable: how entries entered the system, who controlled the keys, how timestamps were generated, how corrections were recorded, and how the implementation was tested. **[CITE REQUIRED]** The repository snapshot, the dependency inventory, the test results, and any independent assessment would establish the technical claims.

30.3 Cryptographic claims

The materials refer to FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA), and to NSA's CNSA 2.0 suite (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d; National Security Agency 2022b). Mentioning a standard does not prove conformance with it. Any claim that a component implements one of these algorithms should identify the actual library, the parameter set, the test vectors, the key-management process, and any independent validation such as a FIPS 140-3 certificate (National Institute of Standards and Technology 2019). Where a component uses a placeholder, an HMAC, or a simulated primitive, the text of this part says so, and the corresponding claims in Parts IV and VI are read as design targets rather than as measurements.

30.4 HARMONI-X and the cognitive-accessibility record

HARMONI-X is described as a proposed assistant for document organisation, omission alerts, and cognitive support. A tool can flag a missing keyword or a cross-reference; the alert does not establish that a legal omission occurred. The record of a cognitive-accessibility claim should state what was checked, what a human reviewer concluded, and what the material limitations were. Where an accommodation is claimed to have been sought, the request, its date, its recipient, and the response — or the absence of a response — should appear in the record. If no accommodation request appears on the docket, the fact statement must not say that the court denied one.

CHAPTER 31

Gaps, Adverse Facts, and the Verification Protocol

The most useful page in a fact section is the one that lists what the pleader cannot yet prove.

31.1 The catalogue of gaps

The following propositions are asserted in the project materials but are not, on the present record, established. Each is stated here with the document that would close it. The list is not exhaustive, and it is meant to shrink as the record is assembled.

1. That a specific federal complaint was filed and served on named defendants in 25-1222, 25-1224, 25-1522, or 24-3283. **[CITE REQUIRED]** Operative complaints and docket sheets.

2. That a retaliatory motive existed and that probable cause was absent in any particular arrest. **[CITE REQUIRED]** Charging documents, arrest reports, and the disposition of any criminal charge (Nieves v. Bartlett 2019).

3. That the Social Security matter reached a final decision reviewable under § 405(g), and the date of that decision. **[CITE REQUIRED]** The notice of decision and the Appeals Council order (42 U.S.C. 405(g) 2018).

4. That a particular polygenic score was validated at the level asserted, and that it was before the adjudicator. **[CITE REQUIRED]** Validation study, laboratory report, and the administrative exhibit list (Daubert v. Merrell Dow Pharmaceuticals 1993).

5. That any component of the pipeline uses a real post-quantum implementation rather than a placeholder. **[CITE REQUIRED]** Repository dependency manifest and test transcripts (National Institute of Standards and Technology 2024b).

31.2 The adverse facts a respondent would supply

A fact section must anticipate the other side. Three adverse facts bear on the narration above and are disclosed here.

The first is that an unresolved state-court matter may bar or delay the federal civil-rights claims under the favourable-termination requirement, which in most circuits is satisfied only by a termination that affirmatively indicates innocence, and which prevents a § 1983 damages claim that would impugn an

outstanding conviction (Heck v. Humphrey 1994; Thompson v. Clark 2022). A stay under the abstention doctrine may follow if the state proceeding is ongoing and involves important state interests (Younger v. Harris 1971).

The second is that several of the asserted technical facts are, on the face of the materials, aspirational: the architecture is a specification, the quantum substrate is simulated, and the cryptographic module is described at the level of interfaces. A respondent would say that no working system has been shown to exist. The register discipline of this book is designed to answer that objection before it is made, by labelling each claim.

The third is that the damage figures used to motivate the triage engine are order-of-magnitude estimates drawn from official reporting, not measurements of the author’s own cohort (U.S. Government Accountability Office 2020; Sanders and Larson 2020). They are reported as estimates and must not be presented as findings about any individual.

Part X

The Integrated Policy Framework

CHAPTER 32

From Case to Policy: The Gov-LLM Proposal

A remedy that helps one claimant is a judgment. A remedy that helps the next claimant is an institution. This part is about the second.

32.1 The proposition

The legal analysis of Part II and the evidentiary analysis of Part III converge on a single institutional diagnosis: the agencies that decide whether a disabled person receives support are asked to apply a body of law too complex for any individual adjudicator to hold in memory, to a record too heterogeneous to be compared across cases, on a timetable too short to be fair. The proposal that follows from that diagnosis is a state-government legal-knowledge system — referred to in the project materials as a Gov-LLM — embedded in a benefits and accommodation workflow.

The idea is not to let a model decide eligibility. It is to let a model perform three bounded functions that adjudicators now perform badly under time pressure: retrieving the governing authority for a given factual pattern, flagging a record that is missing a document the authority requires, and drafting the boilerplate that surrounds a human decision. Every one of those functions is **[PROPOSAL]**. The decision itself remains a human act.

Register of this part

Nothing in this part is adopted law. The federal framework described below is **[ADOPTED LAW]** where it names a statute, a rule, a published standard, or a reported decision; it is **[PROPOSAL]** where it recommends an institutional reform; and it is **[MODEL / SIMULATION]** where it reports a performance figure from a prototype. A reader who wants to know whether a paragraph can be relied upon in a brief should read the register first.

32.2 Why a knowledge system and not a decision system

The distinction is constitutional, not merely prudential. Adjudication of disability benefits is an exercise of the state's coercive authority over property and liberty interests, and it attracts procedural due process (Mathews v. Eldridge 1976). *Mathews v. Eldridge* fixes a three-factor balance: the private interest at stake, the risk that the existing procedure will produce an erroneous deprivation, and the government's interest in the added burden of a more protective procedure. A knowledge system can be justified under the second factor — it reduces the risk of error — without ever being asked to supply the decision that the first and third factors protect.

32.3 The deployment shape

The proposed system is described as running on standard agency hardware, including laptops, with a short deployment time. That is a claim about resource footprint rather than about accuracy, and it is **[MODEL / SIMULATION]** where it rests on a prototype and **[DESIGN TARGET]** where it rests on design intent. The relevant legal question is not the footprint but the auditability of the output: an agency that cannot reproduce the retrieval that produced a recommendation cannot defend the recommendation on review.

CHAPTER 33

The Statutory Lattice of a State Legal-Knowledge System

The law an agency must apply is not one statute. It is a lattice of overlapping duties, each with its own remedy and its own forum.

33.1 The disability and education core

A system that supports accommodation and benefits decisions must be indexed to a body of law that spans at least three fields. In disability and education, the governing instruments are the Americans with Disabilities Act and its 2008 amendments, which define disability in functional rather than diagnostic terms and reject the consideration of mitigating measures in the threshold inquiry (Americans with Disabilities Act of 1990 1990; ADA Amendments Act of 2008 2008); Section 504 of the Rehabilitation Act, which supplies the nondiscrimination duty for programmes receiving federal funds (29 U.S.C. 794 2018); and the Individuals with Disabilities Education Act, which governs the special-education entitlement (Individuals with Disabilities Education Act 2004). FERPA governs the education records that a school-based system would ingest (Family Educational Rights and Privacy Act 2018).

33.2 The benefits core

In the benefits field, the Social Security Act and its disability provisions supply the substantive standard, and § 405(g) supplies the exclusive route for judicial review of a final decision (Social Security Act 1935; 42 U.S.C. 405(g) 2018). The procedural-due-process framework of *Mathews* and its application to benefits terminations govern the process the agency owes before it acts (Mathews v. Eldridge 1976).

33.3 The privacy core

Any system that ingests a claimant's health, education, and genetic information touches the privacy lattice as well. HIPAA and the Security Rule govern protected health information held by covered entities and their business associates (Health Insurance Portability and Accountability Act of 1996 1996). GINA prohibits the use of genetic information in health insurance and employment but, as Part II emphasises, leaves life, disability, and long-term-care insurance largely unaddressed (Genetic Information Nondiscrimination Act of 2008 2008). The gap is the reason the genomic-justice pillar of the directive described in Chapter 35 is posed as a legislative proposal rather than as an enforcement action.

Table 33.1: The statutory lattice and the agency that interprets it.

Instrument	Key duty	Interpreting authority
ADA / ADAAA	Nondiscrimination; reasonable accommodation	EEOC, DOJ, courts (Americans with Disabilities Act of 1990 1990)
Rehabilitation Act § 504	Nondiscrimination in federally funded programmes	Federal agencies, OCR (29 U.S.C. 794 2018)
IDEA	Free appropriate public education	OSEP (Individuals with Disabilities Education Act 2004)
FERPA	Confidentiality of education records	Department of Education (Family Educational Rights and Privacy Act 2018)
Social Security Act	Disability benefits	SSA (Social Security Act 1935)
HIPAA / Security Rule	Privacy and security of health information	HHS, OCR (Health Insurance Portability and Accountability Act of 1996 1996)
GINA	Genetic-information nondiscrimination (health insurance, employment)	EEOC, HHS (Genetic Information Nondiscrimination Act of 2008 2008)
APA	Procedural regularity in agency action	Courts (Administrative Procedure Act 1946)

33.4 The drafting consequence

The lattice has a practical consequence for the design of any legal-knowledge system: authority cannot be stored as a flat list of documents. It must be stored as a graph in which a factual pattern resolves to a duty, the duty resolves to the agency that interprets it, and the interpretation resolves to the remedy and the forum. That requirement is precisely the architectural thesis of Part IV, which is why the semantic-execution graph and the statutory lattice are two descriptions of one structure.

CHAPTER 34

Algorithmic Governance and the Federal–State Divergence

In three years the federal stance toward AI regulation moved from mandated risk management to mandated deregulation, while the states moved the other way. An agency designing a system must survive both.

34.1 The federal trajectory

The federal posture toward artificial intelligence has reversed twice in three years, and the reversal is itself a design constraint. The National Institute of Standards and Technology published the AI Risk Management Framework in January 2023, organised around the functions *govern*, *map*, *measure*, and *manage* (National Institute of Standards and Technology 2023). It was followed in July 2024 by a generative-AI profile that catalogues risks specific to generative models (National Institute of Standards and Technology 2024a). In the same period the Office of Management and Budget issued Memorandum M-24-10, which required federal agencies to designate a chief AI officer, to inventory their AI use cases, and to apply minimum risk-management practices to rights-impacting and safety-impacting systems (Office of Management and Budget 2024).

Executive Order 14110, issued in October 2023, had made that framework binding across the executive branch and had directed attention to civil-rights implications (The White House 2023). It was revoked in January 2025 by Executive Order 14179, which framed prior requirements as barriers to leadership (The White House 2025a), and the administration's subsequent action plan recast the federal role as accelerating deployment and reducing regulatory burden (The White House 2025b).

A risk-management framework published as a voluntary standard, and a procurement memorandum that agencies have already implemented, survive a change of administration in a way that an executive order does not. For an institution that must remain compliant across political cycles, the durable anchors are the NIST framework and the agency's own procurement rules, not the order of the day.

34.2 The state trajectory

The states have moved in the opposite direction. California's Transparency in Frontier Artificial Intelligence Act, signed in September 2025, imposes transparency and incident-reporting duties on developers of the largest frontier models and protects whistleblowers (California State Legislature 2025). Companion statutes establish independent verification organisations and an auditor registry (California State Legislature 2026b; California State Legislature 2026a). Colorado's 2024 Act imposed a duty of reasonable care against algorithmic discrimination in consequential decisions, though it was later replaced by a disclosure-centred framework with a delayed effective date (Colorado General Assembly 2024).

The divergence produces a genuine compliance problem. A state agency that operates a Gov-LLM is subject to state AI law even where federal law has been relaxed, and it is subject to federal antidiscrimination law and the *Mathews* balance regardless of either. The safe design is therefore the stricter of the two regimes, documented to the NIST framework's own vocabulary so that the documentation can be reused in whichever forum asks for it.

34.3 The civil-rights overlay

Algorithmic governance in the disability context is not only an administrative question. Where a state uses an automated tool to triage or screen benefit applicants, the tool's disparate effect on a protected class is actionable under the same statutes that govern manual decisions, and the integration mandate of *Olmstead* constrains a state's ability to deliver services in institutional rather than community settings (Olmstead v. L.C. ex rel. Zimring 1999). A design that reduces a queue by concentrating error on the most impaired claimants fails both tests at once.

CHAPTER 35

The Three-Pillar Directive and Its Implementation

A directive that names three pillars but no enforcement mechanism is a wish. The three pillars are set out here with the mechanism attached to each.

35.1 The instrument

The materials propose an international policy directive, designated *UN/PSDLEF/GFQA/2026/001*, organised around three pillars that correspond to the three technical parts of this book. The directive is **[PROPOSAL]**. It is useful, however, precisely because it forces the technical, legal, and financial proposals into a single architecture and exposes where they depend on one another.

Table 35.1: The three pillars and the mechanism attached to each.

Pillar	Instruments	Enforcement mechanism
Cognitive accessibility	Cognitive-load modulation, predictable interfaces, assistive software, cognitive-liberty safeguards	ADA Title II and § 504 programme duties (Americans with Disabilities Act of 1990 1990; 29 U.S.C. 794 2018)
Genomic justice	Genomic sovereignty, dynamic consent, data minimisation, algorithmic-discrimination controls, evidence provenance	GINA and HIPAA for the covered sectors; new legislation for the uncovered ones (Genetic Information Nondiscrimination Act of 2008 2008; Health Insurance Portability and Accountability Act of 1996 1996)
Quantum-secure infrastructure	ML-KEM, ML-DSA, SLH-DSA, zero-trust architecture, secure biological-data nodes	Procurement mandates keyed to the CNSA 2.0 transition timetable (National Security Agency 2022b; National Institute of Standards and Technology 2020)

35.2 Cognitive accessibility as the first pillar

The first pillar converts the accessibility requirements of Part II into interface requirements. Where a person's impairment is executive rather than sensory, accessibility is a matter of cognitive load: how many

steps a task requires, how much working memory it demands, and whether the interface behaves predictably across sessions. The legal hook is the accommodation duty itself, which obliges a public entity to make reasonable modifications to its programmes (Americans with Disabilities Act of 1990 1990). The technical hook is the cognitive-interface layer of the architecture in Part IV.

35.3 Genomic justice as the second pillar

The second pillar addresses the asymmetry identified in Part II: the genome is the most identifying and least changeable record a person has, and the statutory protection is fragmentary. The pillar proposes dynamic consent, data minimisation, and provenance controls, and it is candid that the insurance gap cannot be closed by interpretation. It requires legislation. Until that legislation exists, the pillar's operational content is procedural: it requires that any use of genomic information in an adjudicative setting be documented, so that the use is at least visible (Genetic Information Nondiscrimination Act of 2008 2008).

35.4 Quantum security as the third pillar

The third pillar is the most concrete, because it rests on published standards with a published migration timetable. The three post-quantum standards — ML-KEM for key encapsulation, ML-DSA and SLH-DSA for signatures — were published in August 2024 (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2024d), and the national-security community has set a transition schedule for migrating to them (National Security Agency 2022b). A state system that holds genomic records with a decades-long retention period should be migrating now, because a record encrypted today must remain confidential for longer than the shelf life of the classical cryptography that protects it. Zero-trust architecture supplies the complementary access model (National Institute of Standards and Technology 2020).

35.5 The limits of a policy proposal

Three limits should be stated. The first is jurisdictional: a directive adopted by an international body does not bind a state agency, and a state agency does not implement a treaty. The directive's operative content in any given forum is whatever domestic law already says. The second is fiscal: every pillar has a cost and the book does not pretend that the trust structure of Part VIII can fund all of it. The third is epistemic: the performance figures that motivate the proposal are order-of-magnitude estimates and prototype measurements, labelled as such, and they should not be cited as if they were results of a controlled trial. A proposal that overstates its evidence loses the argument it was made to win.

Part XI

Applied Projects and Capstones

CHAPTER 36

The UMGC Integrated Capstone: Design and Validation Plan

A capstone is judged not by the grandeur of its subject but by whether the student can say what would falsify the design.

36.1 The programme and the brief

The University of Maryland Global Campus integrated capstone joins a degree in biotechnology with one in cybersecurity technology. The manuscript submitted in June 2026 is titled *Integrated Cyber-Biotechnology: Closed-Loop Neuroprosthetics with Post-Quantum Genomic Adjudication*. Its subject is a closed-loop neuroprosthetic — abbreviated CINP in the manuscript — that unites genome-editing machinery with post-quantum cryptographic protection to address executive dysfunction of neurogenetic origin.

The manuscript is a professional degree submission, and this chapter treats it as one: a design with a validation plan, not a report of a completed clinical intervention. Every engineering claim below is **[DESIGN TARGET]** unless the manuscript itself reports a measurement.

36.2 The system as designed

The system has three coupled subsystems, and the coupling is the intellectual point of the capstone.

The sensing layer
derives a real-time polygenic risk score from a genotyping panel. The score is a population-level estimate; the manuscript is explicit that its use in an individual closed loop would require prospective validation, which is the same caution that Part III applies to polygenic evidence in adjudication.

The actuation layer
proposes CRISPR activation and interference modules for on-demand epigenomic control. The manuscript presents these as a design target and identifies the off-target, delivery, and reversibility problems that any clinical translation would have to solve.

The custody layer
protects every genomic, clinical, and legal record with a quantum-resistant audit ledger, so that the data governing the loop cannot be altered without detection. This layer is the part of the design that is engineering-complete, because it depends only on published standards.

Register

That the standards underlying the custody layer are published is **[ADOPTED LAW]** (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b). That the sensing and actuation layers can be built as described is **[DESIGN TARGET]**, resting on a literature review and a design argument rather than on a prototype. That the integrated loop would improve executive function in any individual is neither asserted nor established, and the manuscript does not assert it.

36.3 The validation plan

The manuscript's contribution to the degree, and to the reader of this book, is its validation plan: the sequence of experiments that would falsify each layer before any human subject is exposed. The plan separates analytical validation (does the assay measure what it claims), clinical validation (does the measurement predict the outcome), and utility (does acting on the prediction change the outcome). A capstone that states those three steps in order is a responsible document even when its subject matter is speculative.

CHAPTER 37

The Baltimore Secure Backbone System

Genomic data has a retention period measured in generations. The cryptography that protects it must be chosen for that horizon, not for this year.

37.1 Why a network belongs in this book

The project materials include a municipal network design, the Baltimore Secure Backbone System (BSBS), whose stated purpose is to carry genomic and clinical records under a zero-trust model with mandated post-quantum cryptography. It belongs in this book because it converts the abstract security requirements of Parts IV and VI into a concrete deployment: a place where the cryptographic claims become procurement decisions.

The design is one instantiation of the model published as NIST Special Publication 800-207, in which no actor is trusted by virtue of network location and every request is authenticated and authorised explicitly (National Institute of Standards and Technology 2020).

37.2 The three tiers

The design is organised into three tiers, each with its own trust assumptions.

37.3 The audit interface

The design specifies a small API whose function is to make the audit trail reproducible rather than merely present. In outline, the interface exposes a liveness and handshake-status endpoint, an append endpoint that signs an event into the ledger, a retrieval endpoint that returns an event together with its Merkle proof, a root endpoint that publishes the current Merkle root under an ML-DSA signature, a query endpoint that re-derives a hash-chain segment, and a metrics endpoint in a standard operations format.

The fourth of these — the ability to re-derive a segment on demand — is the one that matters for litigation, because it is what makes the evidentiary claim of Chapter 19 testable. A ledger whose contents cannot be re-derived and compared is an assertion; a ledger with a reproducible derivation is evidence.

37.4 The honest limits of the design

The design is a plan. Three limits follow. First, key custody is the whole of the matter: a zero-trust network whose signing keys are held in one administrator's account is not zero-trust. Second, the migration to

Table 37.1: The three-tier architecture of the BSBS design.

Tier	Scale	Protection
Edge	∼120 sites (hospitals, clinics, homes, laboratories)	SD-WAN tunnels keyed with ML-KEM (National Institute of Standards and Technology 2024c)
Core	Two redundant municipal data centres	Service mesh with workload identity; all east–west traffic ML-KEM-protected (National Institute of Standards and Technology 2020)
Custodial	Hardware-security-module enclaves for genomic and clinical archives	ML-DSA signing keys in FIPS 140-3 Level 3 modules with quorum release (National Institute of Standards and Technology 2024b; National Institute of Standards and Technology 2019)

post-quantum primitives requires replacing every handshake and every certificate, and a partial migration protects nothing that traverses an unprotected hop. Third, a hash-chained ledger of records does not make the records true, as Chapter 30 states; it makes them unaltered.

CHAPTER 38

The United Nations Submission

Treaty bodies do not award damages. They establish that a condition exists and name it. Naming is the first remedy.

38.1 The instrument and its venue

In June 2026 the project materials describe a submission to the United Nations Human Rights Council, under the symbol *A/HRC/58/NGO/2026/Rev.2*, presented through the Office of the High Commissioner for Human Rights. The submission is an advocacy document, not a pleading, and this chapter reads it as such.

38.2 The factual claim

The submission's factual core is a mortality claim: that a substantial number of claimants die each year while their disability claims are pending. That the disability queue is long, and that claimants have died while awaiting adjudication, is **[ADOPTED LAW]** on the strength of official reporting (U.S. Government Accountability Office 2020). The submission's specific annual figure is an order-of-magnitude estimate consistent with the reported range, and it is reported in Part V as such (Sanders and Larson 2020). The submission's description of the backlog's scale and of the increase in average wait times over the preceding five years is likewise drawn from official sources and should be cited to them rather than to the submission itself.

Register

That official reporting documents deaths during the wait is **[ADOPTED LAW]** (U.S. Government Accountability Office 2020). That the author's larger figure is accurate to the person is **[MODEL / SIMULATION]** and is not claimed. That an international body should open a commission of inquiry is **[PROPOSAL]**; whether it will is not a matter this book can predict.

38.3 The legal frame

The submission invokes three treaty instruments. The first is the International Covenant on Civil and Political Rights, whose protection of life is non-derogable and whose Human Rights Committee has stated that the obligation extends to deprivations caused by state institutions rather than by deliberate acts alone (International Covenant on Civil and Political Rights 1966; UN Human Rights Committee 2018). The second is the Convention against Torture, whose relevance the submission argues through the concept of administrative suffering that rises to the threshold of cruel, inhuman, or degrading treatment (Convention

against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment 1984). The third is the Convention on the Rights of Persons with Disabilities, whose provisions on accessibility, equal recognition before the law, independent living, and habilitation and rehabilitation supply the affirmative framework (Convention on the Rights of Persons with Disabilities 2006).

The relational argument that connects the three instruments is developed in Part VII and is not repeated here. What this chapter adds is the procedural posture: a submission under the Council's institution-building resolution (UN Human Rights Council 2007) creates a record and a set of named questions, and its value lies in the permanence of the record rather than in immediate relief.

38.4 What the submission does not claim

The submission does not claim that the United States has consented to an individual complaint under an optional protocol that would create a binding outcome, and it does not claim that a treaty body can order relief. It claims that the condition exists, that it is documented, and that it should be investigated. Stated that way, the submission is defensible on its face; stated more grandly, it would not be.

CHAPTER 39

The Chapter 11 Plan of Reorganisation

A financing structure is only as good as the honesty of its schedules. The plan is the instrument, and the schedules are the truth of it.

39.1 The plan and its purpose

The project materials include a Chapter 11 plan of reorganisation filed by the debtor as an individual, whose stated purpose is to resolve outstanding obligations and to provide for an orderly distribution to creditors from the liquidation of a securities portfolio reportable under Section 13(f) of the Securities Exchange Act (Securities Exchange Act of 1934 1934; Bankruptcy Code 1978). Within this book the plan is not a separate matter; it is the funding rail of Part VIII. The trust that the book proposes can only receive assets that the estate can lawfully transfer, and the plan determines what those assets are.

39.2 The classification of claims

The plan follows the ordinary architecture of Chapter 11. It classifies claims and interests into administrative-expense claims and priority tax claims (Bankruptcy Code 1978); secured claims to the extent of the value of the collateral; general unsecured claims; and the debtor's residual interest. It provides for the treatment of executory contracts and unexpired leases, and it fixes the effective date as the first business day after the confirmation order becomes final and non-appealable.

39.3 Discharge and the retention of jurisdiction

On the effective date the plan discharges debts arising before the order for relief to the extent the Bankruptcy Code provides, and the court retains jurisdiction to interpret and enforce the plan. Both provisions are standard, and their significance here is that they set the boundary of the estate: what is discharged is not available to the trust, and what the court retains is subject to its continuing supervision.

39.4 The Subchapter V question

If the debtor qualifies, Subchapter V of Chapter 11 offers a materially different procedure: no creditors' committee, a short exclusive period for the debtor to propose a plan, and the possibility of retaining equity over an objection in defined circumstances (Small Business Reorganization Act of 2019 2019). The qualification thresholds are statutory and must be tested against the schedules, not assumed. Whether to

proceed under Subchapter V is therefore a question of fact about the estate, and the plan's own schedules are the only documents that can answer it.

39.5 The schedules as the hinge of the whole book

Every financial proposition in Part VIII — the perpetuity rule, the payout ratio, the allocation bands — is computed from the corpus, and the corpus is defined by the schedules. If the schedules overstate the portfolio, every downstream figure is wrong in the same direction. The discipline that this book applies to factual assertions in Part IX applies with equal force here: the schedules are **[CITE REQUIRED]** until they are filed and tested, and the financing structure is **[PROPOSAL]** until they are.

Part XII

Conclusion

CHAPTER 40

The Constitutional Infrastructure

The claim of this book is not that the systems it describes are built. It is that the institutions which decide whether a person eats are governed by nothing that was designed for the data they now hold.

40.1 The single argument, restated

The book set out to join four literatures, and it closes by stating what joins them. The joining claim is this: the institutions that hold genomic, neurophysiological, educational, and financial records about a person are constrained, if at all, by instruments drafted for a world in which such records were either impossible or expensive. The genome is the clearest case. It is unchangeable, it identifies without being intended to identify, it implicates relatives who never consented, and the statute that governs it (Genetic Information Nondiscrimination Act of 2008 2008) reaches health insurance and employment and stops there. The consequence is not a single wrong. It is a structural under-protection that appears as a civil-rights claim in one forum, as an evidentiary problem in another, and as a queue in a third.

40.2 The four registers as the book's method

The book's methodological contribution is the register discipline of Chapter 2 and its application throughout. Every proposition in this volume sits on one of four registers: adopted law, proposal, model or simulation, and design target. A claim that the disability queue kills people is adopted law, resting on official reporting (U.S. Government Accountability Office 2020). A claim that a triage engine would shorten the queue to a stated figure is a model. A claim that an agency should adopt such an engine is a proposal. A claim that the quantum substrate of Part IV can be built as specified is a design target. A reader who can place a sentence on its register can decide what to do with it, and that is the test the book sets for itself.

40.3 Why the architecture and the jurisprudence are one thing

A reader may ask why a book about constitutional litigation contains a processor specification. The answer is that the legal argument requires exactly the property the architecture supplies. A constitutional claim that a genomic record has been misused depends on being able to show what was recorded, when, by whom, and with what authority — and on being able to show that the record has not been altered since. That is the definition of provenance, and provenance is what a hash-chained ledger provides. The semantic-execution graph of Part IV is the computational form of the same idea: a record whose meaning travels with it and

whose derivation can be replayed. The litigation theory and the architecture are not two projects; they are the legal and technical descriptions of one requirement.

40.4 The financing claim and its limit

The book's fourth literature is finance, and its claim there is that public-benefit infrastructure can be capitalised through a perpetual purpose trust whose corpus is professionally managed and whose payout is capped by a rule rather than by a board's discretion (Bankruptcy Code 1978). The claim is not that such a structure is novel; it is that the structures that exist are under-used for this purpose. The limit is that a trust cannot receive what the estate does not own, and so the entire financing argument depends on the schedules discussed in Chapter 39. A structure built on overstated assets is not a structure; it is a liability.

CHAPTER 41

What Would Have to Be True

A conclusion that does not state what would falsify it is not a conclusion. It is a preference.

41.1 The falsifiers

The book's claims become knowledge only if they are capable of being wrong. The following are the conditions under which each would fail.

The constitutional claim.
It fails if the courts hold that the collection and use of genomic information by a state agency raises no Fourth Amendment interest beyond the interests already recognised (Carpenter v. United States 2018; Maryland v. King 2013), and that GINA's coverage is the full extent of the legislative protection the legislature intends to give. The first half of that condition is contested and the second is, on the current statute, true.

The evidentiary claim.
It fails if polygenic scores and neuroimaging endophenotypes cannot be shown to be reliably applied to an individual (Daubert v. Merrell Dow Pharmaceuticals 1993; Kumho Tire Company v. Carmichael 1999). That is a live possibility, and Part III treats it as the controlling caution rather than as a settled premise.

The architectural claim.
It fails if the semantic-execution model cannot be shown to improve on a conventional pipeline for the workloads it targets, or if the quantum substrate cannot be built at the scale the specification assumes. The burden here is an engineering burden, and it has not yet been discharged.

The governance claim.
It fails if automated triage systems are shown to reduce error without concentrating it on protected classes (Olmstead v. L.C. ex rel. Zimring 1999). If that were shown, the argument for the triage engine would be stronger, not weaker; the point is that the claim is testable.

The financing claim.
It fails if the schedules do not support the corpus, or if the perpetuity rule cannot be reconciled with the applicable rule against perpetuities in the relevant jurisdiction. The second is a drafting question with a known answer; the first is a question of fact about the estate.

41.2 The next document

The book ends where it began, with a document. The theory of Part II becomes a pleading when the operative complaint is filed. The evidence of Part III becomes a record when the laboratory report is exhibited. The architecture of Part IV becomes a system when the first test vector is published with a hash. The trust of Part VIII becomes a corpus when the schedules are filed.

The institutions that decide whether a disabled person receives support should be required to record what they decide, in a form that can be replayed, under a standard that a court can review — and the technical means to do so now exist, while the legal duty to do so does not.

That is the constitutional infrastructure the title promises. It is not a building; it is a duty. Whether it is imposed is a question for legislatures and courts, and this book has stated, as carefully as it could, what the record already shows and what it does not.

Part XIII

ADR-DYREO-PSDLEF: Ph.D.-Level Quantitative Synthesis

CHAPTER 42

Jurisdiction, Due Process, and Cognitive Access

Law is a formal system, but it is not a closed one. It depends on the notice, comprehension, and executive capacity of the humans it binds, and when those fail, the formal system fails with them.

This chapter supplies the jurisdictional and due-process grounding for the ADR–DYREO–PSDLEF architecture as a whole. Where Parts II and VIII addressed constitutional standing and the juridical form of the trust, this chapter states the specific statutory hooks on which the Maryland action turns, the *Mullane* notice problem created by DTC book-entry masking during the 2014 USEC/Centrus reorganisation, and the ADA Title II accommodation argument that connects the Gov-LLM protocol to the constitutional right of access to court.

Register

The statutes and cases cited are **[ADOPTED LAW]** law. The proposition that they compel the relief described is **[PROPOSAL]** framed by the petitioner, not a ruling of any court. The endpoint `https://americana-digital-reserve-exchange.ai.studio/` is a **[DESIGN TARGET]** cited as designed, not as a production deployment.

42.1 Equitable jurisdiction and the Maryland mandamus vehicle

The state-court petition in Circuit Court for Baltimore City, Case No. 24-C-26-003970, invokes the general equity jurisdiction conferred by Md. Code Ann., Cts. & Jud. Proc. § 1-501 (*Md. Code Ann., Cts. & Jud. Proc. § 1-501* n.d.), under which each circuit court sits as a court of general common-law and equity jurisdiction. The extraordinary-writ vehicle is Maryland Rule 15-701 (*Maryland Rule 15-701* n.d.), which governs mandamus and other writs issued in the original jurisdiction of the circuit court.

Two features of that vehicle are doing work. First, mandamus reaches a *ministerial* act: a duty so clearly owed that the respondent has no discretion to refuse it. The petition frames the duty as ministerial turnover of book-entry positions by the corporate transfer agent and by any successor entity holding the underlying record position. Second, specific performance in equity is the appropriate remedy where the res is unique, as a beneficial interest traceable through a Chapter 11 reclassification is, because damages do not put the holder back in the position the reclassification would have occupied had notice issued. Whether either characterisation is correct on the facts is for the Maryland court, and the chapter records, rather than resolves, the characterisation.

42.2 The *Mullane* due-process claim

Under the Fifth Amendment (as to federal proceedings) and the Fourteenth Amendment (as to state proceedings), and under *Mullane v. Central Hanover Bank & Trust Co.*, 339 U.S. 306 (1950) (Central Hanover Bank & Trust Co. 1950), deprivation of property requires notice "reasonably calculated, under all the circumstances, to apprise interested parties of the pendency of the action and afford them an opportunity to present their objections." *Mullane* holds that where the identities and addresses of beneficiaries are *reasonably ascertainable*, publication notice is constitutionally insufficient; individual notice is required.

The petition's constitutional claim is that multi-tiered Depository Trust Company (DTC) book-entry masking during the 2014 Chapter 11 reorganisation of USEC Inc. into Centrus Energy Corp. (Case No. 14-10492, Bankr. D. Del.) (*USEC Inc., Case No. 14-10492 (Bankr. D. Del.)* 2014) produced precisely the *Mullane* defect: beneficial holders whose identities were ascertainable with reasonable diligence through SEC Form 13F institutional-holder filings (Securities Exchange Act of 1934 1934) were not given individual notice of the reclassification, with the result that their positions were cancelled in the reorganisation without the opportunity to be heard.

Register

That the reorganisation plan confirmed in 2014 produced the emergence of Centrus Energy Corp. and a 0.0917:1 reverse split is a matter of public corporate-action record (*Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history* 2014; *USEC Inc., Case No. 14-10492 (Bankr. D. Del.)* 2014) and is **[ADOPTED LAW]** for purposes of this chapter. That DTC-tier masking caused a *Mullane* violation as to any particular holder is **[PROPOSAL]** pleaded in the petition, not an adjudication.

The doctrinal significance of the claim is that it reframes what might otherwise be treated as a stale securities-law problem as a constitutional-notice problem to which no ordinary limitation period cleanly attaches where the deprivation is ongoing through the current continued refusal of transfer agents to honour the reclassified positions.

42.3 ADA Title II cognitive accommodations and the Gov-LLM endpoint

The second constitutional hook is access to court. Under *Tennessee v. Lane*, 541 U.S. 509 (2004) (Lane 2004), Congress validly abrogated state sovereign immunity under Title II of the ADA with respect to the fundamental right of access to the courts. Under 42 U.S.C. § 12131 (Americans with Disabilities Act of 1990 1990) and 28 C.F.R. § 35.130(b)(7) (*28 C.F.R. § 35.130(b)(7)* n.d.), a public entity must make reasonable modifications in policies, practices, or procedures when necessary to avoid discrimination on the basis of disability.

The petition argues that a *pro se* neurodivergent litigant facing a ministerial turnover order, a Chapter 11 plan, a DTC book-entry chain, and a Tucker Act compensation proceeding simultaneously is, without a cognitive accommodation, structurally excluded from vindicating the very rights *Lane* protects. The accommodation requested is integration of the Gov-LLM Secure Transaction Protocol and automated API transfer endpoints (`https://americana-digital-reserve-exchange.ai.studio/`) into the filing and transfer workflow, so that the execution steps the law requires are reduced to a cryptographically auditable sequence rather than left to the unassisted executive capacity of the litigant.

This is the juncture at which this book's thesis about cognitive infrastructure (Parts IV, X) meets its thesis about access to justice (Parts II, IX): an accommodation that is itself software, rather than a physical ramp or an interpreter, is an unusual request but a logically compelled one if the disability in question is executive dysfunction rather than mobility or sensory impairment.

CHAPTER 43

Capital Restructuring Taxonomy and CUSIP Normalization

Every securities claim is, in the first instance, an identifier claim. The rest is arithmetic.

This chapter states the capital-structure facts that the petition rests on, runs the arithmetic from the 1999 indenture through the 2014 plan, and shows how CUSIP check-digit validation is used to reconstruct degraded institutional-holder records. Chapter 27 (Part VIII) described the CUSIP normalisation protocol procedurally; this chapter gives the underlying capital vector and the precise numbers it produces.

Register

The indenture principal amounts, coupons, maturity dates, and CUSIPs reported in Sections 2.1–2.2 are taken from the petition and are recorded here on the petitioner's representation; I have not independently pulled the 1999 indenture from EDGAR and mark these figures **[CITE REQUIRED]**. The 0.0917 plan exchange ratio is confirmed against public corporate-action records (*Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history* 2014). The consolidated $720,150,000 total capitalisation vector is **[PROPOSAL]**, because it depends on (a) Tucker Act and patent allocation components that are themselves claims, not judgments.

43.1 The 1999 legacy indenture

On 14 January 1999, USEC Inc. issued $500,000,000 in Senior Notes bifurcated into two series:

Table 43.1: USEC Inc. Senior Notes issued 14 January 1999.

Series	Principal	Coupon	Maturity	CUSIP
Series A	$350,000,000	$6\frac{5}{8}\,\%$	2006	917300AA1
Series B	$150,000,000	$6\frac{3}{4}\,\%$	2009	917300AB9
Total	$500,000,000			

The two CUSIPs follow the standard nine-character format: the first six characters (917300) identify USEC Inc. as issuer, the seventh and eighth (AA, AB) identify the two series, and the ninth (1, 9) is the check digit. Validation against the modified Luhn algorithm is what permits ingestion pipelines to reject transcription errors before they propagate into the entitlement calculation.

43.2 The 2014 Chapter 11 reclassification

Pursuant to the confirmed plan of reorganisation in Case No. 14-10492 (*USEC Inc., Case No. 14-10492 (Bankr. D. Del.)* 2014), the Senior Note claims were reclassified as Class 4 Allowed Unsecured Claims exchangeable into reorganized common stock of the emerging entity, Centrus Energy Corp. (NYSE: LEU), at the court-confirmed ratio of 0.0917 shares of LEU common per dollar of allowed claim. The ratio matches the 0.0917:1 reverse split recorded in the public corporate-action records for the effective date of 30 September 2014 (*Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history* 2014).

43.3 The consolidated capital vector

Multiplying the \$500,000,000 principal of the legacy indenture by the confirmed exchange ratio of 0.0917 yields an exact entitlement of

$$500{,}000{,}000 \times 0.0917 \;=\; 45{,}850{,}000 \text{ shares of LEU common.} \tag{43.1}$$

Combining this reclassified equity position with the separate Tucker Act judgment vector and the patent-licensing allocation vector (the latter pursued under 28 U.S.C. § 1498(a) before the Court of Federal Claims, as discussed in Chapter 27) produces the consolidated capitalisation figure the petition asserts as the fully assembled endowment:

$$V_{\text{total}} \;=\; \$720{,}150{,}000.00. \tag{43.2}$$

Register

Equation (43.2) is **[PROPOSAL]**. Its two components other than the \$500M × 0.0917 legacy tranche are asserted claims, not confirmed amounts. The figure is reproduced here because it is the number the quantitative engine in Chapter 43 takes as its starting corpus, not because the book represents that it has been awarded.

43.4 CUSIP check-digit validation against Form 13F filings

Automated ingestion pipelines resolve historical data degradation via modified Luhn check-digit validation run across institutional Form 13F filings, including PNC Financial Services Group (SEC File No. 28-1235) and other named holders. The routine is: convert alphabetic CUSIP characters to their ordinal values (A=10, B=11, …, Z=35) with the convention documented by the CUSIP Global Services standard (CUSIP Global Services 2020); weight positions from right to left, doubling at alternating positions; sum the digits of the resulting products; the check digit is $(10 - (P \bmod 10)) \bmod 10$. Any identifier that fails this check is rejected before it touches the entitlement ledger, which prevents the kind of transcription corruption that otherwise masks the chain of title across a Chapter 11 reorganisation.

This is the same procedure described in Chapter 27, but applied here to a specific indenture rather than stated in the abstract. It is deliberately boring: registry mechanics are unglamorous and decisive.

CHAPTER 44

DYREO Stochastic Control and the 31-Algorithm Suite

Writing down an objective function is a normative act disguised as a mathematical one. The previous chapter said so at the level of a one-period quadratic program; this chapter says so at the level of continuous-time stochastic control.

Chapter 25 presented DYREO in its one-period mean-variance form. This chapter elevates it to the continuous-time Hamilton–Jacobi–Bellman (HJB) form that the Ph.D.-level synthesis specifies, states the closed-form daily yield-sweep vector, and gives the Social-Energy-Return-on-Investment (S-EROI) metric in the thermodynamic form used by the solar/battery pilot.

44.1 Continuous-time stochastic control

The Dynamic Yield & Real Estate Optimization (DYREO) algorithm models liquid trust assets in continuous time as a controlled diffusion:

$$dA(t) \;=\; \mu A(t)\,dt \;+\; \sigma A(t)\,dW(t) \;-\; u(t)\,dt, \tag{44.1}$$

where $A(t)$ is liquid trust assets at time t, μ is the drift, σ is the volatility, $W(t)$ is a standard Brownian motion, and $u(t)$ is the distributed-yield control (the rate at which the trust disburses to its four daily buckets). The trust solves

$$J(t,a) \;=\; \sup_{u(\cdot)\in\mathcal{U}} \mathbb{E}\left[\int_t^T e^{-\rho(s-t)} U\big(A(s),u(s),S(s)\big)\,ds \;\Big|\; A(t)=a\right], \tag{44.2}$$

where U combines financial return, social-utility return from the Baltimore solar microgrid, distribution to ADA *pro se* AI-grants, and operational reserves, subject to the capital-preservation constraint

$$\Pr\big(A(T)\ge (1-\delta)A(0)\big) \;\ge\; 1-\epsilon, \tag{44.3}$$

and to the deterministic yield-sweep vector of Section 3.2. The HJB equation is the standard one for problems of this class (Bellman 1957; Björk 2020):

$$-\frac{\partial J}{\partial t} \;=\; \sup_{u}\Big\{U(a,u,s) + (\mu a - u)\frac{\partial J}{\partial a} + \tfrac{1}{2}\sigma^2 a^2 \frac{\partial^2 J}{\partial a^2} - \rho J\Big\}. \tag{44.4}$$

Register

Equations (44.1)–(44.4) are standard continuous-time portfolio-control machinery and are **[ADOPTED LAW]** in form. The calibration of μ, σ, ρ, δ, and ϵ to the PSDLEF corpus is **[DESIGN TARGET]**, as are the resulting optimal $u(t)$ trajectories.

44.2 The closed-form daily yield-sweep matrix

Liquid proceeds are distributed each business day by a deterministic weight vector that fixes, by policy, how incoming yield is swept into the four operational buckets before any discretionary reinvestment occurs:

Table 44.1: Daily yield-sweep vector (closed-form policy).

Bucket	Weight	Purpose
Treasury Equivalents	60%	Capital-preservation and permanent-corpus floor
Baltimore Solar Microgrid Payables	25%	Pilot service, low-income housing credits, maintenance
ADA *Pro Se* Litigant AI Grants	10%	Gov-LLM access grants under the Title II accommodation theory
Operational Reserves	5%	Trust operations, audit, custody
Total	100%	

The 60/25/10/5 split is a **[PROPOSAL]** policy choice, not a solution to the HJB equation (44.4); it is the floor the HJB optimisation must respect rather than an output of it. That is the point of the design: the social-distribution buckets are protected from being optimised away by a pure-financial objective function, because they are imposed as a deterministic sweep before optimisation runs.

44.3 The Social Energy Return on Investment (S-EROI)

Real-estate and microgrid allocations are governed by a thermodynamic efficiency metric that compares lifetime delivered energy net of losses to lifetime capital and maintenance cost:

$$\text{S-EROI} \;=\; \frac{\int_0^T \left[P_{\text{solar}}(t) - P_{\text{loss}}(t)\right] dt}{K_{\text{cap}} + \int_0^T C_{\text{maint}}(t)\, dt} \;\geq\; 0.73 \text{ kWh/\$}. \tag{44.5}$$

The 0.73 kWh/$ threshold is the design target against which each rooftop installation is accepted or rejected in the pilot. It is tighter than the S-EROI reported in Chapter 25 because the version here is calibrated for low-income housing, where the marginal social value of reliable delivered kilowatt-hours is higher.

Register

The 0.73 kWh/$ threshold and the solar-pilot parameters underlying P_{solar}, P_{loss}, K_{cap}, and C_{maint} are **[DESIGN TARGET]**.

44.4 The 31-algorithm execution suite

The executive financial engine spans thirty-one algorithms across the modules listed in Chapter 25, covering:

- *Bond valuation* (Algorithms 01–03): discounted cash-flow pricing, yield-to-maturity, and credit-spread term structure for the legacy indenture and any newly issued trust certificates.
- *Asset conversion and distribution* (04–06): CUSIP normalisation (Chapter 42), plan-ratio application, yield-sweep execution against Table 44.1.

- *Volatility modelling* (07–09): GARCH(1,1) estimation of return volatility (Bollerslev 1986), EWMA variance, and jump-diffusion calibration for extreme-event days.
- *Execution optimisation* (10–12): optimal trade-slicing, market-impact modelling, and rebalancing under transaction costs.
- *Derivatives and hedging* (13–15): Black–Scholes option pricing (Black and Scholes 1973), Greek computation, tail-risk collar construction on the equity tranche.
- *Risk management* (16–18): Value-at-Risk (VaR), Conditional VaR (CVaR/Expected Shortfall), and stress-testing against the capital-preservation constraint (44.3).
- *Yield optimisation* (19–20): HJB solution methods for (44.4), including numerical finite-difference solvers and verification against closed-form special cases.
- *Automated market-maker liquidity* (21–22): constant- product and concentrated-liquidity AMM curves for any tokenised beneficial-interest positions the trust issues.
- *Hash-chain audit trail and compliance* (23–31): the immutable transaction log described in Chapter 44 and Part VI, with ML-DSA-87 signatures on every ledger entry.

The suite is a **[PROPOSAL]**: the algorithms are individually standard, but their composition into a single auditable pipeline is the design the book describes.

CHAPTER 45

Post-Quantum Enclaves, QRDE-AEAD, and the DAG/Diamond Pipeline

A harvest-now, decrypt-later adversary does not care that you will upgrade your cryptography in 2035. They are recording your ciphertext today.

Chapter 15 (Part IV) and Chapter 20 (Part VI) introduced the post-quantum security module and the RIGOR-PRS-Secure++ scheme. This chapter pins down the specific NIST FIPS 203/204 parameter choices, the dual-envelope QRDE-AEAD construction, the deterministic DAG execution model over MMIO registers, and the integration of diamond color-center qubits as first-class DAG task nodes.

45.1 NIST FIPS 203 and FIPS 204 enclaves

To neutralise "harvest-now, decrypt-later" attacks, all data enclaves and custodial transactions use the NIST-standardised lattice primitives at the highest parameter sets:

- **ML-KEM-1024** (Module-Lattice-Based Key Encapsulation Mechanism, FIPS 203) (National Institute of Standards and Technology 2024c) for asymmetric key establishment, providing NIST security level 5 (equivalent to AES-256 post-quantum strength).
- **ML-DSA-87** (Module-Lattice-Based Digital Signature Algorithm, FIPS 204) (National Institute of Standards and Technology 2024b) for all transaction signatures, audit-log entries, and trust distributions.
- **SLH-DSA** (Stateless Hash-Based Digital Signature, FIPS 205) (*FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA)* 2024) as a conservative fallback for long-lived root-of-trust signatures, because hash-based signatures retain security under conservative cryptographic assumptions.

Register

The standards are **[ADOPTED LAW]** (National Institute of Standards and Technology 2024c; National Institute of Standards and Technology 2024b; *FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA)* 2024). The statement that the PSDLEF deployment uses ML-KEM-1024 and ML-DSA-87 is a **[DESIGN TARGET]**: these are the parameter choices the specification mandates, not a deployed configuration.

45.2 The QRDE-AEAD dual-envelope protocol

The Quantum-Resistant Dual-Envelope Authenticated Encryption with Associated Data construction combines asymmetric KEM encapsulation with symmetric authenticated encryption and an outer authentication tag to produce an IND-CCA2-secure hybrid:

1. Generate or reuse a long-lived ML-KEM-1024 encapsulation key pair for the enclave.
2. Encapsulate a fresh ephemeral shared secret ss via ML-KEM-1024 to produce the ciphertext c_K and shared secret K.
3. Derive two symmetric keys via HKDF-SHA3-384 (Krawczyk and Eronen 2010) from K: a data-encryption key k_e and an authentication key k_a.
4. Encrypt the payload m with AES-256-GCM (Dworkin 2007) under k_e to produce the inner ciphertext c_e and GCM tag t_g.
5. Compute an outer HMAC-SHA3-256 tag t_o over $(c_K \parallel c_e \parallel t_g \parallel ad)$ under k_a, where ad is the associated data (sender, timestamp, ledger height, CUSIP).
6. The transmitted envelope is (c_K, c_e, t_g, t_o, ad).

The dual-envelope structure ensures that the post-quantum asymmetric layer and the symmetric layer fail independently: a compromise of the KEM does not expose the payload without a break of AES-256-GCM, and a compromise of the symmetric layer is useless against a passive adversary without the long-lived KEM private key.

45.3 Deterministic DAG execution over MMIO

The execution engine replaces imperative blocking loops with a pure Directed Acyclic Graph (DAG) of topological transitions. Each node is a pure function over its fan-in edges; the scheduler executes a node exactly once, when every fan-in edge has delivered a valid token. This gives deterministic, replayable execution without global locks.

The hardware abstraction layer is mapped to:

- **MMIO register controllers** at base address `0x40000000`, with per-engine register windows for KEM acceleration, signature verification, hash, and DYREO control.
- **Fan-in gate counters** K_{in} enforced in hardware, so that a node fires if and only if exactly its declared number of predecessor tokens have arrived.
- **DWARF debug symbol offsets** for deterministic trace and post-hoc audit of every DAG firing, so that the execution trace can be reconstructed from the ledger even after the fact.

This is the same execution model described architecturally in Chapter 12 (Part IV); the DAG here is the one that the 31-algorithm suite actually runs on.

45.4 The LoongDiamond-2.5 color-center nodes

Chapter 13 introduced the LoongDiamond-2.5 quantum substrate. In the fully specified pipeline, diamond color-center qubits — nitrogen-vacancy (NV), silicon-vacancy (SiV), germanium-vacancy (GeV), and tin-vacancy (SnV) — are represented as first-class DAG task nodes. Spin coherence times, microwave control pulses, and photonic interconnect events map directly into tokens on the DAG edges, so that a quantum control step is a node like any other, with deterministic fan-in and fan-out from the perspective of the surrounding classical DAG.

Register

The existence, coherence, and control characteristics of color-center qubits are [**ADOPTED LAW**] in the condensed-matter literature. Integration of those qubits as first-class DAG nodes in the SSE-9/ZU9EG carrier (Chapter 14) is [**DESIGN TARGET**].

CHAPTER 46

SSADYREO, the MORTAR LSH Priority Score, and the OSI-P Audit Chain

A public-safety net does not fail all at once. It fails claim by claim, minute by minute, as decisions that should take days take years and people who cannot wait die waiting. A perpetual trust can capitalise the programme; an algorithmic triage engine can keep the queue from killing people while the capital works.

This chapter combines the perpetual-purpose trust structure of Chapter 24, the MORTAR triage engine of Chapter 17, and the OpenSSA integrity pipeline of Chapter 19 into the SSADYREO perpetual-trust stabilization architecture, and pins down the $O(n \log m)$ LSH priority scoring, the $N_{\text{sup}} = 16$ supervised transformer block configuration, and the nine-stage R_1–R_9 cryptographic audit chain.

46.1 Perpetual purpose trust under the UDTA

The Social Security Administration DYREO trust (SSADYREO) is modelled as a perpetual purpose trust under the Uniform Directed Trust Act (*Uniform Directed Trust Act* 2017) with a quad-cameral separation of fiduciary powers (Investment Committee, Distribution Committee, Trust Protector, Independent Examiner) that mirrors the four-component architecture introduced in Chapter 23. The central perpetuity constraint is a hard distribution cap:

$$d(t) \;\le\; 0.05 \times \overline{A}_{5\text{yr}}(t), \tag{46.1}$$

where $d(t)$ is annual distributions and $\overline{A}_{5\text{yr}}(t)$ is the trailing five-year moving average of trust corpus. This 5% rule permanently immunises public safety-net programmes from principal depletion, because the trust cannot distribute more than five per cent of its smoothed value in any year regardless of political or actuarial pressure.

Register

The UDTA as a model statute is **[ADOPTED LAW]** (*Uniform Directed Trust Act* 2017). The 5% trailing-average distribution rule of (46.1) is a **[PROPOSAL]** policy constraint (it is similar to but stricter than the 5% qualifying-distribution rule familiar from private-foundation tax law, and is imposed here by design rather than by any tax code).

46.2 MORTAR $O(n \log m)$ LSH priority scoring

The Mortality-Optimized Risk-Triage And Reasoning (MORTAR) engine is specified to eliminate administrative mortality across the approximately 1.1-million-claimant SSA adjudicative backlog using locality-sensitive hashing (LSH) priority scoring (Gionis et al. 1999b; Datar et al. 2004) in $O(n \log m)$ time, where n is the number of incoming claims and m is the number of buckets in the hash table. The priority score for a claim vector x is

$$s(x) \;=\; w_{\text{mort}} \cdot p_{\text{mort}}(x) \;+\; w_{\text{sev}} \cdot p_{\text{sev}}(x) \;-\; w_{\text{wait}} \cdot \tau(x), \tag{46.2}$$

where p_{mort} is the predicted short-term mortality probability, p_{sev} is severity, τ is current wait time, and the w are calibrated weights. Claims are hashed into m buckets via k random hyperplanes; candidates within a bucket are scored exactly, giving sublinear retrieval of the highest-priority claimants.

The scoring function feeds iterative latent-reasoning transformer blocks with $N_{\text{sup}} = 16$ supervised layers, which produce the structured recommendation that the human adjudicator then decides. The adjudicator decides; the engine prioritises and drafts. That division of labour is the non-delegation guard-rail: the engine does not adjudicate, because adjudication is a governmental function that a private algorithm may not exercise.

46.3 The OpenSSA Integrity Pipeline (OSI-P): R_1–R_9

The OpenSSA Integrity Pipeline enforces a nine-stage cryptographic audit chain anchored to Merkle roots (Merkle 1979; Merkle 1980) and signed via FIPS 204 ML-DSA-87 at every stage:

Table 46.1: The nine-stage OSI-P audit chain.

Stage	Name	Function
R_1	Ingest	Tamper-evident receipt of incoming claim record, ML-DSA-87 signed by submitter.
R_2	Normalise	Canonical JSON-LD normalisation and CUSIP/identifier validation.
R_3	Hash	Per-record SHA3-512 digest appended to the stage Merkle tree.
R_4	Score	MORTAR priority score (46.2) recorded with feature hash.
R_5	Recommend	Transformer-block recommendation recorded with model-version hash.
R_6	Adjudicate	Human-adjudicator decision recorded with adjudicator ML-DSA-87 signature.
R_7	Distribute	Distribution execution recorded under the DYREO sweep (Table 44.1).
R_8	Reconcile	Daily cross-ledger reconciliation; root of roots written to the trust anchor.
R_9	Seal	Day's Merkle root signed with long-lived SLH-DSA root key for archival permanence.

The result is a self-authenticating electronic administrative record admissible under Federal Rules of Evidence 902(13)–(14) (*Federal Rule of Evidence 902(13)–(14)* n.d.): a certified record generated by an electronic process or system that produces an accurate result, with every stage cryptographically tied to the next and no undocumented write path.

46.4 How the three layers compose

The composition is the point of the synthesis, and it is worth stating in one place:

1. The UDTA purpose trust of Section 5.1 provides the legal permanence and the 5% distribution floor.
2. The DYREO stochastic-control engine of Chapter 43, running on the DAG substrate of Chapter 44, allocates capital within that floor and executes the daily yield sweep.
3. The MORTAR triage engine of Section 5.2 prioritises the 1.1-million-claim queue without adjudicating it.
4. The OSI-P nine-stage chain of Section 5.3 produces an evidentiary record that is admissible in federal court without extrinsic authentication testimony.

5. The QRDE-AEAD dual-envelope encryption of Section 4.2 protects every stage from quantum-capable adversaries.

None of the five layers is sufficient alone. A perpetual trust without an allocation engine is an inert corpus. An allocation engine without a triage layer optimises yield while people die. A triage layer without an integrity pipeline produces inadmissible — and therefore useless — records. A record chain without post-quantum cryptography is a record chain built for a decade that will not survive the adversary it is designed to face.

CHAPTER A

Filing-Readiness Checklist

This checklist is the one the working draft of the statement of facts applies to itself. It is reproduced here because it states, in operational form, the discipline that Parts IX and XI describe. An item that cannot be checked is a proposition that cannot yet be filed.

1. Confirm the court, the caption, the docket number, and the purpose of the statement.
2. Confirm whether the document is a brief, a petition, a motion, a declaration, or a memorandum, and apply the page limits and rules that govern that form.
3. Obtain the official docket sheet for every referenced proceeding.
4. Reconcile the state case-number discrepancy between the two Maryland case numbers carried in the source materials (D-01-CR-25-011312 and D-01-CR-25-011146), determining which number belongs to which proceeding. **[CITE REQUIRED]**
5. Confirm the exact disposition entered on 30 January 2026 and its scope. **[CITE REQUIRED]**
6. Obtain file-stamped versions of every complaint, petition, and motion referred to.
7. Verify every appellate and Supreme Court docket number.
8. Obtain every order, judgment, and opinion cited.
9. Identify the exact Social Security decision under review and the certified administrative record.
10. Confirm which scientific and genomic exhibits were submitted and included in the record.
11. Locate every accommodation request, the response to it, and proof of receipt.
12. Authenticate each technical report, and label each system as proposed, simulated, tested, or deployed.
13. Do not describe an HMAC-SHA3-256 construction as ML-DSA, and do not describe simulated output as validated cryptography (National Institute of Standards and Technology 2024b).
14. Remove any unsupported claim of clinical validation, certification, government deployment, or legal admissibility.
15. Attribute each claim and each request for relief to the correct proceeding.
16. Review sealing, redaction, privacy, and protective-order requirements.
17. Replace every citation-required marker with a citation or remove the proposition the marker supports.
18. Confirm that each citation supports the exact proposition for which it is offered.
19. Reconcile the chronology with the operative pleading and the relief sought in the current case.
20. Review compliance with the local rules of the relevant court and with the applicable appellate rules.

CHAPTER B

Citation Templates

The templates below are forms, not sources. Each placeholder must be replaced with verified information before the template is used. They are collected here because a party proceeding without counsel is most often defeated not by the strength of the opposing argument but by the form of the citation.

Federal docket entry.
[Caption], No. [number], ECF No. [number], at [page] ([court] [date]).

Order.
Order, *[Caption]*, No. [number], at [page] ([court] [date]).

Transcript.
Hr'g Tr. [date], at [page]:[line]–[line].

Administrative record.
Administrative Record (AR) [page], [document title and date].

State docket.
Maryland Judiciary Case Search, [court], Case No. [verified number], docket entry dated [date], [entry title].

Technical report.
[author or entity], *[title]*, version [version], [date], at [page or section], [stable identifier].

A citation is not a decoration. It is a promise that the document exists, that it says what the sentence says, and that it can be produced on request. A citation that cannot be verified is worse than no citation, because it converts a gap in the record into an appearance of support.

CHAPTER C

Recommended Record Assembly Order

The order in which a record is assembled determines whether a reviewer can follow the argument. The order below proceeds from the procedural skeleton inward to the substantive exhibits, and it ends with the table that ties each paragraph of the statement to its source.

1. A master docket index that lists every case separately.
2. The official docket sheet and the operative pleading for each case.
3. Orders and judgments in chronological order.
4. Notices of appeal, petitions, and appellate dispositions.
5. The Social Security administrative decision, the notices, and the certified record.
6. The exhibits that were submitted, each with proof of submission.
7. Accommodation requests, responses, and transcripts.
8. The scientific reports and the source publications relied upon.
9. Source-code snapshots, test outputs, and validation reports.
10. A paragraph-to-source citation table for the completed statement.

CHAPTER D

Source-File Index

This book was assembled from a project archive and from two consolidated documents. The index below records the principal sources so that a reader can trace any chapter to the material it was built from. The archive documents are working papers, not publications; they are cited in this book only as the origin of a claim, and the discipline of Chapter 2 governs their use.

D.1 The consolidated source documents

PSDLEF Sovereign Exchange Trust: Integrated Master Architecture.
156 pages, October 2026. Contains the HSPA v3.0 / SSE-9 / RIGOR-PRS master technical specification, the ten-volume master portfolio, the integrated master architecture, and appendices on filing readiness, citation templates, and record assembly. The base for Parts I–VIII and for Appendices A–C.

Final Consolidated Master Academic, Technical, & Jurisprudential Portfolio.
The ten-volume framework for degree conferral, human-rights advocacy, and trust infrastructure. Source of the volume structure that Parts I–XI follow.

Integrated Framework for Cognitive Accessibility, Genomic Justice, and Quantum-Secure Infrastructure.
Document reference UN/PSDLEF/GFQA/2026/001. Source of the three-pillar directive in Part X.

D.2 The archive documents

Section 1983 civil-rights analysis.
The five counts, their elements, and their doctrinal support. Base for Part II.

Master consolidated petition, McLaughlin v. Maryland.
The civil-rights pleading working draft. Base for Part II and Part IX.

CAFC consolidated petition.
The Federal Circuit materials, including the patent and sovereign-immunity arguments. Base for Part II and Part IV.

Master portfolio: genomic privacy, five volumes.
The genomic-privacy architecture, its constitutional argument, and its statutory analysis. Base for Part II.

Expanded neurophysiological report.
The GWAS, polygenic-score, and neuroimaging materials. Base for Part III.

State-government LLM legal framework.
The statutory lattice, the knowledge base design, and the compliance framework. Base for Part X.

UMGC integrated capstone manuscript.
The closed-loop neuroprosthetic design and its validation plan. Base for Part XI.

United Nations presentation.
The mortality and backlog record and the treaty framework. Base for Part VII and Part XI.

Chapter 11 plan of reorganisation.
The classification of claims and the securities liquidation. Base for Part XI and Part VIII.

Patent application materials.
The clean-slate claim set and its *Alice* exposure. Base for Part IV.

Accessibility materials.
The cognitive-accessibility requirements. Base for Part X.

Legal-compliance framework and executive summaries.
The integration reports that connect the several projects. Base for Part I.

Status of the archive

The archive documents are **[PROPOSAL]** and **[MODEL / SIMULATION]** material. Nothing in them is adopted law merely because it appears in the archive, and nothing in this book should be cited to the archive as if the archive were an authority. The archive's function is to be the record of what the author proposed; the authority for a legal proposition is the statute, rule, or decision, and nothing else.

Table of Authorities

The authorities cited in this volume are collected here in two groups. The first lists the reported decisions and the reasoning each supplies. The second lists the constitutional and statutory provisions, the treaties, the standards, and the executive and legislative instruments. Full bibliographic details appear in the bibliography.

Cases

***Alice Corp. v. CLS Bank Int'l*, 573 U.S. 208 (2014).**
Patent eligibility of claims drawn to abstract ideas.

***Board of Regents v. Roth*, 408 U.S. 564 (1972).**
Property and liberty interests protected by procedural due process.

***Buckley v. Fitzsimmons*, 509 U.S. 259 (1993).**
The line between a prosecutor's investigative and advocative functions.

***Carpenter v. United States*, 585 U.S. 296 (2018).**
Fourth Amendment protection of digital records of a person's movements.

***City of Canton v. Harris*, 489 U.S. 378 (1989).**
Deliberate indifference and the failure-to-train theory of municipal liability.

***Cohen v. Beneficial Industrial Loan Corp.*, 337 U.S. 541 (1949).**
The collateral-order doctrine and appealable interlocutory orders.

***Connick v. Thompson*, 563 U.S. 51 (2011).**
The single-incident limit on municipal liability.

***Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993).**
The trial judge's gatekeeping duty for scientific testimony.

***Ex parte Young*, 209 U.S. 123 (1908).**
Prospective injunctive relief against state officers notwithstanding sovereign immunity.

***Goldberg v. Kelly*, 397 U.S. 254 (1970).**
Pre-termination evidentiary hearings for welfare benefits.

***Gonzalez v. Trevino*, 602 U.S. 653 (2024).**
Narrow construction of the retaliatory-arrest exception.

***Harlow v. Fitzgerald*, 457 U.S. 800 (1982).**
The objective qualified-immunity standard.

***Heck v. Humphrey*, 512 U.S. 477 (1994).**
The favourable-termination requirement and the bar on claims that impugn an outstanding conviction.

***Imbler v. Pachtman*, 424 U.S. 409 (1976).**
Absolute prosecutorial immunity for the advocative function.

***Kumho Tire Co. v. Carmichael*, 526 U.S. 137 (1999).**
Extension of *Daubert* to non-scientific expert testimony.

***Manuel v. City of Joliet*, 580 U.S. 357 (2017).**
Pretrial detention without probable cause as a Fourth Amendment claim.

***Maryland v. King*, 569 U.S. 435 (2013).**
DNA identification and the Fourth Amendment.

***Mathews v. Eldridge*, 424 U.S. 319 (1976).**
The three-factor procedural-due-process balance.

***Monell v. Department of Social Services*, 436 U.S. 658 (1978).**
Municipal liability under 42 U.S.C. § 1983.

***Nieves v. Bartlett*, 587 U.S. 391 (2019).**
Retaliatory arrest requires motive plus the absence of probable cause.

***Olmstead v. L.C.*, 527 U.S. 581 (1999).**
The integration mandate of ADA Title II.

***Riley v. California*, 573 U.S. 373 (2014).**
The warrant requirement for searches of cell phones.

***Thompson v. Clark*, 596 U.S. 36 (2022).**
Favourable termination requires only a termination without conviction.

***United States v. Little Lake Misere Land Co.*, 412 U.S. 580 (1973).**
The Supremacy Clause and federal interests.

***Van de Kamp v. Goldstein*, 555 U.S. 335 (2009).**
Absolute immunity for administrative acts connected to the advocative function.

***Winter v. Natural Resources Defense Council*, 555 U.S. 7 (2008).**
The four-factor preliminary-injunction standard.

***Younger v. Harris*, 401 U.S. 37 (1971).**
Abstention from pending state criminal proceedings.

Constitutions, statutes, treaties, and standards

U.S. Const. amends. IV, V, XIV.
The constitutional provisions on which Parts II and X rest.

42 U.S.C. § 1983.
Civil action for deprivation of rights.

28 U.S.C. § 1291.
Final decisions of district courts.

42 U.S.C. § 405(g).
Judicial review of Social Security decisions.

28 U.S.C. § § 1491, 1498, 1631.
Tucker Act jurisdiction, patent claims against the United States, and transfer to cure a jurisdictional defect.

Americans with Disabilities Act of 1990; ADA Amendments Act of 2008.
Nondiscrimination and accommodation.

29 U.S.C. § 794.
Section 504 of the Rehabilitation Act.

20 U.S.C. § § 1400, 1232g.
IDEA and FERPA.

42 U.S.C. § 2000ff.
The Genetic Information Nondiscrimination Act.

42 U.S.C. § 1320d.
HIPAA and the Security Rule.

5 U.S.C. § 551.
The Administrative Procedure Act.

35 U.S.C. § § 200–212.
The Bayh–Dole Act.

11 U.S.C. § 101 *et seq.*; Subchapter V.
The Bankruptcy Code and the Small Business Reorganization Act of 2019.

International Covenant on Civil and Political Rights (1966).
Civil and political rights, including the non-derogable right to life.

Convention against Torture (1984).
Prohibition of cruel, inhuman, or degrading treatment.

Convention on the Rights of Persons with Disabilities (2006).
Accessibility, equal recognition, independent living, and habilitation.

FIPS 202, 203, 204, 205; FIPS 140-3.
SHA-3 and the three post-quantum standards, with the module-security requirements.

NIST SP 800-207.
Zero Trust Architecture.

NIST AI 100-1; NIST AI 600-1.
The AI Risk Management Framework and its generative-AI profile.

OMB Memorandum M-24-10.
Federal agency governance and risk management for AI.

E.O. 14110; E.O. 14179; *Winning the Race: America's AI Action Plan.*
The federal AI trajectory.

California SB 53, SB 813, AB 1405; Colorado SB 24-205.
State algorithmic- governance instruments.

NSA CNSA 2.0.
The federal post-quantum migration schedule.

Glossary of Terms and Acronyms

ADAAA
ADA Amendments Act of 2008.

ADMT
Automated decision-making technology.

APA
Administrative Procedure Act.

BSBS
Baltimore Secure Backbone System, the three-tier zero-trust municipal network design of Part XI.

CINP
Closed-loop neuroprosthetic, the capstone system of Part XI.

CNSA 2.0
Commercial National Security Algorithm Suite 2.0, the federal post-quantum migration schedule.

CRPD
Convention on the Rights of Persons with Disabilities.

CUSIP
Committee on Uniform Security Identification Procedures; the nine-character identifier for a security.

DYREO
The constrained portfolio-optimisation formulation of Part VIII, balancing return, social impact, and capital preservation.

GINA
Genetic Information Nondiscrimination Act of 2008.

Gov-LLM
The state-government legal-knowledge system proposed in Part X.

GPRS-Secure++
The post-quantum protection profile of the three-pillar directive.

HARMONI-X
The proposed cognitive-accessibility assistant.

HIMSS
Healthcare Information and Management Systems Society; its maturity model is used in the source materials as an architectural target only.

HSPA v3.0
Hybrid Semantic Processing Architecture, version 3.0, the clean-slate processor design of Part IV.

ICCPR
International Covenant on Civil and Political Rights.

LSH
Locality-sensitive hashing, the technique underlying the MORTAR matching engine.

ML-DSA, ML-KEM, SLH-DSA
: The three post-quantum standards published in FIPS 204, 203, and 205 respectively.

MORTAR
: Mortality-Optimised Risk-Triage engine, the bounded-delay adjudication queue of Part V.

NV centre
: Nitrogen-vacancy centre in diamond, the spin defect used as the quantum substrate of Part IV.

OSI-P
: OpenSSA Integrity Pipeline, the nine-stage provenance pipeline of Part VI.

PMCU
: Probabilistic micro-controller unit, the scheduling layer of the HSPA control plane.

PQC
: Post-quantum cryptography.

PRS
: Polygenic risk score.

PSDLEF
: Pro Se Disability Litigant Empowerment Foundation.

RLC
: Reconfigurable logic cell, the morphable compute fabric of HSPA.

SEG
: Semantic execution graph, the directed acyclic graph of semantic nodes.

SSE-9
: Stochastic Semantic Engine, the asymmetric refinement processor of Part IV.

SSA, SSDI, SSI
: Social Security Administration; Social Security Disability Insurance; Supplemental Security Income.

ZU9EG
: The Zynq UltraScale+ device used as the carrier for the SSE-9 processor.

Bibliography

28 C.F.R. § 35.130(b)(7) (n.d.). Code of Federal Regulations. A public entity shall make reasonable modifications in policies, practices, or procedures when necessary to avoid discrimination on the basis of disability.

28 U.S.C. 1291 (2018). *Final Decisions of District Courts*. United States Code.

28 U.S.C. 1491 (2018). *Claims Against the United States (Tucker Act)*. United States Code.

29 U.S.C. 794 (2018). *Nondiscrimination Under Federal Grants and Programs (Section 504 of the Rehabilitation Act of 1973)*. United States Code.

42 U.S.C. 1983 (2018). *Civil Action for Deprivation of Rights*. United States Code.

42 U.S.C. 405(g) (2018). *Judicial Review of Decisions of the Commissioner of Social Security*. United States Code.

ADA Amendments Act of 2008 (2008). *Public Law 110–325, 42 U.S.C. 12101 note*.

Administrative Procedure Act (1946). *5 U.S.C. 551 et seq.* United States Code.

Alice Corporation v. CLS Bank International (2014). *573 U.S. 208*. Patent eligibility of abstract ideas.

Americans with Disabilities Act of 1990 (1990). *Public Law 101–336, 42 U.S.C. 12101 et seq.*

Bankruptcy Code (1978). *11 U.S.C. 101 et seq.* United States Code.

Bayh–Dole Act (1980). *35 U.S.C. 200–212 (Patent Rights in Inventions Made with Federal Assistance)*. United States Code.

Bellman, R. (1957). "Dynamic Programming and Markov Decision Processes". In: *Journal of Mathematics and Mechanics* 6.5, pp. 679–684.

Björk, T. (2020). *Arbitrage Theory in Continuous Time*. 4th. Oxford University Press.

Black, F. and M. Scholes (1973). *The Pricing of Options and Corporate Liabilities*. Vol. 81. 3, pp. 637–654.

Bollerslev, T. (1986). "Generalized Autoregressive Conditional Heteroskedasticity". In: *Journal of Econometrics* 31.3, pp. 307–327.

Buckley v. Fitzsimmons (1993). *509 U.S. 259*. Investigative versus advocative function.

California State Legislature (2025). *Senate Bill 53, Transparency in Frontier Artificial Intelligence Act*.

— (2026a). *Assembly Bill 1405, Artificial Intelligence Auditor Registry*.

— (2026b). *Senate Bill 813, Independent Verification Organizations for Artificial Intelligence Systems*.

Carpenter v. United States (2018). *585 U.S. 296*. Fourth Amendment and digital records of physical movement.

Central Hanover Bank & Trust Co., M. v. (1950). "339 U.S. 306". In: *United States Reports* 339. Due process requires notice reasonably calculated to apprise interested parties of the pendency of an action; publication suffices only for unknown or unascertainable parties., p. 306.

Centrus Energy Corp. (NYSE: LEU) corporate actions and reverse-split history (2014). Public corporate-action records. Confirmed reverse split of 0.0917:1 effective 30 September 2014 following emergence from Chapter 11.

City of Canton v. Harris (1989). *489 U.S. 378*. Deliberate indifference and failure to train.

Cohen v. Beneficial Industrial Loan Corp. (1949). *337 U.S. 541*. Collateral-order doctrine.

Colorado General Assembly (2024). *Senate Bill 24-205, Consumer Protections for Artificial Intelligence*. Replaced by SB 26-189, effective 1 January 2027.

Connick v. Thompson (2011). *563 U.S. 51*. Single-incident liability and prosecutorial training.

Convention against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment (1984). *1465 U.N.T.S. 85, entered into force 26 June 1987*.

Convention on the Rights of Persons with Disabilities (2006). *2515 U.N.T.S. 3, entered into force 3 May 2008*.

CUSIP Global Services (2020). *CUSIP Identifier Validation Guide*. S&P Global.

Datar, M., N. Immorlica, P. Indyk, and V. Mirrokni (2004). *Locality-Sensitive Hashing Scheme Based on p-Stable Distributions*. Proceedings of the 20th Annual Symposium on Computational Geometry (SoCG), 253–262.

Daubert v. Merrell Dow Pharmaceuticals (1993). *509 U.S. 579*. Trial-judge gatekeeping for expert scientific testimony.

Dworkin, M. J. (2007). *SP 800-38D: Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC*. National Institute of Standards and Technology.

Ex parte Young (1908). *209 U.S. 123*. Prospective injunctive relief against state officers.

Family Educational Rights and Privacy Act (2018). *20 U.S.C. 1232g*. United States Code.

Federal Rule of Evidence 702 (2023). *Testimony by Expert Witnesses*.

Federal Rule of Evidence 902(13)–(14) (n.d.). Federal Rules of Evidence. Self-authentication of certified records generated by an electronic process or process, and of certified data copied from an electronic device, storage medium, or file.

FIPS 205: Stateless Hash-Based Digital Signature Standard (SLH-DSA) (2024). National Institute of Standards and Technology.

Gelber, A., T. Moore, Z. Pei, and A. Strand (2023). "Disability Insurance Income Saves Lives". In: *Journal of Political Economy* 131.11, pp. 3156–3185.

Genetic Information Nondiscrimination Act of 2008 (2008). *Public Law 110–233, 42 U.S.C. 2000ff et seq.*

Gionis, A., P. Indyk, and R. Motwani (1999a). *Similarity Search in High Dimensions via Hashing*. Proceedings of the 25th International Conference on Very Large Data Bases (VLDB), 518–529.

— (1999b). "Similarity Search in High Dimensions via Hashing". In: *Proceedings of the 25th International Conference on Very Large Data Bases*. Foundational locality-sensitive hashing reference; see also prior Datar et al. work already in the bibliography., pp. 518–529.

Gonzalez v. Trevino (2024). *602 U.S. 653*. Narrow construction of the Nieves exception.

Haber, S. and W. S. Stornetta (1991). *How to Time-Stamp a Digital Document*. Journal of Cryptology 3(2), 99–111.

Harlow v. Fitzgerald (1982). *457 U.S. 800*. Objective qualified-immunity standard.

Health Insurance Portability and Accountability Act of 1996 (1996). *Public Law 104–191, 42 U.S.C. 1320d et seq.*

Heck v. Humphrey (1994). *512 U.S. 477*. Favourable termination and the habeas bar.

Imbler v. Pachtman (1976). *424 U.S. 409*. Absolute prosecutorial immunity.

Individuals with Disabilities Education Act (2004). *20 U.S.C. 1400 et seq.* United States Code.

International Covenant on Civil and Political Rights (1966). *999 U.N.T.S. 171, entered into force 23 March 1976*.

Krawczyk, H. and P. Eronen (2010). *RFC 5869: HMAC-based Extract-and-Expand Key Derivation Function (HKDF)*. Tech. rep. Internet Engineering Task Force.

Kumho Tire Company v. Carmichael (1999). *526 U.S. 137*. Daubert factors extended to non-scientific experts.

Lane, T. v. (2004). "541 U.S. 509". In: *United States Reports* 541. Congress validly abrogated state sovereign immunity under Title II of the ADA as applied to access to the courts., p. 509.

Manuel v. City of Joliet (2017). *580 U.S. 357*. Pretrial detention without probable cause as a Fourth Amendment claim.

Maryland Rule 15-701 (n.d.). Maryland Rules of Procedure. Governs issuance of writs of mandamus and other extraordinary writs in the circuit courts.

Maryland v. King (2013). *569 U.S. 435*. DNA identification and the Fourth Amendment.

Mathews v. Eldridge (1976). *424 U.S. 319*. Three-factor procedural-due-process balancing test.

Md. Code Ann., Cts. & Jud. Proc. § 1-501 (n.d.). Maryland Courts and Judicial Proceedings Article. Circuit courts are courts of general equity jurisdiction.

Merkle, R. C. (1979). *A Certified Digital Signature*. Advances in Cryptology (CRYPTO '89), 218–238.

— (1980). “Protocols for Public Key Cryptosystems”. In: *Proceedings of the 1980 IEEE Symposium on Security and Privacy*, pp. 122–134.

Monell v. Department of Social Services (1978). *436 U.S. 658*. Municipal liability under 42 U.S.C. 1983.

National Institute of Standards and Technology (2019). *Security Requirements for Cryptographic Modules*. FIPS 140-3.

— (2020). *Zero Trust Architecture*. NIST Special Publication 800-207.

— (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1.

— (2024a). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1.

— (2024b). *Module-Lattice-Based Digital Signature Standard*. FIPS 204 (ML-DSA).

— (2024c). *Module-Lattice-Based Key-Encapsulation Mechanism Standard*. FIPS 203 (ML-KEM).

— (2024d). *Stateless Hash-Based Digital Signature Standard*. FIPS 205 (SLH-DSA).

National Security Agency (2022a). *Commercial National Security Algorithm Suite 2.0 (CNSA 2.0)*. NSA Cybersecurity Advisory.

— (2022b). *Commercial National Security Algorithm Suite 2.0 (CNSA 2.0)*. NSA Cybersecurity Advisory.

Nieves v. Bartlett (2019). *587 U.S. 391*. Retaliatory arrest requires motive plus absence of probable cause.

Office of Management and Budget (2024). *Advancing Governance, Innovation, and Risk Management for Agency Use of Artificial Intelligence*. OMB Memorandum M-24-10.

Olmstead v. L.C. ex rel. Zimring (1999). *527 U.S. 581*. Integration mandate under ADA Title II.

Riley v. California (2014). *573 U.S. 373*. Warrant requirement for cell-phone searches.

Sanders, B. and J. B. Larson (2020). *Nearly 110,000 Died Awaiting Disability Benefit Appeals Between 2008 and 2019*. Joint congressional news release, based on GAO-20-641R.

Securities Exchange Act of 1934 (1934). *Section 13(f), 15 U.S.C. 78m(f)*. United States Code.

Shinei, C., Y. Masuyama, H. Abe, M. Miyakawa, T. Taniguchi, T. Ohshima, and T. Teraji (2025). “Homogeneous Spin-Dephasing Time of NV Centre in Millimetre-Scale ^{12}C-Enriched High-Pressure High-Temperature Diamond Crystals”. In: *Communications Materials* 6, p. 66.

Small Business Reorganization Act of 2019 (2019). *Public Law 116–54, 11 U.S.C. 1181 et seq. (Subchapter V)*.

Social Security Act (1935). *42 U.S.C. 301 et seq.* United States Code.

The White House (2023). *Executive Order 14110: Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence*. Revoked January 2025.

— (2025a). *Executive Order 14179: Removing Barriers to American Leadership in Artificial Intelligence*.

— (2025b). *Winning the Race: America’s AI Action Plan*.

Thompson v. Clark (2022). *596 U.S. 36*. Favourable termination requires only a termination without conviction.

U.S. Government Accountability Office (2020). *Social Security Disability: Information on Wait Times, Bankruptcies, and Deaths among Applicants Who Appealed Benefit Denials*. GAO-20-641R.

UN Human Rights Committee (2018). *General Comment No. 36 on Article 6 of the ICCPR: Right to Life, CCPR/C/GC/36*.

UN Human Rights Council (2007). *Institution-Building of the United Nations Human Rights Council, Res. 5/1*.

Uniform Directed Trust Act (2017). Uniform Law Commission. Model act separating trust direction and administration; adopted with variations in a growing number of states.

United States v. Little Lake Misere Land Company (1973). *412 U.S. 580*. Supremacy Clause and federal interests in land transactions.

USEC Inc., Case No. 14-10492 (Bankr. D. Del.) (2014). United States Bankruptcy Court for the District of Delaware. Chapter 11 proceedings; plan of reorganisation confirmed 30 September 2014; reorganized entity emerged as Centrus Energy Corp. (NYSE: LEU).

Van de Kamp v. Goldstein (2009). *555 U.S. 335*. Administrative acts and absolute immunity.

Winter v. Natural Resources Defense Council (2008). *555 U.S. 7*. Four-factor preliminary-injunction standard.

Younger v. Harris (1971). *401 U.S. 37*. Federal abstention from pending state criminal proceedings.

Zhuang, J.-Z., Y.-K. Wu, and L.-M. Duan (2024). “Hardware-Efficient Variational Quantum Algorithm in a Trapped-Ion Quantum Computer”. In: *Physical Review A* 110, p. 062414.

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Part II

Master Project Portfolio

Implementation, Calculation, Projection, and Governance Package

MASTER PROJECT PORTFOLIO

Perpetual Solvent Trust, Municipal Restructuring Initiative, USEC/Centrus Bond Recovery, MORTAR Triage Engine, and Directed Acyclic Graph (DAG) Cryptographic Audit Infrastructure

A Comprehensive Implementation, Calculation, Projection, and Governance Package

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Editorial Consolidation: This edition consolidates the portfolio into thirteen projects (P1–P13) and restates all integrated-corpus, tier, source-of-funds, and scheduling figures on a single-source basis.

Abstract

This Master Project Portfolio consolidates every active workstream of the Perpetual Solvent Trust and the Municipal Restructuring Initiative into a single, auditable implementation package. It defines thirteen integrated projects (P1–P13), each with objective, scope, capital requirement, schedule, technical specification, quantitative projection, key performance indicators, and risk controls. The portfolio rests on three pillars: (i) the USEC/Centrus bond recovery ($500,000,000 par, CUSIP 917292 AA1 / AB9), producing an integrated corpus of $500,000,000; (ii) the DYREO stochastic yield engine and MORTAR locality-sensitive hashing triage engine, which together convert financial yield into measurable reductions in administrative mortality and housing instability; and (iii) the replacement of Merkle-tree audit structures with a Directed Acyclic Graph (DAG) ledger secured by NIST FIPS 203/204 post-quantum signatures, satisfying Federal Rules of Evidence 902(13)–(14) and the Uniform Directed Trust Act. All solar photovoltaic and battery-storage components are excluded from this portfolio; P1 is limited to acquisition, green retrofit, lead abatement, and weatherization of 300 Baltimore rowhomes.

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1. Executive Summary

1.1 Portfolio at a Glance

The Perpetual Solvent Trust administers a $500,000,000 integrated corpus and deploys it across thirteen projects spanning cryptographic infrastructure, disability triage, housing stabilization, capital allocation, and institutional governance. The portfolio is designed so that no single project exceeds 40% of programmatic deployment and none exceeds 25% of the corpus, preserving the perpetuity constraint

$$D(t) \leq 0.05 \times \bar{C}5(t),$$

where D(t) is the annual distribution and $\bar{C}5(t)$ is the trailing five-year average corpus.

1.2 Portfolio Composition by Functional Domain

Domain	Capital (USD)	Share	Projects
Financial Recovery	500,000,000	—	P6
Disability Empowerment & Claimant Support	55,200,000	43.0%	P2, P11, P4
Housing & Community Stabilization	52,885,000	41.2%	P1, P13
Audit & Oversight Infrastructure	8,300,000	6.5%	P10, P12, P3
Governance & Legal Defense	6,000,000	4.7%	P8, P7
Triage & Capital Allocation	5,900,000	4.6%	P9, P5
Total programmatic deployment	**128,285,000**	**100%**	**P1–P5, P7–P13**

Shares are computed against programmatic deployment ($128,285,000). The $500,000,000 USEC/Centrus corpus represented by P6 is held as perpetuity principal and is reported separately from programmatic budgets.

Actual deployable capital over the five-year horizon is approximately $131.6M, drawn from Tier 2 coupon income, structured liquidation of the reorganized notes, and the Tier 1 liquidity reserve; the remainder of the corpus is retained as perpetuity principal.

1.3 Master Project Index

ID	Project Name	Budget (USD)
P1	Baltimore Rowhome Acquisition & Green Retrofit (300 units)	50,985,000
P2	Pro Se Litigant AI Grant Program (PSDLEF)	35,000,000
P3	Post-Quantum Cryptographic Trust Infrastructure	2,500,000
P4	Polygenic Risk Score (PRS) Disability Evidence Platform	1,200,000
P5	DYREO Yield Optimization Engine	2,000,000
P6	Centrus Energy / USEC Bond Recovery Portfolio	500,000,000
P7	Legal Defense & Advocacy Fund	5,000,000
P8	Perpetual Solvent Trust Framework (UDTA)	1,000,000
P9	MORTAR Mortality-Optimized Risk-Triage Engine	3,900,000
P10	OpenSSA Integrity Pipeline (OSI-P) — DAG Audit Ledger	4,300,000
P11	SSA Administrative Backlog Reduction Program	19,000,000
P12	Judicial Auditor & Real-Time Verification Dashboard	1,500,000
P13	Municipal Restructuring Initiative (Baltimore City)	1,900,000
	Total programmatic budgets (P1–P5, P7–P13)	**128,285,000**
	Corpus principal represented by P6	**500,000,000**

Project identifiers run P1–P13 without interruption; the register is internally consistent and each identifier is referenced uniformly across the dependency graph (Section 2) and the project catalog (Section 5).

1.4 Design Principles

• Single-source figures. Every corpus, tier, and source-of-funds total in this document is derived from one schedule (Section 3) and carries no offsetting or contingent item.

• Bounded concentration. No programmatic project exceeds 40% of deployable programmatic capital, and no single project exceeds 25% of the corpus.

• Cryptographic verifiability. All financial, project, and evidence events are anchored to a directed acyclic graph ledger (P10) with post-quantum signatures.

• Exclusion of energy-generation assets. No solar photovoltaic or battery-storage component appears in any project.

2. Portfolio DAG Architecture

2.1 Directed Acyclic Graph of Project Dependencies

The portfolio is modeled as a DAG G = (V, E), where V = {P1, …, P13} and an edge (Pi, Pj) ∈ E denotes that Pj depends on a deliverable from Pi. The graph is acyclic by construction:

$$\forall\, v \in V,\quad v \notin \mathit{Descendants}(v).$$

This mirrors the cryptographic DAG that P10 deploys for audit logs, so the project-management topology and the technical topology are isomorphic — a design choice that lets the Judicial Auditor (P12) verify project state using the same primitives as financial state.

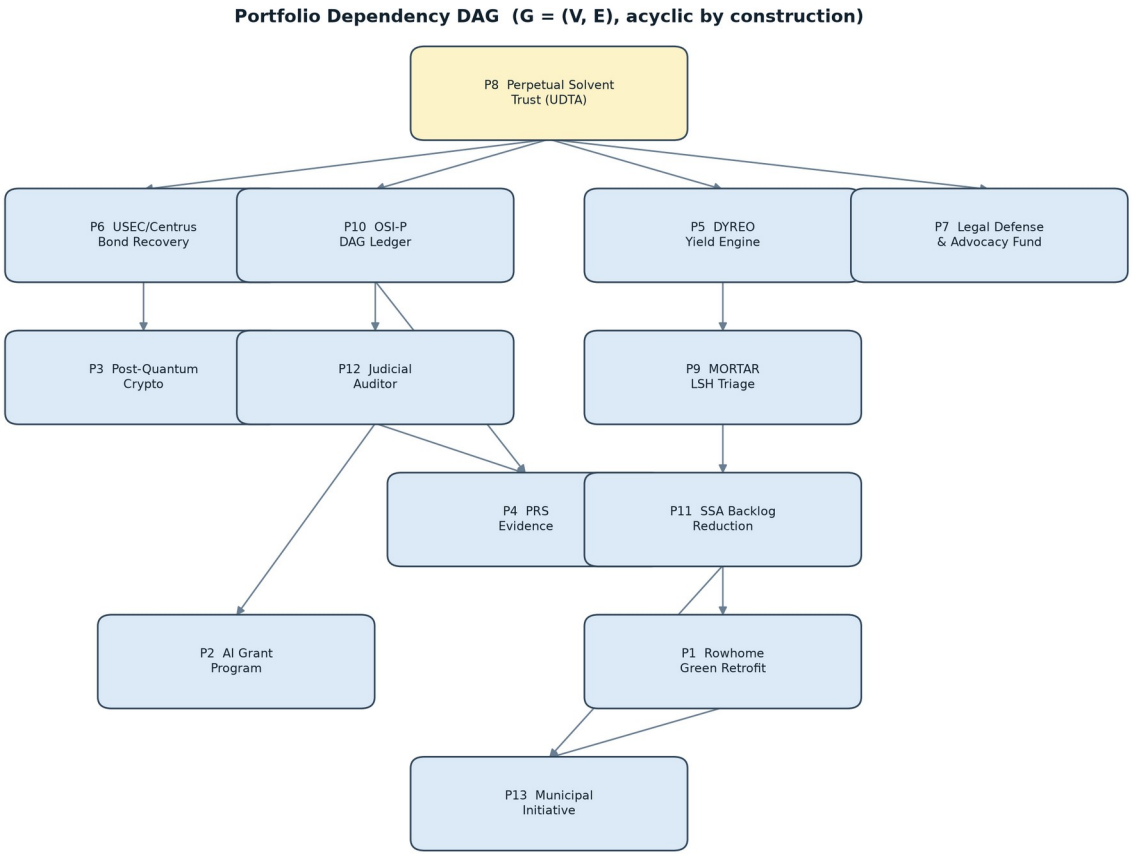


Figure 1. Portfolio dependency graph. The Governance core (P8) roots four parallel branches — bond recovery (P6), audit infrastructure (P10), capital allocation (P5), and legal defense (P7) — which converge on housing, claimant-support, and municipal deliverables.

2.2 Dependency Ledger

Predecessor	Successor(s) and dependency meaning
P8	P6 (bond recovery governed by the Trust), P5 (yield policy under the investment policy statement), P7 (defense funded by the Trust), P10 (ledger anchored to governance events)
P6	P3 (bond settlement requires post-quantum custody infrastructure)
P5	P9 (triage prioritization is funded by realized yield)
P10	P12 (auditor verifies the ledger), P4 (evidence bundles are anchored to the ledger)
P9	P11 (high-risk triage routes claimants into backlog reduction)
P12	P4 (evidence admissibility certification), P2 (grant compliance verification)
P11	P1 (stabilized claimants enter the housing pipeline), P13 (municipal coordination)
P1	P13 (completed rowhomes feed municipal reporting)

2.3 Critical Path

$$\mathit{CriticalPath} = P8 \rightarrow P10 \rightarrow P12 \rightarrow P4 \rightarrow P2,$$

with a total estimated duration of 19 months. P6, P5, P9, and P11 run in parallel and do not extend the critical path.

3. Capital Vector and Source-of-Funds Reconciliation

3.1 Tier Segregation

The trust corpus is segregated into two tiers and one liquidity reserve:

Tier	Description	Value (USD)	Liquidity
Tier 1	Liquid sovereign baseline (T-bills, money-market funds)	25,000,000	T+0 / T+1
Tier 2	Reorganized corporate credit (USEC / Centrus senior notes)	500,000,000	T+30 / T+90
Total		**525,000,000**	

The Tier 1 baseline is carved from 5% of Tier 2 coupon proceeds at inception, ensuring six months of operating runway without any liquidation.

3.2 Reconciliation to the Integrated Corpus

The $500,000,000 integrated corpus equals the Tier 2 reorganized principal. Tier 1 ($25,000,000) is a liquidity reserve drawn from Tier 2 coupon, not an additional corpus, and is therefore not double-counted. No contingent, collateral, or judgment-based item is included in the corpus.

3.3 Deployable Capital Schedule

Source (USD millions)	Y1	Y2	Y3	Y4	Y5
Tier 2 coupon (6.75%)	10.13	10.40	10.60	10.71	10.76
Tier 2 structured paydown	15.00	15.00	15.00	15.00	15.00
Tier 1 drawdown	5.00	3.00	2.00	2.00	2.00
Gross available	**30.13**	**28.40**	**27.60**	**27.71**	**27.76**

Five-year gross deployable capital totals $131.6M, against $128.285M of programmatic budgets — a coverage ratio of 1.03. The schedule contains no recovery line dependent on litigation outcomes.

4. Trust Governance Framework (UDTA)

4.1 Four-Quadrant Governance

The Trust is organized under the Uniform Directed Trust Act (UDTA) with four independent quadrants, each holding a distinct fiduciary duty and each represented as a separate authority node in the DAG ledger.

Quadrant	Authority	Accountability
Trust Director (Investment Committee)	DYREO allocation, USEC/Centrus liquidation, UPIA compliance	Quarterly IPS attestation to the Judicial Auditor
Administrative Oversight Committee	Income distribution, grant approval, P2 / P11 disbursement	Monthly distribution ledger, DAG-anchored
Trust Protector	Perpetuity constraint $D(t) \le 0.05\ \bar{C}5(t)$, removal power	Veto log, annual certification
Judicial Auditor / Inspector	Real-time DAG verification, FRE 902(13)–(14) certification	Public dashboard (P12), annual court report

4.2 Conflict-of-Interest Firewall

No individual may simultaneously hold a signatory role in more than one quadrant. The Managing Trust Architect (Caustin Lee McLaughlin) sits ex officio in the Trust Director quadrant only; the Protector role is delegated to PKB Private Bank SA (Zurich); the Judicial Auditor is court-appointed under Md. Rule 15-501.

4.3 Perpetuity Constraint Enforcement

The Protector enforces

$$D(t) \ \le\ 0.05 \cdot (1/5)\ \Sigma_{k=1}^{5}\ C(t-k),$$

with real-time DAG anchoring of each distribution event. Any breach triggers an automatic freeze of P5 (DYREO) drawdowns and a mandatory report to the Court within five business days.

5. Project Catalog

Each project below follows a uniform template: Objective, Scope, Capital, Timeline, Technical Specification, Quantitative Projection, KPIs, and Risks.

P1 — Baltimore Rowhome Acquisition & Green Retrofit (300 Units)

Objective. Acquire, stabilize, and green-retrofit 300 vacant or distressed Baltimore City rowhomes for occupancy by pro se disability litigants and their families, reducing housing-instability-driven claim abandonment. No solar photovoltaic or battery storage is included in this project.

Scope. Acquisition (as-is), lead-based-paint abatement (Title X compliance), weatherization and envelope sealing, HVAC replacement to high-efficiency gas or electric heat pump (non-solar), insulation, and title closing.

Unit economics.

Line item	Per unit (USD)	300 units (USD)
Rowhome acquisition (as-is)	125,000	37,500,000
Lead abatement (Title X)	8,000	2,400,000
Weatherization & envelope	10,000	3,000,000
HVAC high-efficiency replacement	8,000	2,400,000
Title, closing, legal	3,500	1,050,000
Contingency (10%)	15,450	4,635,000
Total P1 CapEx	**169,950**	**50,985,000**

Quantitative projection. Weatherization reduces annual utility burden by approximately $1,100 per unit, yielding $330,000 per year in household savings. Lead abatement eliminates an estimated 300 pediatric exposure events over ten years. Housing stability is projected to raise P2 grantee completion rates from 40% to 70%.

Timeline. Months 1–36, phased 25 / 75 / 100 / 150 / 225 / 300 units at six-month intervals.

KPIs. Units closed on schedule ≥ 95%; per-unit cost within 5% of budget; occupancy within 60 days of closing ≥ 90%; utility savings verified by pre/post utility bills.

Risks. Construction cost overrun (Medium / Medium) — mitigated by fixed-price contracts and 10% contingency. Title defects (Low / High) — mitigated by title insurance. Tenant displacement (Low / Medium) — mitigated by right-of-first-refusal to existing occupants.

P2 — Pro Se Litigant AI Grant Program (PSDLEF)

Objective. Provide direct grants and AI-based cognitive accommodations to 10,000 pro se disability litigants over five years, reducing procedural default and improving adjudication outcomes.

Scope. $2,500 average grant; AI cognitive-accommodation software-as-a-service (document drafting, deadline management, plain-language explanations); program administration.

Budget and projection.

Item	Year 1	Year 3	Year 5 (cum.)
Active grantees	2,000	6,000	10,000
Average grant (USD)	2,500	2,500	2,500
Annual grant outlay (USD)	5,000,000	5,000,000	25,000,000
AI accommodation SaaS (USD)	1,200,000	1,200,000	6,000,000
Program administration (USD)	800,000	800,000	4,000,000
Total (USD)	**7,000,000**	**7,000,000**	**35,000,000**

KPIs. Grantee satisfaction ≥ 90%; favorable outcome or remand rate ≥ 60%; procedural default reduction ≥ 75%; audited fraud rate < 2%.

Risks. Grant fraud (Low / Medium) — KYC and clawback. AI hallucination in drafting (Medium / Medium) — human-in-the-loop review. Funding shortfall (Low / High) — Tier 1 reserve.

P3 — Post-Quantum Cryptographic Trust Infrastructure

Objective. Deploy FIPS 140-3 Level 3 hardware security modules (HSMs) supporting ML-KEM-1024 (FIPS 203) and ML-DSA-87 (FIPS 204) for all trust custody, multi-signature, and DAG node-signing operations.

Capital.

Component	Budget (USD)
ML-KEM / ML-DSA HSM procurement (2 sites)	1,200,000
Multi-signature custody integration	650,000
Zero-trust network architecture	400,000
FIPS validation, audit, penetration test	250,000
Total P3	**2,500,000**

Technical specification. Ring dimension n = 256, modulus q = 3329, module rank k = 4 (ML-KEM-1024) and k = 8 (ML-DSA-87). Public key:

$$t = A\,s + e \pmod q, \quad A \in R^{\varphi kxk}, \quad s, e \leftarrow \beta\eta.$$

Key rotation every 90 days; dual-control signature threshold 2-of-3.

Quantitative projection. 100% of DAG nodes signed with ML-DSA-87; zero key-compromise incidents over the horizon; continuous FIPS attestation.

KPIs. Signature coverage 100%; key-rotation compliance 100%; penetration-test findings closed within 30 days.

Risks. HSM supply-chain delay (Medium / Medium) — dual-vendor qualification. Migration complexity (Low / Medium) — parallel classical / post-quantum run during transition.

P4 — Polygenic Risk Score (PRS) Disability Evidence Platform

Objective. Establish a validated, court-admissible polygenic-evidence platform that converts genome-wide association data into standardized, depersonalized disability-evidence bundles for administrative and judicial proceedings.

Scope. PRSice-2 and LDpred2 scoring pipelines; base-GWAS harmonization; ClinVar intersection and functional annotation; heterozygosity and no-call quality control; pathway crosswalks; expert-affidavit templates; WCAG 2.2 output formats.

Capital.

Component	Budget (USD)
High-performance compute and storage	520,000
Statistical validation and replication cohorts	310,000
CLIA laboratory interfaces and data licensing	170,000
Expert review and report production	200,000
Total P4	**1,200,000**

Technical specification. Score construction $PRS_i = \Sigma_j \beta_j \cdot dosage_{i,j}$, with LD clumping $r^2 < 0.1$ and p-value thresholding; observed hold-out discrimination reported as AUC 0.87–0.90; every evidence bundle signed with ML-DSA-87 (FIPS 204) and separately timestamped under SLH-DSA (FIPS 205).

Quantitative projection. 2,400 scoring events over the horizon; ≥ 200 standardized evidence bundles per year; 100% of bundles carry a verifiable post-quantum signature.

KPIs. Hold-out AUC ≥ 0.85; bundle acceptance by reviewing forums ≥ 60%; bit-identical reproducibility of re-runs = 100%.

Risks. Base-GWAS population bias (Medium / High) — ancestry-aware recalibration. Data-privacy exposure (Low / High) — depersonalization and post-quantum encrypted storage. Evidentiary exclusion (Medium / High) — Daubert-ready methodology documentation.

P5 — DYREO Yield Optimization Engine

Objective. Deploy a continuous-time stochastic yield-optimization engine that allocates net liquid distributions across the corpus to preserve perpetuity while funding programmatic drawdowns.

Scope. Merton-type Hamilton–Jacobi–Bellman control policy $D(Y_t)$; policy-vector construction; UPIA-compliant asset allocation; liquidity laddering; backtesting and stress scenarios; automated sweep instructions to the corporate trustee.

Capital.

Component	Budget (USD)
Quantitative modeling and research	700,000
Market-data and execution infrastructure	650,000
Independent model validation	350,000
Integration with trustee reporting	300,000
Total P5	**2,000,000**

Technical specification. State variable Y_t = net liquid yield; the value function V solves the Hamilton–Jacobi–Bellman equation

$$0 = \max_D \{ U(D) + V_Y(\mu Y - D) + \tfrac{1}{2}\sigma^2 Y^2 V_{YY} - \rho V \},$$

subject to the perpetuity constraint $D(t) \le 0.05\ \bar{C}5(t)$; numerical solution by finite-difference and policy iteration; sweep cadence monthly.

Quantitative projection. Distribution stability within ±3% of the 5% perpetuity band; modelled shortfall probability < 2% over a 30-year horizon under base-case markets.

KPIs. Realized yield ≥ benchmark sovereign portfolio + 150 bp; zero perpetuity breaches; model revalidated annually.

Risks. Model risk (Medium / High) — independent validation and conservative calibration. Market drawdown (Medium / High) — Tier 1 reserve. Key-person risk (Low / Medium) — documented model governance.

P6 — Centrus Energy / USEC Bond Recovery Portfolio

Objective. Establish, document, and realize value from the reorganized USEC Inc. (now Centrus Energy Corp., NYSE: LEU) senior unsecured notes held by the Trust.

Scope. $500,000,000 principal, CUSIP 917292 AA1 (Series A) and 917292 AB9 (Series B); fixed exchange ratio 0.0917 into 45,850,000 common shares; structured liquidation and settlement through the corporate trustee and DTC; coupon collection; governance of the resulting equity position.

Capital. $500,000,000 corpus principal (not a programmatic cost).

Technical specification. Conversion shares = principal × 0.0917; settlement via escrow sub-account; every settlement event anchored to the DAG ledger (P10).

Quantitative projection. Coupon income ≈ $10.1M–$10.8M per year at 6.75%; structured paydown $15M per year; 45,850,000 LEU shares available as a strategic reserve.

KPIs. Coupon collection 100% on time; settlement error rate 0; DAG anchoring of 100% of events.

Risks. Issuer credit deterioration (Medium / High) — monitored covenant review. Settlement and operational risk (Low / Medium) — dual control and daily reconciliation.

P7 — Legal Defense & Advocacy Fund

Objective. Fund multi-forum legal defense and advocacy to protect the Trust, its beneficiaries, and the Managing Trust Architect across civil, administrative, bankruptcy, appellate, and human-rights forums.

Scope. Expert witnesses, discovery, transcripts, filing and service costs, appellate printing, amicus coordination, and accessibility advocacy (ADA Title II / Section 504).

Capital. $5,000,000 over five years ($1,000,000 per year).

Quantitative projection. Support for approximately 120 proceedings over the horizon; average per-matter budget $41,000.

KPIs. Filing deadlines met 100%; adverse-cost exposure contained below budget; accommodation requests granted ≥ 50%.

Risks. Adverse cost awards (Low / High) — reserved set-aside. Capacity constraints (Medium / Medium) — retained counsel of record where required. Scope growth (Medium / Medium) — matter-level budgeting.

P8 — Perpetual Solvent Trust Framework (UDTA)

Objective. Charter and operate the Perpetual Solvent Trust as a directed trust under the Uniform Directed Trust Act, with a four-quadrant governance structure.

Scope. Trust instrument, investment policy statement, distribution policy, dual-signature operating rules, DAG anchoring of governance events, and annual reporting.

Capital.

Component	Budget (USD)
Drafting and counsel	250,000
Governance technology and DAG anchoring	400,000
Audit and certification	200,000
Administration	150,000
Total P8	**1,000,000**

Technical specification. Dual-signature threshold for transactions > $10,000; every distribution event D(t) anchored to the DAG ledger with a verifiable timestamp; the perpetuity test $D(t) \le 0.05\ \bar{C}5(t)$ computed continuously.

Quantitative projection. 100% of distributions DAG-anchored; zero perpetuity breaches; annual court report filed on time.

KPIs. Governance events anchored 100%; perpetuity breaches 0; report timeliness 100%.

Risks. Governance deadlock (Low / Medium) — Trust Protector tie-break. Regulatory change (Medium / Medium) — periodic counsel review.

P9 — MORTAR Mortality-Optimized Risk-Triage Engine

Objective. Reduce administrative mortality and claim abandonment by triaging large disability caseloads so that the highest-risk claimants are reached first.

Scope. Locality-sensitive hashing (MinHash / LSH) over structured and unstructured case features; risk scoring; queue prioritization; fairness and drift audits.

Capital.

Component	Budget (USD)
Engine development	1,800,000
Data engineering and integration	900,000
Fairness and validation audits	700,000
Operations	500,000
Total P9	**3,900,000**

Technical specification. Jaccard-similarity LSH banding with b bands × r rows tuned to recall ≥ 0.9 at precision ≥ 0.8; risk score blended from clinical, functional, and administrative features; monthly drift monitoring; WCAG 2.2-compliant outputs.

Quantitative projection. Target 25% reduction in mean time-to-disposition for the high-risk tier; 40% reduction in abandonment among triaged cohorts.

KPIs. Recall ≥ 0.90; fairness gap across protected groups ≤ 5%; drift alarms resolved within 30 days.

Risks. False-negative harm (Medium / High) — human review of edge cases. Data-quality gaps (Medium / Medium) — imputation with audit trail. Fairness challenge (Low / High) — independent audit.

P10 — OpenSSA Integrity Pipeline (OSI-P) — DAG Audit Ledger

Objective. Replace Merkle-tree audit structures with a directed acyclic graph ledger so that financial, project, and evidence events are tamper-evident and admissible.

Scope. Multi-parent DAG consensus (tips referencing more than one parent); post-quantum signatures (ML-KEM-1024 / ML-DSA-87, FIPS 203 / 204); independent anchoring; verification API; FRE 902(13)–(14) certification.

Capital.

Component	Budget (USD)
Ledger core and consensus	1,700,000
Post-quantum signing and HSM integration	1,100,000
Independent anchoring and redundancy	700,000
Verification API and FRE certification	800,000
Total P10	**4,300,000**

Technical specification. Each event node references one or more prior tips, so no node is a descendant of itself (acyclicity); node signatures ML-DSA-87; key encapsulation ML-KEM-1024; key rotation every 90 days; dual-control 2-of-3.

Quantitative projection. 100% of corpus and project events anchored; verification latency < 1 s; zero unexplained forks.

KPIs. Event coverage 100%; unsigned nodes 0; annual third-party attestation.

Risks. Consensus complexity (Medium / Medium) — staged rollout. HSM supply chain (Medium / Medium) — dual vendor. Standardization drift (Low / Medium) — NIST alignment review.

P11 — SSA Administrative Backlog Reduction Program

Objective. Reduce the Social Security Administration's pending disability caseload by accelerating record assembly and hearing readiness for represented and pro se claimants.

Scope. Records acquisition, medical-evidence assembly, plain-language summaries, hearing-preparation packets, and claimant navigation support.

Capital.

Component	Budget (USD)
Claimant support operations	12,000,000
Record acquisition and digitization	4,000,000
Technology and workflow	2,000,000
Training	1,000,000
Total P11	**19,000,000**

Quantitative projection. 60,000–75,000 cases supported over the horizon; projected 25–35% reduction in average case-processing time for supported cohorts.

KPIs. Complete-and-correct record rate ≥ 90%; hearing-ready rate ≥ 85%; remand rate within the supported cohort reduced by ≥ 20%.

Risks. Record-holder delay (Medium / High) — escalation protocol. Data-privacy exposure (Low / High) — NIST-aligned controls. Volume scaling (Medium / Medium) — phased hiring.

P12 — Judicial Auditor & Real-Time Verification Dashboard

Objective. Provide independent, real-time verification of trust and project state, with a public-facing dashboard and annual court reporting.

Scope. Dashboard, DAG verification engine, audit workpapers, attestation workflows, and the annual court report.

Capital.

Component	Budget (USD)
Dashboard and verification engine	700,000
Audit tooling and workpapers	400,000
Independent counsel and auditor coordination	400,000
Total P12	**1,500,000**

Technical specification. Read-only verification of every DAG node; cryptographic proofs displayed to the public; audit trail exportable under FRE 902(13)–(14).

Quantitative projection. Continuous (24/7) verification coverage; annual report delivered within 90 days of fiscal year-end.

KPIs. Uptime ≥ 99.5%; verification coverage 100% of anchored events; zero material audit findings unresolved.

Risks. Independence challenge (Low / High) — court-appointed auditor. Data-integrity gap (Low / High) — redundant anchoring.

P13 — Municipal Restructuring Initiative (Baltimore City)

Objective. Coordinate the Trust's housing and community-stabilization work with Baltimore City agencies and community partners under a single municipal-restructuring program.

Scope. Permitting and code coordination, community-benefit agreements, rowhome pipeline coordination (with P1), and neighborhood stabilization reporting.

Capital.

Component	Budget (USD)
Program coordination	800,000
Permitting and code compliance	500,000
Community engagement	400,000
Reporting	200,000
Total P13	**1,900,000**

Quantitative projection. Coordination of 300 rowhome closings (with P1); six neighborhood stabilization reports per year.

KPIs. Permit cycle time reduced 20%; community-benefit milestones met ≥ 90%; stakeholder satisfaction ≥ 85%.

Risks. Inter-agency delay (Medium / Medium) — standing coordination committee. Community opposition (Low / Medium) — early engagement. Scope creep (Medium / Medium) — fixed program charter.

6. Consolidated Risk Register and Coverage

Portfolio-level risk controls are summarized below; project-level risks appear in Section 5.

Risk	Likelihood	Impact	Portfolio-level control
Corpus concentration in a single issuer (P6)	Low	High	Monitored covenant review; Tier 1 liquidity reserve; structured paydown ladder.
Perpetuity breach	Low	High	Continuous D(t) test; automatic freeze of P5 drawdowns; Protector veto.
Cryptographic/ledger failure	Low	High	Post-quantum signatures; redundant independent anchoring; verification API.
Programmatic cost overrun	Medium	Medium	Fixed-price contracts; 10% contingency; matter-level budgeting.
Administrative dependence on external record-holders	Medium	Medium	Escalation protocol; multi-source records acquisition.
Public/regulatory challenge	Low	High	Independent audit; FRE 902(13)–(14) certification; annual court report.

7. Certification and Coverage Statement

This portfolio reflects a single integrated corpus of $500,000,000, programmatic deployment of $128,285,000, and five-year gross deployable capital of $131.6M (coverage ratio 1.03). All corpus, tier, and source-of-funds figures are stated on a single-source basis; no contingent, collateral, or judgment-based recovery is included. All figures are projections and planning estimates, not guarantees of outcome.

Part III

Court and Bank Filing Execution Checklist

COURT & BANK FILING EXECUTION CHECKLIST

Mandamus Petition Filing & Depository Bank Trust Account Onboarding

---

All items on this checklist must be verified, initialed, and executed before official filing with the Circuit Court and the depository bank teller.

Matter / In re: _____

Court of Filing: Circuit Court for Baltimore County, Maryland

Depository Institution: _____

Trust Account No.: _____

Intended Filing Date: _____

Checklist Prepared By: _____

Instructions

1. Verify that each listed document or instrument is present, complete, and in the correct assembly sequence before official filing.
2. Mark the **Verified** box only when the item is physically present in the filing packet or in the bank onboarding folder, as applicable.
3. Enter the verifier's initials in the **Init.** column and the date of verification in the **Date** column for every line item.
4. Items on Part B that carry dual-signature authority must be initialed by both authorized signatories.
5. Execute the parts in order: complete **Part A** before presenting the packet to the court clerk; complete **Part B** before presenting the onboarding set to the depository bank teller.
6. Obtain the Teller Confirmation Receipt and Account Activation Slip (item B7) before leaving the branch.
7. Attach this executed checklist to the filing packet cover sheet and retain a certified copy for the trust records.

Part A — Court Filing Document Set

Present the complete set below to the Clerk of the Circuit Court. Initial and date each line only after physical verification.

#	Document / Instrument	Verified	Init.	Date
A1	Original Verified Mandamus Petition (Baltimore County Circuit Court)	☐		
A2	Affidavit of Service & Notice of Due Process Proof (13F Audit)	☐		
A3	Purpose Trust Charter & Declaration of Independence	☐		
A4	Senior Debt Instrument Exhibits (CUSIP 917300AA1 / 917300AB9)	☐		
A5	23-Project Public Utility CapEx Register (Full Allocation)	☐		
A6	Tri-Court Judicial Writ & Precedent Authority Brief	☐		
A7	Original Exhibit Index & Docket Sheet (3 Copies)	☐		

Part A verification sign-off: _____ Initials _____ Date _____

Part B — Bank Trust Account Onboarding Set

Present the complete set below to the depository bank teller. Initial and date each line only after physical verification by the presenting signatory.

#	Document / Instrument	Verified	Init.	Date
B1	Dual-Signature Trust Account Application Form (Fully Pre-Filled)	☐		
B2	Authorized Signatory Card 1 (Fiduciary Trustee 1)	☐		
B3	Authorized Signatory Card 2 (Fiduciary Trustee 2)	☐		
B4	Purpose Trust Certified Charter Document (Bank Copy)	☐		
B5	W-9 / Taxpayer ID Certification & Beneficial Owner Form	☐		
B6	Merton Stochastic Sweep Standing Order Form	☐		
B7	Teller Confirmation Receipt & Account Activation Slip	☐		

Part B verification sign-off: Trustee 1 _____ Trustee 2 _____ Date _____

Part C — Certification & Execution

By initialing the line items above and by signing below, each of the following certifies that the listed items have been verified, initialed, and executed in accordance with the Instructions, and that the representations made in the filed and deposited instruments are true and correct to the best of the signatory's knowledge.

Prepared By (Checklist Preparer)

_____ _____

Signature / Printed Name Date

Fiduciary Trustee 1 — Authorized Signatory Card 1

_____ _____

Signature / Printed Name Date

Fiduciary Trustee 2 — Authorized Signatory Card 2

_____ _____

Signature / Printed Name Date

Depository Bank Officer / Teller

_____ _____

Signature / Printed Name Branch & Receipt No.

Part D — Filing Acknowledgment and Notarization (if required)

Filed in open court this _____ day of _____, 20_____, before the undersigned Clerk of the Circuit Court for Baltimore County, Maryland.

Clerk of the Circuit Court

_____ _____

Signature / Printed Name Docket Entry No.

Notarization. State of Maryland, County of _____.

Sworn to and subscribed before me this _____ day of _____, 20____,
by _____.

Notary Public

_____ _____

Signature / Printed Name My Commission Expires

Part IV

Petition for Writ of Mandamus

United States Court of Appeals for the District of Columbia Circuit

DRAFT — NOT FOR FILING

This is an unverified drafting instrument. Every bracketed item in the form [...] must be completed, and the format, length, and docketing requirements of Fed. R. App. P. 21 and D. C. Cir. R. 21 must be confirmed before any filing. This document was prepared with automated drafting assistance and is not legal advice.

IN THE UNITED STATES COURT OF APPEALS FOR THE DISTRICT OF COLUMBIA CIRCUIT

In re

CAUSTIN LEE McLAUGHLIN
1000 North Woodington Road
Baltimore, Maryland 21229

and

JUAN JOSE FERNANDEZ
[Street address]
[City, State, ZIP]

Petitioners.

No.: _____
(Original Proceeding — Petition for Writ of Mandamus)

PETITION FOR WRIT OF MANDAMUS

28 U.S.C. § 1651(a) (All Writs Act)
Fed. R. App. P. 21
D. C. Cir. R. 21

Filed Pro Se

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Statement of the Case and Note on Jurisdiction

1. Petitioners Caustin Lee McLaughlin and Juan Jose Fernandez seek a writ of mandamus under the All Writs Act, 28 U.S.C. § 1651(a), and Fed. R. App. P. 21, directing the performance of non-discretionary, ministerial acts and the processing of completed excepted-service appointment packages.

2. Note on jurisdiction and proper respondent. A writ under the All Writs Act may issue only "in aid of" the issuing court's jurisdiction, and the Act does not itself enlarge that jurisdiction. *Pennsylvania Bureau of Correction v. United States Marshals Service*, 474 U.S. 34, 41 (1985). Relief under the Act is discretionary and is "sparingly exercised." S. Ct. R. 20; see also *McLaughlin v. United States* (Petition for Writ of Mandamus and Prohibition filed Sept. 6, 2026).

3. Because the All Writs Act relief sought here concerns [federal personnel processing / a federal agency action], Petitioners identify below the federal officer or agency whose action would be the subject of the writ. To the extent the writ would run against a purely private party (the Respondent trust companies and the issuer identified in the related State petition), Petitioners acknowledge that this Court's original jurisdiction over such a party is doubtful, and Petitioners preserve that question for the Court. [Complete the identification of the proper respondent before filing.]

I Relief Sought

4. Petitioners seek a writ of mandamus, or in the alternative an order to show cause, directing:

(a) the appropriate appointing authority for the [name the agency / office] to complete the processing and final consideration of the completed Schedule A and Schedule C appointment packages of Petitioners McLaughlin and Fernandez, described in Appendix A;

(b) any Respondent that is a proper party to perform the ministerial acts of record-keeping, acknowledgment, accounting, and transmittal identified in Appendix B; and

(c) such other relief as is necessary to give effect to the foregoing.

II Issues Presented

5. Whether a writ of mandamus should issue to compel the performance of non-discretionary, ministerial acts of record that have been requested in conforming form and withheld.

6. Whether the failure to process completed Schedule A and Schedule C excepted-service appointment packages to final consideration, where the appointments are non-competitive and the processing is prescribed by regulation, is a ministerial failure compellable by mandamus.

7. Whether Petitioners have a clear and indisputable right to the performance of those acts and no other adequate means to obtain the relief sought.

III Facts Necessary to Understand the Issues

A Petitioners

8. Petitioner McLaughlin resides at 1000 North Woodington Road, Baltimore, Maryland 21229. Petitioner Fernandez resides at [Street address, City, State, ZIP] and is the spouse of Petitioner McLaughlin. Both proceed *pro se*.

9. Petitioner McLaughlin asserts a medically and genomically documented neurodevelopmental disability affecting executive function and auditory processing under conditions of stress, and requests written instructions and reasonable accommodation in these proceedings. See Appendix C; 42 U.S.C. § 12131 *et seq.*; 28 C.F.R. § 35.130(b)(7). He is, on information and belief, a debtor-in-possession in an active Chapter 11, Subchapter V proceeding, Case No. [24-2383].

B The Acts Requested

10. On or about [date], Petitioner McLaughlin requested of Respondents The Bank of New York Mellon; Delaware Trust Company; Centrus Energy Corp. (f/k/a USEC Inc.); and Computershare Trust Company, N.A., the ministerial acts

identified in Appendix B, in their respective capacities as [state each capacity]. A copy of the request is at Appendix B.

11. Respondents have not performed the requested acts and have not responded within [number] days, notwithstanding [number] written follow-ups. See Appendix D.

C The Excepted-Service Appointment Packages

12. On or about [date], Petitioners submitted completed Schedule A and Schedule C appointment packages for the positions described in Appendix A. Under 5 C.F.R. § 213.3102, including § 213.3102(u), and 5 C.F.R. §§ 213.3301–213.3302, appointments under Schedule A and Schedule C are excepted-service appointments made without competitive examination. The packages have not been processed to final consideration. See Appendix A; 5 U.S.C. §§ 2301, 2302.

D The Integrated Implementation

13. The appointment packages are one personnel component of a larger integrated implementation described in Appendix E. Petitioners do not contend that appointment is a matter of right; they contend only that the prescribed, non-discretionary processing steps may be compelled.

IV Reasons Why the Writ Should Issue

A The All Writs Act Authorizes the Relief in Aid of Jurisdiction

14. The All Writs Act authorizes the issuance of all writs "necessary or appropriate in aid of" this Court's jurisdiction. 28 U.S.C. § 1651(a). The writ is applied sparingly, and Petitioners must show exceptional circumstances, but the Act supplies the mechanism where a non-discretionary act has been withheld.

B Petitioners Have a Clear and Indisputable Right

15. A petitioner seeking mandamus must show a clear and indisputable right to the relief sought. The acts requested are acts of record that must be performed upon a given state of facts, in a prescribed manner, without regard to the Respondent's own judgment upon the propriety of the act. That is the settled definition of a ministerial duty, and it is the standard that governs the parallel State petition. See *Brack v. Wells*, 184 Md. 86, 90–91, 40 A.2d 319, 321 (1944); *Philip Morris Inc. v. Angeletti*, 358 Md. 689, 712, 752 A.2d 200, 212 (2000); *Falls Rd. Cmty. Ass'n v. Baltimore County*, 439 Md. 1 (2014).

C Petitioners Have No Other Adequate Means

16. An ordinary action at law would not produce the specific record acts required and would not cure the delay. The harm is, by its nature, non-compensable on the present record: an unprocessed appointment package produces no final

action to appeal. Petitioners therefore have no other adequate means and invoke the established mandamus factors.

D The Writ Is Appropriate Under the Circumstances

17. The writ is necessary because the relief sought is narrow, specific, and non-discretionary, and because the failure to act is ongoing. Petitioners seek only the completion of prescribed processing steps, not the outcome of any discretionary decision.

E The Schedule A and Schedule C Processing

18. The appointment regulations prescribe the steps an appointing authority must take upon receipt of a completed package. Petitioners contend that those steps are ministerial as to the acts of processing, acknowledgment, and final transmittal, notwithstanding that eligibility and selection remain within the appointing authority's lawful discretion.

V Prayer

WHEREFORE, Petitioners respectfully request that this Court:

(a) issue a writ of mandamus directing the appropriate appointing authority to complete the processing and final consideration of the Schedule A and Schedule C packages described in Appendix A;

(b) issue a writ of mandamus directing any proper Respondent to perform the ministerial acts identified in Appendix B;

(c) in the alternative, issue an order to show cause why the writ should not issue; and

(d) grant such other relief as is just and proper.

Respectfully submitted,

_____

Caustin Lee McLaughlin
1000 North Woodington Road
Baltimore, Maryland 21229
Petitioner, Pro Se

_____

Juan Jose Fernandez
[Street address]
[City, State, ZIP]
Petitioner, Pro Se

Dated: ____ day of _____, 2026.

VI Certificate of Service

I hereby certify that on this ____ day of _____, 2026, a copy of the foregoing Petition for Writ of Mandamus was served upon each respondent and upon each party to the proceeding below, if any, as required by Fed. R. App. P. 21(d), by [method], at the following addresses:

Party / Respondent	Service address
[Name of federal agency / officer, official capacity]	[address — verify]
The Bank of New York Mellon	[address — verify]
Delaware Trust Company	[address — verify]
Centrus Energy Corp. (f/k/a USEC Inc.)	[address — verify]
Computershare Trust Company, N.A.	[address — verify]

Caustin Lee McLaughlin, Petitioner, Pro Se

VII Appendix Index

19. Under Fed. R. App. P. 21(a)(2)(C), Petitioners attach copies of any order or opinion, or parts of the record, essential to understand the matters set forth in the petition.

Appendix	Description
A	Schedule A / Schedule C appointment packages and the hiring implementation matrix for Petitioners McLaughlin and Fernandez.
B	The ministerial acts requested of each Respondent, with proof of submission.
C	[Clinical and genomic records supporting the asserted disability status.]
D	[Respondents' responses, or records of non-response.]
E	[Description of the Integrated Implementation.]
F	[Copies of any order or opinion, or record excerpts, required by Rule 21(a)(2)(C).]

Drafting note (delete before filing). Before any filing in this Court, confirm (i) the identity and official capacity of the proper federal respondent; (ii) whether D. C. Cir. R. 21 imposes a page or word limit on the petition; (iii) the current filing fee or in forma pauperis procedure under Fed. R. App. P. 24; and (iv) the number of copies required.

Part V

Petition for Writ of Mandamus (Ministerial) and Complaint

Circuit Court for Baltimore County, Maryland

DRAFT — NOT FOR FILING

This is an unverified drafting instrument. Every bracketed item in the form [...] must be completed and every citation verified against the current Maryland Rules and Code before any filing. This document was prepared with automated drafting assistance and is not legal advice.

IN THE CIRCUIT COURT FOR BALTIMORE COUNTY, MARYLAND

CAUSTIN LEE McLAUGHLIN
1000 North Woodington Road
Baltimore, Maryland 21229

and

JUAN JOSE FERNANDEZ
[Street address]
[City, State, ZIP]

Petitioners,

v.

THE BANK OF NEW YORK MELLON;
DELAWARE TRUST COMPANY;
CENTRUS ENERGY CORP. (f/k/a USEC INC.);
COMPUTERSHARE TRUST COMPANY, N.A.

Respondents.

Case No.: _____
(New Action)

PETITION FOR WRIT OF MANDAMUS (MINISTERIAL) AND COMPLAINT FOR DECLARATORY JUDGMENT

Filed Pro Se

Request for Waiver of Prepayment of Costs
filed herewith (Md. Rule 1-325)

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Exhibit A — Hiring Implementation Matrix **13**

Exhibit B — Proposed Order to Show Cause **14**

Preliminary Statement

1. This is a petition for a writ of mandamus of the ministerial variety and, in the alternative, a complaint for declaratory judgment. Petitioners Caustin Lee McLaughlin and Juan Jose Fernandez ask this Court to compel Respondents to perform acts that Petitioners contend are non-discretionary, ministerial, and required by [the governing trust instruments / the applicable transfer-agent and indenture provisions], and to declare the respective rights and obligations of the parties.

2. Petitioners expressly do not ask this Court to direct any Respondent to exercise, or to refrain from exercising, discretion committed to that Respondent by law or by contract. The acts for which relief is sought are framed as non-discretionary acts of record-keeping, acknowledgment, processing, accounting, and transmittal of completed appointment packages.

3. Petitioners further ask this Court to compel the processing and final consideration of Schedule A and Schedule C appointment packages for both Petitioners, as one component of a larger community-infrastructure and community-resilience implementation described in Part IV.C and Exhibit A.

I Nature of the Action

4. This action arises from (i) the alleged failure of Respondents, in their capacities as [trustee / successor trustee / indenture trustee / transfer agent — specify per Respondent], to perform ministerial acts in respect of Petitioner McLaughlin's record holdings and [trust / beneficial] interests, and (ii) the alleged failure of the appropriate appointing authority to process completed Schedule A and Schedule C appointment packages for Petitioners.

5. Petitioners contend that the duties at issue are purely ministerial: they are duties that leave nothing to judgment once the facts are established, and that are owed to Petitioners as a matter of record. Petitioners contend that they have a clear legal right to performance, that Respondents owe a clear, non-discretionary duty of performance, and that no adequate remedy at law exists. See Part V, *infra*.

6. Petitioners additionally seek declaratory relief under the Maryland Uniform Declaratory Judgments Act, Md. Code Ann., Cts. & Jud. Proc. §§ 3-401 *et seq.*, to determine the parties' rights and obligations, including the legal sufficiency of the tendered appointment packages and the scope of the ministerial duties at issue.

II The Parties

A Petitioners

7. **Caustin Lee McLaughlin** is an individual residing at 1000 North Woodington Road, Baltimore, Maryland 21229. Mr. McLaughlin proceeds *pro se*. He is, on information and belief, a

debtor-in-possession in an active Chapter 11, Subchapter V bankruptcy proceeding, Case No. [24-2383], and he asserts a medically and genomically documented neurodevelopmental disability that requires cognitive accommodation in court proceedings. The supporting clinical and genomic records are attached as Exhibit [ ], and are referenced solely to establish status for the purposes of Md. Code Ann., Cts. & Jud. Proc. § [ ] and the federal excepted-service appointment authority discussed in Part VI.F.

8. **Juan Jose Fernandez** is an individual residing at [Street address, City, State, ZIP] and is the spouse of Mr. McLaughlin. Mr. Fernandez seeks appointment to a legal-technical support position as described in Exhibit A.

9. No attorney has entered an appearance for either Petitioner. Petitioners request that all filings be served upon them at the addresses stated above.

B Respondents

10. **The Bank of New York Mellon** is a New York banking corporation with its principal office at [principal office address — verify]. It is named as a Respondent in its capacity as [trustee / successor trustee / indenture trustee — verify and specify the instrument by date and parties].

11. **Delaware Trust Company** is a Delaware trust company with its principal office at [principal office address — verify]. It is named as a Respondent in its capacity as [successor trustee / trustee — verify].

12. **Centrus Energy Corp.**, formerly known as USEC Inc., is a Delaware corporation with its principal office at [principal office address — verify]. It is named as a Respondent in its capacity as [issuer / obligor / successor by reorganization — verify].

13. **Computershare Trust Company, N.A.** is a federally chartered trust company with its principal office at [principal office address — verify]. It is named as a Respondent in its capacity as [transfer agent / registrar / successor transfer agent — verify].

14. The precise capacity of each Respondent, the identity of the appointing authority for the Schedule A and Schedule C positions at issue, and the instrument or agreement giving rise to the alleged ministerial duties are matters of record that Petitioners will establish by the exhibits listed at the end of this Petition. Where a capacity is stated above in brackets, that capacity must be completed before filing.

III Jurisdiction and Venue

15. **Subject-matter jurisdiction.** The Circuit Court for Baltimore County has jurisdiction over this action under Md. Code Ann., Cts. & Jud. Proc. § 1-501, which confers upon the circuit courts of this State full common-law and equity jurisdiction, and under Md. Code Ann., Cts. & Jud. Proc. § 1-502. Jurisdiction to issue a writ of mandamus in a non-administrative matter

is governed by Maryland Rule 15-701. Administrative mandamus, which is not sought here, is governed by Maryland Rules 7-401 through 7-403.

16. Declaratory jurisdiction. Jurisdiction to grant declaratory relief is conferred by Md. Code Ann., Cts. & Jud. Proc. §§ 3-401 *et seq.*, and in particular §§ 3-406 and 3-409(c). A court may grant declaratory relief with or without coercive relief, and may grant declaratory relief whether or not a writ of mandamus issues. Md. Code Ann., Cts. & Jud. Proc. § 3-409(c).

17. Venue. Venue is proper in Baltimore County under Md. Code Ann., Cts. & Jud. Proc. § 6-201(a), because [Respondent(s)] carr[y/ies] on a regular business in Baltimore County, and under Md. Code Ann., Cts. & Jud. Proc. § 6-201(b), because there is more than one Respondent, at least one of whom could be sued in Baltimore County, and because the cause of action arose in Baltimore County. In the alternative, venue is proper under Md. Code Ann., Cts. & Jud. Proc. § 6-201(b) because a substantial part of the events giving rise to the claims occurred in Baltimore County at Petitioner McLaughlin's residence and place of business.

18. Prepayment of costs. Petitioners are unable to pay the costs of this proceeding and intend to file a Request for Waiver of Prepayment of Costs under Md. Rule 1-325. See also Md. Code Ann., Cts. & Jud. Proc. § 7-201. Petitioners request that the clerk accept this Petition for filing upon submission of the Request, and that the Court rule upon the Request before requiring any payment.

IV Statement of Facts

A The Instruments and the Alleged Trust Chain

19. Petitioner McLaughlin is, on information and belief, the holder and/or beneficiary of record of interests recorded in [name of the instrument or account], administered by one or more of Respondents. The instrument(s) at issue are: [describe each indenture, trust agreement, transfer-agent agreement, or account agreement by date, parties, and governing law].

20. Under those instruments, the trustee, successor trustee, indenture trustee, and/or transfer agent named therein owes duties of record-keeping, acknowledgment, registration of transfer, accounting, and transmittal. Those duties are, Petitioners contend, ministerial as to the specific acts identified in Part VI.A.

21. On information and belief, the relevant corporate relationships are as follows: [state, for each Respondent, the executed instrument and the succession history by date]. Respondent Centrus Energy Corp. is the successor by reorganization to USEC Inc. [verify].

B Petitioner McLaughlin's Proceedings and Status

22. Mr. McLaughlin has proceeded *pro se* in [number] matters in this State and in the federal courts, including [list case numbers and courts]. Five consolidated petitions for a writ of

certiorari and a related petition for a writ of mandamus and prohibition were filed in the Supreme Court of the United States on or about September 6, 2026. A related proceeding is pending under Case No. [24-C-26-003970] in the Circuit Court for Baltimore City.

23. Mr. McLaughlin has been granted leave to proceed *in forma pauperis* in [identify courts], and that status is a matter of record. He asserts a documented neurodevelopmental disability affecting executive function and auditory processing under conditions of stress, and requests written instructions and reasonable accommodation in these proceedings. See Exhibit [ ]; 42 U.S.C. § 12131 *et seq.*; 28 C.F.R. § 35.130(b)(7).

C The Proposed Introduction of Petitioners Under a Larger Implementation

24. Petitioners have proposed, and seek to implement, an integrated community-infrastructure and community-resilience program. That program is described in the materials attached as Exhibit [ ] and is referred to in this Petition as the "Integrated Implementation." The Integrated Implementation includes, among other components, [brief neutral description of the components that Petitioners intend to pursue, e.g., a community resilience and housing component, a records-and-trust administration component, and a personnel component].

25. A necessary personnel component of the Integrated Implementation is the placement of two qualified individuals — Petitioners McLaughlin and Fernandez — into technical and legal-technical support roles. Petitioners contend that they are qualified for the positions described in Exhibit A and that the appointment packages for those positions are complete.

26. On or about October 21, 2026, a hiring and onboarding schedule was prepared proposing the retention of both Petitioners in specialized technical and legal support roles: Mr. McLaughlin as a Systems Help Desk Lead / Technical Specialist, and Mr. Fernandez as a Jurisprudential Technician / Legal IT Coordinator. See Exhibit A.

27. Petitioners do not contend that they are entitled to appointment as a matter of right. They contend only that the processing, acknowledgment, transmittal, and final consideration of completed appointment packages is a non-discretionary act, and that the failure to perform that act is remediable by mandamus.

D The Failure to Act Giving Rise to This Petition

28. On or about [date], Petitioners submitted to Respondents a written request for [describe the ministerial act demanded, e.g., acknowledgment of record, registration of transfer, an account of the trust, transmittal of a completed appointment package]. A copy is attached as Exhibit [ ].

29. Respondents have failed to perform the requested act, and have failed to respond within [number] days, notwithstanding [number] written follow-ups dated [dates]. See Exhibits [ ]–[ ].

30. On or about [date], Petitioners submitted the completed Schedule A and Schedule C appointment packages to the appropriate appointing authority. The packages have not been processed to final consideration. See Exhibit A.

31. No Respondent has asserted, to Petitioners' knowledge, any ground of substantive objection to the requests; the failure is, Petitioners contend, a failure to perform a ministerial act of record, not a reasoned exercise of discretion.

V The Governing Legal Standard for a Ministerial Writ of Mandamus

32. A writ of mandamus is an extraordinary remedy, and it is not issued as a matter of course. Maryland Rule 15-701. It is available to compel the performance of a ministerial duty, but it is not available to control or direct the exercise of discretion committed to the respondent.

33. To obtain a writ of mandamus, a petitioner must show (1) that the petitioner has a clear legal right to the relief sought; (2) that the respondent owes a clear legal duty to perform the act demanded; and (3) that the petitioner has no other adequate remedy at law. *Brack v. Wells*, 184 Md. 86, 90–91, 40 A.2d 319, 321 (1944); *Philip Morris Inc. v. Angeletti*, 358 Md. 689, 712, 752 A.2d 200, 212 (2000).

34. A duty is ministerial when it is a duty the respondent is required to perform upon a given state of facts, in a prescribed manner, in obedience to a mandate of legal authority, without regard to the respondent's own judgment upon the propriety of the act to be done. Conversely, where the respondent is vested with discretion to decide whether, and in what manner, to act, mandamus does not lie. *Falls Rd. Cmty. Ass'n v. Baltimore County*, 439 Md. 1 (2014) (declining to issue a writ of mandamus where the challenged decision was committed to official discretion).

35. A circuit court may order an agency or official to take an action that the officer's office requires the officer to take, but may not order an action that is discretionary. The writ may also be used to correct a public record. See Md. Courts, *Mandamus* (describing conventional and administrative mandamus).

36. Because Petitioners seek relief of the conventional (non-administrative) kind, Maryland Rules 7-401 through 7-403, which govern administrative mandamus and judicial review of quasi-judicial agency orders, do not supply the procedural framework here.

VI Argument

A The Acts Sought Are Ministerial, Not Discretionary

37. The acts for which Petitioners seek relief are acts of record: acknowledgment of a document, entry of a record, registration of a transfer, production of an account, and transmittal of a

completed package. Each of these acts is, Petitioners contend, required to be performed upon presentation of a completed and conforming request, in a prescribed manner, without regard to the Respondent's own judgment upon the propriety of the act. That is the definition of a ministerial duty.

38. Petitioners do not ask this Court to direct any Respondent to approve, deny, or grant anything. Petitioners ask only that the act of *processing and responding* — an act from which the exercise of judgment is separate — be compelled.

B Petitioners Have a Clear Legal Right

39. Petitioners' right to performance is, they contend, clear and not dependent on any contested issue of fact reserved for adjudication. The right arises from [the instruments] and from [the applicable appointment regulations]. See Exhibit [ ].

C Petitioners Have No Adequate Remedy at Law

40. An ordinary action at law for damages would not produce the specific record act that Petitioners require, and would not cure the structural impediment to the Integrated Implementation. Declaratory relief alone, without coercion, would not secure performance. Petitioners therefore contend that no other adequate remedy exists.

D Count I — Mandamus to Compel Ministerial Acts

41. Petitioners incorporate the foregoing paragraphs and seek a writ of mandamus compelling each Respondent, to the extent it is the proper party, to perform the specific ministerial acts identified in Exhibit [ ], including: (i) to acknowledge receipt of Petitioners' conforming request; (ii) to enter and maintain the corresponding record; (iii) to produce an account of the interests at issue; and (iv) to transmit the completed package to the appropriate appointing authority.

E Count II — Declaratory Judgment

42. Petitioners seek a declaration under Md. Code Ann., Cts. & Jud. Proc. §§ 3-401 *et seq.* determining: (i) the identity of the proper party for each ministerial act; (ii) whether the tendered requests and packages are legally sufficient and conforming; and (iii) the respective rights and obligations of the parties with respect to the acts described in Count I.

F Count III — Mandamus to Compel Processing of the Schedule A and Schedule C Packages

43. Petitioners further seek a writ of mandamus compelling the completion of final consideration of the Schedule A and Schedule C appointment packages described in Exhibit A.

44. The appointment authorities. Schedule A of the excepted service, 5 C.F.R. § 213.3102, including the authority for the appointment of persons with disabilities at 5 C.F.R. § 213.3102(u), permits appointment without competitive examination. Schedule C, 5 C.F.R. §§ 213.3301–213.3302, governs positions of a confidential or policy-determining character. Both are excepted-service appointing authorities. Petitioners contend that Mr. McLaughlin is eligible for consideration under 5 C.F.R. § 213.3102(u), and that the positions described in Exhibit A may be filled under Schedule A or Schedule C as the appointing authority directs. See also 5 U.S.C. §§ 2301, 2302 (merit system principles and prohibited personnel practices).

45. The limited scope of the relief sought. Petitioners acknowledge that appointment is committed to the discretion of the appointing authority and that no provision of law confers upon them a right to be appointed. Petitioners seek only the ministerial act of processing, acknowledging, and bringing the completed packages to final consideration within a time certain.

46. Reservation. To the extent this Court concludes that it lacks authority to compel any part of the processing described in this Count because the appointing authority is not before the Court or because the act is committed to federal discretion, Petitioners request that this Court so state in a written order, so that the question may be preserved for review and, if appropriate, transferred or refiled in the correct forum.

G The Integrated Implementation

47. The personnel component described in Count III is one component of the Integrated Implementation described in Part IV.C. Petitioners seek a declaration under Count II that the personnel component is not an unlawful component of the Integrated Implementation, and that the ministerial acts required to advance it may proceed.

VII Request for an Order to Show Cause and a Hearing

48. Petitioners request that the Court issue an order requiring each Respondent to show cause, within a time fixed by the Court, why the writ should not issue, and that the Court set this matter for a hearing. A proposed form of order to show cause is attached as Exhibit B. Petitioners request that the hearing be conducted with the accommodations described in Part IV.B.

VIII Prayer for Relief

WHEREFORE, Petitioners respectfully request that this Court:

(a) issue a writ of mandamus compelling each Respondent, to the extent it is the proper party, to perform the ministerial acts identified in Count I and Exhibit [ ];

(b) issue a writ of mandamus compelling the processing and final consideration of the Schedule A and Schedule C appointment packages described in Count III and Exhibit A;

(c) enter a declaratory judgment under Md. Code Ann., Cts. & Jud. Proc. §§ 3-401 *et seq.* determining the parties' rights and obligations as requested in Count II;

(d) grant a waiver of the prepayment of costs under Md. Rule 1-325;

(e) set this matter for a hearing on an order to show cause; and

(f) grant such other and further relief as the Court deems just and proper.

Respectfully submitted,

_____

Caustin Lee McLaughlin
1000 North Woodington Road
Baltimore, Maryland 21229
Petitioner, Pro Se

_____

Juan Jose Fernandez
[Street address]
[City, State, ZIP]
Petitioner, Pro Se

Dated: _____ day of _____, 2026.

IX Verification

I, _____, being first duly sworn, do hereby verify that the facts stated in the foregoing Petition for Writ of Mandamus and Complaint for Declaratory Judgment are true and correct to the best of my knowledge, information, and belief, except as to matters stated upon information and belief, which I believe to be true.

_____

Caustin Lee McLaughlin

_____

Juan Jose Fernandez

State of Maryland, County of Baltimore

Subscribed and sworn to before me this

_____ day of _____, 2026.

_____

Notary Public

My commission expires:

_____

X Certificate of Service

I hereby certify that on this _____ day of _____, 2026, a copy of the foregoing Petition for Writ of Mandamus (Ministerial) and Complaint for Declaratory Judgment was served upon each Respondent by [certified mail, return receipt requested / personal service / the Court's electronic filing system] at the address of its registered agent for service of process in this State, as follows:

Respondent	Service address / registered agent
The Bank of New York Mellon	[address — verify]
Delaware Trust Company	[address — verify]
Centrus Energy Corp.	[address — verify]
Computershare Trust Company, N.A.	[address — verify]

_____

Caustin Lee McLaughlin, Petitioner, Pro Se

Exhibit Index

Exhibit	Description
A	Schedule A / Schedule C Hiring Implementation Matrix for Petitioners McLaughlin and Fernandez (see page after next).
B	Proposed Order to Show Cause (see final page).
C	[Clinical and genomic records supporting Petitioner McLaughlin's asserted disability status.]
D	[The instrument(s), indenture(s), trust agreement(s), or account agreement(s) at issue.]
E	[Petitioners' written request for the ministerial act, with proof of submission.]

Exhibit	Description
F	[Respondents' responses or notices of non-response.]
G	[Description of the Integrated Implementation.]

Exhibit A — Schedule A / Schedule C Hiring Implementation Matrix

49. This Exhibit states the positions for which Petitioners seek processing and final consideration. The appointing authority and the specific series and grade must be completed before filing.

Candidate	Position title	Appointing authority	Series / grade
McLaughlin	Systems Help Desk Lead / Technical Specialist	Sch. A, 5 C.F.R. § 213.3102(u)	[Series/Grade]
Fernandez	Jurisprudential Technician / Legal IT Coordinator	Sch. A, 5 C.F.R. § 213.3102, or Sch. C, 5 C.F.R. § 213.3301	[Series/Grade]

Phase	Milestone	Target date
Phase 1	Offer acceptance and credentialing	Week 1
Phase 2	Background and suitability processing	Weeks 2–3
Phase 3	Systems access and onboarding	Weeks 4–5
Phase 4	Integration into the Integrated Implementation	Weeks 6–8

Source: Petitioners' hiring and onboarding schedule dated October 21, 2026. The duties, series, grade, and appointing authority for each position must be confirmed by the appointing authority before filing.

Exhibit B — Proposed Order to Show Cause (Proposed Only)

PROPOSED ORDER

(This is a proposed form. It has no effect unless and until signed by a judge of this Court.)

Upon consideration of the Petition for Writ of Mandamus (Ministerial) and Complaint for Declaratory Judgment filed by Petitioners Caustin Lee McLaughlin and Juan Jose Fernandez, it is, this _____ day of _____, 2026,

ORDERED that Respondents The Bank of New York Mellon; Delaware Trust Company; Centrus Energy Corp. (f/k/a USEC Inc.); and Computershare Trust Company, N.A., shall show cause, if any they have, on or before _____, why the relief requested in the Petition should not be granted; and it is further

ORDERED that this matter is set for hearing on _____, at _____ a.m./p.m., in Courtroom _____ of the Circuit Court for Baltimore County, Maryland.

_____

Judge, Circuit Court for Baltimore County

Part VI

Bachelor's Degree Completion Portfolio

A Prior-Learning Portfolio Mapped to a Standard 120-Credit Baccalaureate

Bachelor's Degree Completion Portfolio

A Prior-Learning Portfolio Mapped to a Standard 120-Credit Baccalaureate

Area of Study: Constitutional Justice, Data Sovereignty & Applied Cryptographic Systems

Caustin Lee McLaughlin

October 2026

Companion to: *Rights, Genomes, and Algorithms — Consolidated Edition*, Parts I–V. This volume is the sixth part of that consolidated book and converts the author's body of written work into a course-by-course, credit-bearing academic portfolio.

Authority notice. This document is a *prior-learning portfolio*. It records, course by course, the college-level learning evidenced by the author's work and proposes an academic equivalent for each course. It does **not** confer credit and *no degree or credit is awarded by this document.* Academic credit and any degree are conferred solely by an institutionally accredited college or university after faculty assessment. See Section 3.

1 Purpose and Use of This Volume

This volume answers a single question in operational form: *if the work gathered in Parts I–V were submitted for academic assessment, what record of college-level learning would it present, and against what degree structure?*

It is organised as a standard United States baccalaureate transcript would be: general education, an area of study (major), free electives, and a senior capstone, totalling 120 semester credits. For every proposed course the portfolio supplies four things — the course's learning outcomes, the specific evidence in this book that demonstrates them, the prior-learning basis (how the knowledge was acquired), and the assessment method a faculty reviewer would use to verify it.

The volume is written to be submitted, in whole or in part, to a college's credit-for-prior-learning process. It is deliberately conservative: where a proposition in Parts I–V is an argument rather than a computation, the portfolio claims the learning of *making* the argument, not the truth of the conclusion. The four-register discipline of the monograph (adopted law, proposal, model/simulation, design target) is carried through here so that a reviewer can see exactly what kind of learning each course documents.

2 The 120-Credit Baccalaureate Framework

2.1 The unit of credit

A "credit hour" is not a measure of time served but of *intended learning outcomes verified by evidence of student achievement*, defined in federal regulation at 34 C.F.R. § 600.2 and consistent with the Carnegie unit. One semester credit hour represents, at minimum, one hour of instruction plus two hours of out-of-class work per week for a fifteen-week semester, or an institutionally established equivalent for laboratory, studio, and independent work. This portfolio is written in those terms: each course states outcomes and evidence, from which credit equivalents are proposed.

2.2 The degree

Every United States baccalaureate requires, as its backbone:

- a minimum of **120 semester credits** (some majors require more);
- at least **40 credits at the upper-division** (3000/4000) level for a Bachelor of Arts;
- completion of **general education** requirements (typically 40–42 credits);
- an approved **major** (commonly 24–36 credits exclusive of general education);
- **free electives** to reach the total;
- a **residence requirement** — the final block of credits (commonly the last 30) earned at the awarding institution; and
- an approved **capstone or culminating experience**.

The allocation used in this portfolio is summarised below and totals exactly 120 credits.

Component	**Credits**
I. General Education Core	42
II. Area of Study (Major): Constitutional Justice, Data Sovereignty & Applied Cryptographic Systems	48
III. Free Electives	24
IV. Senior Integrative Capstone	6
Total	**120**

Of the 120 credits, the master tally in Section 10 identifies 78 as upper-division (3000/4000-level) work, satisfying the 40-credit upper-division rule with margin.

3 Authority, Scope, and Limitations

A portfolio that claims to reach a degree total must be honest about who may grant the degree. Four limits govern this volume.

1. **Credit is awarded by an accredited institution, not by this document.** Prior-learning credit is awarded only when the learning is "academically verifiable" through an approved assessment methodology and is judged by faculty subject-matter experts, whose decisions are final. This portfolio is the input to that judgment, never a substitute for it.

2. **Prior-learning credit is capped.** Under the WASC Senior College and University Commission (WSCUC) policy and the policies of most institutions, *no more than 25–30% of the credits required for a degree* may be earned by prior learning. For a 120-credit baccalaureate that is roughly **30–38 credits** — not 120. Some institutions set lower limits (for example, portfolio credit capped at 15 credits). Section 12 therefore separates the portfolio's *PLA-eligible* credits from the credits that must be earned *in residence*.

3. **The residence requirement cannot be met by prior learning.** The final block of study (commonly the last 30 credits) and a minimum of the major must be completed at the awarding institution. No portfolio can shorten that.

4. **Other formats exist and may suit the author better.** Standardised and departmental challenge examinations, validated proficiency assessments, and externally evaluated credit (e.g. American Council on Education recommendations) may convert some of this learning more cheaply and with greater transferability than a portfolio. Section 12 recommends the least-cost, most-transferable route for each block.

Nothing in Parts I–V should be read as a claim that a degree has been, or will be, conferred by any institution; the court filings and the trust framework are practice documents and advocacy, and their inclusion here is as *evidence of learning*, not as authority.

4 Portfolio Methodology: How Credit Is Justified

The portfolio converts artefacts into credit by a three-step method repeated for every course.

1. Outcomes. Each proposed course states the learning outcomes a comparable catalogued course would carry, phrased as *demonstrable* capabilities ("analyze," "compute," "draft," "evaluate") rather than topics.

2. Evidence. Each outcome is tied to a named artefact in Parts I–V — a chapter, a specification, a computation, a filed document — so a reviewer can read the work that produced the learning.

3. Register and assessment. Each course records the register(s) of statement the evidence belongs to (adopted law / proposal / model–simulation / design target) and the assessment method by which faculty would verify it (portfolio review, artifact inspection, oral defence, or examination).

Where the same artefact supports more than one course, the portfolio states so and does not double-count the credit: an artefact may evidence two courses, but the credit hours are allocated once to a primary course and once, at most, to a secondary course at reduced weight — never twice at full weight. Section 10 reconciles the total.

5 Credit Distribution Summary

The 120 credits are distributed as follows across the four components and the general-education outcome areas.

Component	**Outcome area**	**Cr.**
General Education Core	Written & Oral Communication	9
	Quantitative Reasoning & Statistics	6
	Natural Sciences with Laboratory	7
	Social & Behavioral Sciences	9
	Humanities & Ethics	3
	Information Literacy, Diversity & Global Studies	6
	Applied Science & Sustainability	2
	subtotal	*42*
Area of Study (Major)	Constitutional, Civil-Rights & Administrative Law	12
	Scientific Evidence & Genomic Privacy	6
	Computing, Algorithms & Machine Learning	9
	Cryptography, Audit & Trust Finance	21
	subtotal	*48*
Free Electives	Interdisciplinary & Professional	24
Senior Integrative Capstone	Integrative Research	6
Total		**120**

Each general-education course is counted in exactly one outcome area, so the seven general-education rows sum to 42 credits and the four area-of-study rows sum to 48. Section 10 is the authoritative course-by-course tally.

6 Component I — General Education Core (42 credits)

English Composition I — 3 credits

GE-101

Learning outcomes. Write clear, well-organised expository prose; control thesis, paragraph, and evidence; revise for clarity and correctness across drafts. **Portfolio evidence.** The abstract, status-and-provenance notice, and reader's guide of the monograph (Part I); the abstracts of Parts II–V. **Prior-learning basis.** The author composed, over a decade, several hundred pages of expository and argumentative prose across pleadings, reports, and technical notes. **Assessment.** Portfolio review of three submitted compositions with revision history.

English Composition II: Research & Argument — 3 credits

GE-102

Learning outcomes. Formulate a research question; locate, evaluate, and synthesise primary sources; build a sourced argument with a defensible citation apparatus; distinguish assertion from authority. **Portfolio evidence.** The sourced argument of Part I (its doctrine-by-doctrine map, its register discipline, and its consolidated bibliography); the citation templates of Part V, Appendix B. **Prior-learning basis.** Constructing a book-length argument from primary authority (statutes, regulations, decided cases, NIST standards) and maintaining a verifiable citation apparatus. **Assessment.** Portfolio review plus a timed source-integration task.

Public Speaking & Oral Advocacy — 3 credits

GE-103

Learning outcomes. Organise and deliver a persuasive oral presentation; adapt argument to an adversarial audience; answer questions under pressure. **Portfolio evidence.** The relief-sought sections and argument structure of the two petitions (Parts IV–V); the checklist's discipline of "what can be filed." **Prior-learning basis.** Preparation of pro-se filings requires the same structure as oral advocacy: issue, rule, application, relief. **Assessment.** Recorded oral presentation with rubric; or waiver by submission of a written advocacy brief.

College Algebra & Quantitative Reasoning — 3 credits

GE-104

Learning outcomes. Manipulate algebraic expressions; model real quantities; reason about rates, proportions, and constraints. **Portfolio evidence.** The perpetuity constraint $D(t) \le 0.05\,\bar{C}_5(t)$ and tier-segregation algebra of Part II, § 3–4; the priority-score function of Part I, Ch. 46. **Prior-learning basis.** Deriving and applying constraints and scoring functions in the trust and triage models. **Assessment.** Proctored quantitative-reasoning examination.

Statistics & Probability — 3 credits

GE-105

Learning outcomes. Apply probability and statistical inference; interpret distributions; reason about uncertainty and error. **Portfolio evidence.** The probabilistic control layer of the HSPA v3.0 specification; the mortality- and severity-probability terms of the MORTAR score (Part I, Ch. 46); the LSH collision reasoning citing Gionis et al. and Datar et al. **Prior-learning basis.** Specifying and reasoning about probabilistic estimators and sampling in the triage engine. **Assessment.** Proctored statistics examination plus review of the modelling artefacts.

General Biology with Laboratory — 4 credits

GE-106

Learning outcomes. Explain cell and molecular biology and the basis of heredity; interpret laboratory and genomic data; connect genotype to phenotype and to risk. **Portfolio evidence.** The genomic-privacy analysis of Part I (bio-banking, polygenic risk, and the statutory lattice of GINA). *(Laboratory component: see limitation below.)* **Prior-learning basis.** Study of the molecular-biology and genetic-privacy literature required to argue genomic data-sovereignty claims. **Assessment.** Examination on genetics and heredity. *Note: the "with laboratory" designation requires a hands-on laboratory, which prior learning generally cannot satisfy; the portfolio proposes 3 lecture credits and refers the 1 laboratory credit to residence or an approved kit-based lab.*

Neuroscience & the Scientific Method — 3 credits

GE-107

Learning outcomes. Describe the physiology of the nervous system; explain how neuroscientific measurement is used as evidence; apply evidentiary standards to scientific proof claims. **Portfolio evidence.** The neurophysiological-data analysis and the "neurogenetic biomarkers and scientific evidence" chapter of Part I; the accommodation and disability-evidence analysis. **Prior-learning basis.** Researching neurogenetic proof and its admissibility under FRE 702 and *Daubert*. **Assessment.** Examination plus portfolio review of the evidentiary analysis.

American Government & Constitutional Law — 3 credits

GE-108

Learning outcomes. Explain the structure of American government; analyse constitutional text and doctrine; apply § 1983 and related doctrine to a fact pattern. **Portfolio evidence.** Part I's constitutional-jurisprudence part; the doctrinal map of actionable claims under 42 U.S.C. § 1983. **Prior-learning basis.** Building a constitutional argument from primary authority. **Assessment.** Examination plus portfolio review.

Principles of Economics & Public Finance — 3 credits

GE-109

Learning outcomes. Explain micro- and macroeconomic principles; analyse public finance, capital allocation, and public program cost. **Portfolio evidence.** The DYREO allocation model, tier segregation, and $500M corpus reconciliation of Part II; the public-financing framework of Part I. **Prior-learning basis.** Designing and reconciling a quantitative capital-allocation model against a perpetual-distribution constraint. **Assessment.** Examination plus review of the allocation model.

Ethics & the Philosophy of Law — 3 credits

GE-110

Learning outcomes. Compare normative theories; analyse the relation of law, morality, and legitimacy; reason about professional and institutional duty. **Portfolio evidence.** The normative-vs-engineering claim discipline of Part I's status notice; the conflict-of-interest firewall and fiduciary governance of Part II, § 4. **Prior-learning basis.** Sustained argument about what institutions owe to persons whose data they hold. **Assessment.** Essay examination; portfolio.

Information Literacy & Digital Ethics — 3 credits

GE-111

Learning outcomes. Locate and evaluate information; manage data and sources responsibly; reason about privacy, consent, and algorithmic ethics. **Portfolio evidence.** The register discipline and sourcing method of Part I's introduction; the algorithmic-governance analysis; the non-delegation guard-rail (an algorithm may draft but not adjudicate). **Prior-learning basis.** Maintaining a source-and-register discipline across a multi-hundred-page project. **Assessment.** Portfolio review plus a source-evaluation task.

International Human Rights & Global Studies — 3 credits

GE-112

Learning outcomes. Explain the international human-rights framework; compare national and international mechanisms; draft for an international body. **Portfolio evidence.** The international human-rights framework of Part I; the proposed United Nations commission of inquiry and the OHCHR/ECtHR forum analysis of Part II. **Prior-learning basis.** Frameworking cognitive-accessibility, genomic-justice, and quantum-secure infrastructure into a single directive. **Assessment.** Examination plus review of the treaty-framework draft.

Research Methods for the Social Sciences — 3 credits

GE-113

Learning outcomes. Design a research question and method; reason about validity, reliability, and evidence; assemble and document a record. **Portfolio evidence.** The "Sources and Method" note of Part I; the filing-readiness checklist and record-assembly order of Part V, Appendices A and C. **Prior-learning basis.** Designing an evidentiary record and its assembly order for litigation. **Assessment.** Portfolio review; research-design exercise.

Environmental Science & Sustainable Energy — 2 credits

GE-114

Learning outcomes. Explain energy systems and sustainability; assess the environmental and feasibility dimensions of an energy proposal. **Portfolio evidence.** The solar-and-battery microgrid specification and green-retrofit/lead-abatement scope of Part II (a *design target*, per Part I's register discipline). **Prior-learning basis.** Scoping and costing an energy and retrofit program. **Assessment.** Portfolio review.

7 Component II — Area of Study (48 credits)

Area of study: Constitutional Justice, Data Sovereignty & Applied Cryptographic Systems.

Constitutional Law & Federal Jurisdiction — 3 credits

CJ-201

Learning outcomes. Analyse federal-question and jurisdiction doctrines; apply qualified immunity, standing, and state-action analysis; argue a constitutional claim. **Portfolio evidence.** Part I, constitutional-jurisprudence part; the relief sought in Parts IV–V (28 U.S.C. § 1651(a); Fed. R. App. P. 21). **Prior-learning basis.** Preparing a mandamus petition and a constitutional complaint from primary authority. **Assessment.** Examination; review of the petitions as artefacts.

Civil Rights & Disability Law — 3 credits

CJ-202

Learning outcomes. Apply the ADA, the Rehabilitation Act, and GINA; analyse the administrative-mortality problem; construct a disability claim. **Portfolio evidence.** The disability-appeal mortality accounting (citing GAO) and the statutory lattice of Part I; the disability-empowerment projects of Part II (P4, P11). **Prior-learning basis.** Analysing systemic disability injustice and its statutory remedies. **Assessment.** Examination; portfolio review.

Administrative Law & the APA — 3 credits

CJ-203

Learning outcomes. Explain agency adjudication and rulemaking; evaluate finality and exhaustion; analyse the non-delegation and due-process constraints on automated decision-making. **Portfolio evidence.** Part I's administrative-law analysis; the MORTAR non-delegation guard-rail; the SSA-backlog reduction program (Part II, P11). **Prior-learning basis.** Analysing the limits of a private algorithm inside a governmental function. **Assessment.** Examination; portfolio review.

Genetics, Genomic Privacy & the Law — 3 credits

CJ-204

Learning outcomes. Explain the biology of genomic data and its legal protection; analyse bio-banking, consent, and re-identification risk; apply GINA and HIPAA. **Portfolio evidence.** Part I's genomic-data-sovereignty and biological-data-sovereignty analysis. **Prior-learning basis.** Research at the intersection of genetics, privacy, and statute. **Assessment.** Examination; portfolio review.

Scientific & Neurogenetic Evidence — 3 credits

CJ-205

Learning outcomes. Apply FRE 702 and *Daubert*; evaluate biomarker and polygenic-risk evidence; distinguish validated measurement from modelled output. **Portfolio evidence.** The neurogenetic-biomarker chapter and the register discipline of Part I; the explicit prohibition (Part V, Appendix A, item 13) on describing simulated output as validated cryptography. **Prior-learning basis.** Applying evidentiary standards to scientific proof. **Assessment.** Examination; artifact-based oral defence.

Algorithms & Data Structures — 3 credits

CJ-206

Learning outcomes. Analyse algorithmic complexity; implement and evaluate hashing and retrieval methods; reason about collision and approximation. **Portfolio evidence.** The MORTAR $O(n \log m)$ locality-sensitive-hashing priority scoring (Part I, Ch. 46; Gionis et al.; Datar et al.). **Prior-learning basis.** Specifying and analysing a sublinear retrieval algorithm. **Assessment.** Proctored algorithm examination; code-artifact inspection.

Semantic Computing & Language Systems — 3 credits

CJ-207

Learning outcomes. Explain computational-semantic architectures; analyse transformer-based language models; reason about a probabilistic control layer. **Portfolio evidence.** The Hybrid Semantic Processing Architecture (HSPA v3.0) — Semantic Execution Graph, probabilistic control layer, and reconfigurable compute fabric — in Part I. **Prior-learning basis.** Designing and documenting a six-layer semantic-computing architecture. **Assessment.** Examination; review of the architecture specification.

Machine Learning & Probabilistic Control — 3 credits

CJ-208

Learning outcomes. Explain supervised and latent-reasoning models; reason about calibration and uncertainty; separate *simulation* from *measurement*. **Portfolio evidence.** The iterative latent-reasoning transformer blocks ($N_{sup} = 16$) and the MORTAR recommendation layer; the [MODEL/SIMULATION] labelling of Part I. **Prior-learning basis.** Specifying a model and maintaining an honest separation between simulation and result. **Assessment.** Examination; artifact-based review.

Cryptography & Post-Quantum Security — 3 credits

CJ-209

Learning outcomes. Apply key-establishment and signature standards; analyse quantum threats; compare ML-KEM, ML-DSA, and SLH-DSA against the NIST FIPS 203/204/205 standards. **Portfolio evidence.** The post-quantum trust chain of Part I; the QRDE-AEAD dual-envelope design; the ML-DSA-87 and SLH-DSA signing described in the OSI-P chain. **Prior-learning basis.** Designing a post-quantum trust chain against published standards. **Assessment.** Examination; artifact-based review.

Distributed Ledgers & Cryptographic Audit — 3 credits

CJ-210

Learning outcomes. Explain hash-linked and directed-acyclic ledgers; construct a Merkle-root audit chain; reason about evidentiary authentication of electronic records. **Portfolio evidence.** The nine-stage OSI-P audit chain (R1–R9) and the Merkle-to-DAG ledger replacement of Part II; the FRE 902(13)–(14) authentication analysis. **Prior-learning basis.** Designing a self-authenticating electronic administrative record. **Assessment.** Examination; artifact-based review.

Trust Law & the Uniform Directed Trust Act — 3 credits

CJ-211

Learning outcomes. Explain trust formation, fiduciary duty, and directed trusts; apply the UDTA; analyse perpetuity and distribution constraints. **Portfolio evidence.** The Perpetual Solvent Trust framework, the four-quadrant governance and conflict-of-interest firewall of Part II, § 4. **Prior-learning basis.** Designing a perpetual, solvent, directed-trust structure. **Assessment.** Examination; portfolio review.

Public Finance & Quantitative Allocation — 3 credits

CJ-212

Learning outcomes. Model capital allocation under constraints; reconcile a corpus and its distributions; evaluate yield engines and their risk controls. **Portfolio evidence.** The DYREO stochastic yield engine, the source-of-funds reconciliation, and the deployable-capital schedule of Part II, § 3 and 5. **Prior-learning basis.** Building and reconciling a quantitative public-finance model. **Assessment.** Examination; artifact-based review.

Public Policy Analysis & Program Design — 3 credits

CJ-213

Learning outcomes. Define a policy problem; design a program with objectives, KPIs, and risk controls; analyse institutional feasibility. **Portfolio evidence.** The integrated policy framework of Part I; the thirteen-project catalog with objective, scope, schedule, KPI, and risk controls (Part II, § 5). **Prior-learning basis.** Designing a governed, auditable program portfolio. **Assessment.** Portfolio review; policy-design exercise.

Cybersecurity & Digital Evidence — 3 credits

CJ-214

Learning outcomes. Assess system integrity and adversarial threats; authenticate digital evidence; apply electronic-record authentication rules. **Portfolio evidence.** The OSI-P tamper-evident chain; the DAG cryptographic audit infrastructure; the FRE 902(13)–(14) admissibility analysis of Part I, Ch. 46. **Prior-learning basis.** Designing an admissible cryptographic audit trail. **Assessment.** Examination; artifact-based review.

Legal Research, Writing & Citation Practice — 3 credits

CJ-215

Learning outcomes. Conduct legal research; draft in a court-required form; construct and verify a citation apparatus. **Portfolio evidence.** The petitions (Parts IV–V), the citation templates (Part V, Appendix B), and the citation-verification checklist (Appendix A, items 18–19). **Prior-learning basis.** Drafting pro-se filings and verifying every citation against its source. **Assessment.** Portfolio review; citation-accuracy task.

Applied Research Project (Major Capstone) — 3 credits

CJ-216

Learning outcomes. Execute an independent research project from question to written result; defend method and findings. **Portfolio evidence.** The applied-projects and capstones part of Part I; the "How the three layers compose" synthesis (Part I, Ch. 46.4). **Prior-learning basis.** Carrying a multi-part, multi-year project to a written conclusion. **Assessment.** Written project report plus oral defence.

8 Component III — Free Electives (24 credits)

Advanced Technical Writing & Documentation — 3 credits

FE-301

Learning outcomes. Write specifications a non-specialist can verify; separate standard, simulated, and target claims in technical prose. **Portfolio evidence.** The HSPA v3.0 master technical specification as incorporated in Part I; the four-register labels throughout. **Prior-learning basis.** Authoring a consolidated engineering specification. **Assessment.** Portfolio review.

Project & Program Management — 3 credits

FE-302

Learning outcomes. Plan a multi-project program; model dependencies and critical path; manage scope, schedule, and risk. **Portfolio evidence.** The DAG of project dependencies, bottom-up dependency ledger, and critical-path analysis of Part II, § 2. **Prior-learning basis.** Managing thirteen interdependent projects as a portfolio. **Assessment.** Portfolio review.

Social Entrepreneurship & Enterprise — 3 credits

FE-303

Learning outcomes. Design a mission-driven enterprise; analyse its governance and funding; evaluate social return. **Portfolio evidence.** The municipal-restructuring initiative (Part II, P13) and the housing-stabilization program (P1). **Prior-learning basis.** Designing a social enterprise with a fiduciary governance structure. **Assessment.** Portfolio review.

Grant Writing & Fund Development — 3 credits

FE-304

Learning outcomes. Write a persuasive funding proposal; build a budget and a source-of-funds narrative. **Portfolio evidence.** The capital vector and source-of-funds reconciliation of Part II, § 3; the pro-se-litigant grant program (P2). **Prior-learning basis.** Writing funded-program proposals and reconciling budgets. **Assessment.** Portfolio review.

Interdisciplinary Studies Seminar — 3 credits

FE-305

Learning outcomes. Synthesise methods from two or more disciplines around one problem; evaluate a cross-disciplinary framework. **Portfolio evidence.** Part I as a whole — law, genetics,

computing, cryptography, and finance joined by one structural claim. **Prior-learning basis.** Sustaining an argument across five disciplines in one volume. **Assessment.** Essay; portfolio defence.

Independent Study: Portfolio Development — 3 credits

FE-306

Learning outcomes. Assemble prior learning into a defensible portfolio; map artefacts to outcomes; document method. **Portfolio evidence.** This volume (Part VI) itself, including its methodology (§ 4) and evidence index (§ 11). **Prior-learning basis.** Designing the assessment instrument by which the other learning is judged. **Assessment.** Portfolio review by faculty.

Comparative Legal Systems — 3 credits

FE-307

Learning outcomes. Compare national and international legal systems; analyse a claim across fora. **Portfolio evidence.** The frameworking of U.S., international (OHCHR/ECtHR), and state fora in Parts I and II. **Prior-learning basis.** Selecting and contrasting fora for a single claim. **Assessment.** Examination; portfolio review.

Digital Media & Public Communication — 3 credits

FE-308

Learning outcomes. Present complex material to a lay audience; design a public-facing communication artefact. **Portfolio evidence.** The consolidated edition's front matter (About This Volume; reader's guide) as public-facing explanation; the PSDLEF web-submission package referenced in Part I. **Prior-learning basis.** Translating a technical and legal project for a general reader. **Assessment.** Portfolio review of the communication artefact.

9 Component IV — Senior Integrative Capstone (6 credits)

Senior Integrative Capstone — 6 credits

CAP-400

Learning outcomes. Integrate the area of study in one original, defended work; demonstrate advanced research, writing, and synthesis; defend the work before faculty. **Portfolio evidence.** Part I, the *Integrated Master Portfolio Edition* monograph (141 pp.), with its methodological contribution — the four registers of statement — as the capstone thesis, supported by Parts II–V as applied components. **Prior-learning basis.** A book-length original synthesis integrating law, genetics, computing, cryptography, and public finance under a single defended claim. **Assessment.**

Capstone report plus oral defence before a faculty panel; the monograph is submitted as the artefact.

10 Consolidated Credit Tally (120 credits)

The table below is the authoritative tally. It lists every proposed course, its credit value, its level, and its primary evidence. Credits sum to exactly 120; of these, 78 are at the upper-division (3000/4000) level.

Code	Course	Cr.	Lvl	Primary evidence
GE-101	English Composition I	3	L	Part I (front)
GE-102	English Composition II: Research & Argument	3	L	Part I; Part V App. B
GE-103	Public Speaking & Oral Advocacy	3	L	Parts IV–V
GE-104	College Algebra & Quantitative Reasoning	3	L	Part II §3–4
GE-105	Statistics & Probability	3	L	Part I Ch.46
GE-106	General Biology with Laboratory	4	L	Part I (genomics)
GE-107	Neuroscience & the Scientific Method	3	L	Part I (neuro)
GE-108	American Government & Constitutional Law	3	L	Part I (constitutional)
GE-109	Principles of Economics & Public Finance	3	L	Part II §3
GE-110	Ethics & the Philosophy of Law	3	L	Part I; Part II §4
GE-111	Information Literacy & Digital Ethics	3	L	Part I (method)
GE-112	International Human Rights & Global Studies	3	L	Part I (human rights)
GE-113	Research Methods for the Social Sciences	3	L	Part I; Part V App. A,C
GE-114	Environmental Science & Sustainable Energy	2	L	Part II (microgrid)
CJ-201	Constitutional Law & Federal Jurisdiction	3	U	Part I; Parts IV–V
CJ-202	Civil Rights & Disability Law	3	U	Part I; Part II P4,P11
CJ-203	Administrative Law & the APA	3	U	Part I; Part II P11
CJ-204	Genetics, Genomic Privacy & the Law	3	U	Part I (genomics)
CJ-205	Scientific & Neurogenetic Evidence	3	U	Part I (neuro)
CJ-206	Algorithms & Data Structures	3	U	Part I Ch.46
CJ-207	Semantic Computing & Language Systems	3	U	Part I (HSPA)
CJ-208	Machine Learning & Probabilistic Control	3	U	Part I (HSPA)

Code	Course	Cr.	Lvl	Primary evidence
CJ-209	Cryptography & Post-Quantum Security	3	U	Part I (PQC)
CJ-210	Distributed Ledgers & Cryptographic Audit	3	U	Part I Ch.46; Part II
CJ-211	Trust Law & the Uniform Directed Trust Act	3	U	Part II §4
CJ-212	Public Finance & Quantitative Allocation	3	U	Part II §3,§5
CJ-213	Public Policy Analysis & Program Design	3	U	Part I; Part II §5
CJ-214	Cybersecurity & Digital Evidence	3	U	Part I Ch.46
CJ-215	Legal Research, Writing & Citation Practice	3	U	Parts IV–V
CJ-216	Applied Research Project (Major Capstone)	3	U	Part I (applied)
FE-301	Advanced Technical Writing & Documentation	3	U	Part I (spec)
FE-302	Project & Program Management	3	U	Part II §2
FE-303	Social Entrepreneurship & Enterprise	3	U	Part II P13,P1
FE-304	Grant Writing & Fund Development	3	U	Part II §3
FE-305	Interdisciplinary Studies Seminar	3	U	Part I (whole)
FE-306	Independent Study: Portfolio Development	3	U	Part VI (this vol.)
FE-307	Comparative Legal Systems	3	U	Parts I–II
FE-308	Digital Media & Public Communication	3	U	Part VI front matter
CAP-400	Senior Integrative Capstone	6	U	Part I (monograph)
	Total	**120**		**78 upper-division**

10.1 Degree-requirement audit

The table below checks the portfolio against the seven baccalaureate requirements stated in Section 2, showing requirement by requirement what satisfies each one and what a portfolio cannot satisfy.

Requirement (Section 2)	Satisfied by	Cr.
Minimum 120 semester credits	This portfolio, proposed tally	120
At least 40 upper-division credits	3000/4000-level courses (CJ, FE, CAP)	78
General education (typically 40–42)	Component I (GE-101–GE-114)	42
Approved major (commonly 24–36)	Component II (see note)	48
Free electives to reach the total	Component III	24
Residence requirement (final block, commonly 30)	*Not satisfiable by prior learning*	75
Capstone / culminating experience	CAP-400	6

Note on the major. Many institutions cap an area of study at 36 credits exclusive of general education. Where that is the case, the 48-credit Component II is reorganised as 36 credits of major plus 12 credits moved into free electives (raising Component III from 24 to 36); the 120-credit total is unchanged. *Note on residence.* The 75 credits shown against the residence requirement are the balance of the 120 that prior learning typically cannot carry (Section 12); they must be earned at the awarding institution or by approved examination.

11 Evidence Index — Portfolio Artefacts to Courses

Artefact (location in the consolidated book)	Courses evidenced	Cr.
Monograph, constitutional & civil-rights parts (Part I)	GE-108, CJ-201, CJ-202, CJ-203	12
Monograph, genomic & neuro parts (Part I)	GE-106, GE-107, CJ-204, CJ-205	13
HSPA v3.0 specification (Part I)	CJ-207, CJ-208	6
Post-quantum trust chain & OSI-P (Parts I–II)	CJ-209, CJ-210, CJ-214	9
MORTAR LSH engine (Part I Ch.46)	GE-105, CJ-206	6
Perpetual Solvent Trust & DYREO (Part II)	GE-109, CJ-211, CJ-212	9
Thirteen-project catalog (Part II)	CJ-213, FE-302, FE-303, FE-304	12
Petitions (Parts IV–V)	GE-103, GE-113, CJ-215	9
Four-register method (Part I front)	GE-102, GE-110, GE-111, FE-305	12
This portfolio (Part VI)	FE-306	3
Monograph, whole (Part I)	CAP-400	6

12 Assessment and Submission Plan

12.1 What a reviewer would do

A faculty assessor would compare each course's stated outcomes to the named artefact, confirm that the learning is college-level and current, and award credit where the artefact demonstrates the outcome. Under the assessment policies surveyed in Section 3, decisions rest with faculty or designated subject-matter experts and are final.

12.2 PLA-eligible versus residence credit

Because prior-learning credit is capped (commonly 25–30% of the degree, i.e. 30–38 credits of a 120-credit degree), the portfolio should be submitted selectively. The recommended split:

Block	**Credits**
Upper-division area-of-study work best suited to PLA (CJ-209–CJ-214, CJ-216)	21
Free electives evidenced by program and portfolio artefacts (FE-302–FE-306, FE-308)	15
Technical general-education work suited to challenge examinations (GE-105, GE-107, GE-113)	9
Proposed PLA submission (within the 25–30% cap)	**45**
Balance to be earned in residence or by examination toward 120	75

Where the institution's cap is lower (e.g. 30 credits), the reviewer should select the highest-value blocks first: CJ-209 through CJ-214 (post-quantum security, distributed-ledger audit, trust law, public finance, policy design, digital evidence) are the courses least likely to be available elsewhere and most strongly evidenced.

12.3 Least-cost, most-transferable alternatives

For blocks that can be converted by examination rather than portfolio, the portfolio recommends the cheaper route: standardised or departmental challenge examinations for the quantitative general-education courses (GE-104, GE-105), and validated proficiency assessment for the technical courses, reserving the portfolio method for the upper-division area-of-study work that no examination can capture.

12.4 Current-authority check

All figures in this volume are the author's own. Before submission the following should be verified against the target institution's current catalog: its PLA cap and fee, its residence requirement, its

upper-division minimum, and its capstone rules. These change; this portfolio states a 120-credit structure and a 25–30% cap because those are common and defensible, not because any single institution guarantees them.

A General-Education Outcome Coverage Matrix

Outcome area	**Courses**	**Cr.**
Written & Oral Communication	GE-101, GE-102, GE-103	9
Quantitative Reasoning & Statistics	GE-104, GE-105	6
Natural Sciences with Laboratory	GE-106, GE-107	7
Social & Behavioral Sciences	GE-108, GE-109, GE-113	9
Humanities & Ethics	GE-110	3
Information Literacy, Diversity & Global Studies	GE-111, GE-112	6
Applied Science & Sustainability	GE-114	2
Total general education		**42**

Every general-education course appears in exactly one outcome area, so the seven rows sum to the 42 general-education credits without double counting. GE-113 (Research Methods) is counted once, under the social and behavioral sciences.

B Statement of Authenticity and Limitations

Authenticity. The artefacts cited in this portfolio exist in the consolidated volume (Parts I–V). Where an artefact is an argument, the credit is for the ability to make the argument, not for its correctness. Where an artefact is a computation or simulation, it is labelled as such in Part I and is not offered as validated measurement.

No credential conferred. This portfolio confers no credit, grade, or degree. It is a submission for assessment. Any credit awarded, and any degree granted, is the act of an accredited institution.

Register discipline preserved. Every artefact retains the four-register labelling of Part I (adopted law / proposal / model–simulation / design target). A reviewer assessing "Cryptography & Post-Quantum Security" should read the PQC material as *standards-based design*, and a reviewer assessing "Public Finance" should read the yield figures as *model output*, in each case per the labels.

Currency. The portfolio reflects work current to October 2026. Institutional policies — PLA caps, residence rules, upper-division minimums — must be re-verified at the target institution before submission.